



US012408915B2

(12) **United States Patent**
Souls et al.

(10) **Patent No.:** **US 12,408,915 B2**

(45) **Date of Patent:** **Sep. 9, 2025**

(54) **APPLICATOR INSTRUMENTS FOR DISPENSING SURGICAL FASTENERS, THE APPLICATOR INSTRUMENTS BEING CONTROLLED BY SURGICAL ROBOTS**

(71) Applicant: **Ethicon, Inc.**, Raritan, NJ (US)

(72) Inventors: **Doug Souls**, Andover, NJ (US); **Simon Cohn**, Lebanon, NJ (US); **Michael Cardinale**, Morristown, NJ (US); **Jianxin Guo**, Livingston, NJ (US); **Mark D. Kenyon**, Ringoes, NJ (US); **Jared Schneider**, Union, NJ (US); **Danial Paul Ferreira**, Woodbridge, CT (US); **Xuelin Guo**, Livingston, NJ (US); **Lindsay Epstein**, Belle Mead, NJ (US); **Michael Nordmeyer**, Pittstown, NJ (US)

(73) Assignee: **Ethicon, Inc.**, Raritan, NJ (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **17/965,488**

(22) Filed: **Oct. 13, 2022**

(65) **Prior Publication Data**

US 2023/0035583 A1 Feb. 2, 2023

Related U.S. Application Data

(63) Continuation of application No. 15/494,012, filed on Apr. 21, 2017, now Pat. No. 11,490,888.

(Continued)

(51) **Int. Cl.**

A61B 17/068 (2006.01)

A61B 17/00 (2006.01)

(Continued)

(52) **U.S. Cl.**

CPC **A61B 17/068** (2013.01); **A61B 17/00234** (2013.01); **A61B 17/064** (2013.01);

(Continued)

(58) **Field of Classification Search**

CPC **A61B 2017/07271**; **A61B 17/00234**; **A61B 17/072**

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

3,708,095 A 1/1973 Briggs

3,740,994 A 6/1973 De Carlo

(Continued)

FOREIGN PATENT DOCUMENTS

CA 2464287 A1 5/2003

CN 1280472 A 1/2001

(Continued)

OTHER PUBLICATIONS

Declaration, Declaration of Non-Establishment of International Search Report issued in the corresponding International Application No. PCT/US2017/061284, dated Apr. 19, 2018, 2 pages.

(Continued)

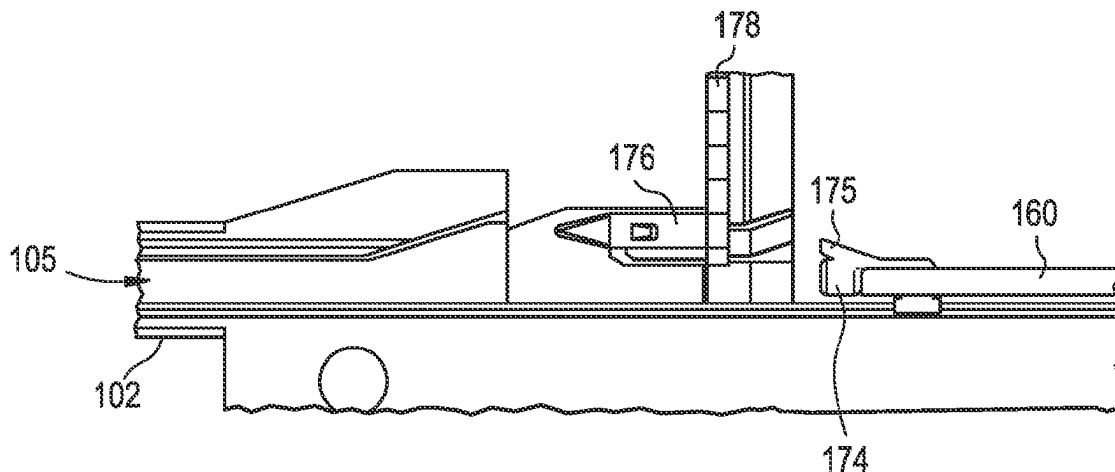
Primary Examiner — Gloria R Weeks

(74) *Attorney, Agent, or Firm* — Amir Bishara

(57) **ABSTRACT**

A method of dispensing surgical fasteners includes providing an applicator instrument having a housing with a cartridge opening, an elongated shaft extending from the housing, a firing system disposed within the housing for dispensing surgical fasteners, and an actuator for activating the firing system. The method includes forming a surgical opening in tissue, inserting a distal end of the elongated shaft into the surgical opening, inserting a cartridge containing a plurality of first surgical fasteners into the cartridge opening, and engaging the actuator for dispensing one of the first

(Continued)



surgical fasteners from the distal end of the elongated shaft. While maintaining the distal end of the elongated shaft inside the surgical opening, the cartridge is removed from the cartridge opening, a second cartridge containing a plurality of second surgical fasteners is inserted into the cartridge opening, and the actuator is engaged for dispensing one of the second surgical fasteners from the distal end of the elongated shaft.

24 Claims, 56 Drawing Sheets

Related U.S. Application Data

(60) Provisional application No. 62/431,355, filed on Dec. 7, 2016.

(51) Int. Cl.

A61B 17/064 (2006.01)
A61F 2/02 (2006.01)
A61B 17/072 (2006.01)
A61B 17/29 (2006.01)
A61B 34/30 (2016.01)
A61B 34/37 (2016.01)
A61B 90/00 (2016.01)
A61B 90/92 (2016.01)

(52) U.S. Cl.

CPC *A61B 17/0682* (2013.01); *A61F 2/02* (2013.01); *A61B 2017/003* (2013.01); *A61B 2017/00314* (2013.01); *A61B 2017/00327* (2013.01); *A61B 2017/00407* (2013.01); *A61B 2017/0053* (2013.01); *A61B 2017/0688* (2013.01); *A61B 17/07207* (2013.01); *A61B 2017/291* (2013.01); *A61B 2017/2923* (2013.01); *A61B 2017/2925* (2013.01); *A61B 2017/2927* (2013.01); *A61B 2034/306* (2016.02); *A61B 34/37* (2016.02); *A61B 2090/0808* (2016.02); *A61B 90/92* (2016.02)

(56) References Cited

U.S. PATENT DOCUMENTS

4,127,227 A 11/1978 Green
 4,152,920 A 5/1979 Green
 4,325,376 A 4/1982 Klieman et al.
 4,471,780 A 9/1984 Menges et al.
 4,478,220 A 10/1984 Di Giovanni et al.
 D286,180 S 10/1986 Korthoff
 D286,442 S 10/1986 Crossley et al.
 4,724,839 A 2/1988 Bedi et al.
 4,832,245 A 5/1989 Terayama et al.
 4,932,960 A 6/1990 Green et al.
 4,951,860 A 8/1990 Peters et al.
 5,174,487 A 12/1992 Rothfuss et al.
 5,190,560 A 3/1993 Woods et al.
 5,203,864 A 4/1993 Phillips
 5,271,544 A 12/1993 Fox et al.
 5,290,297 A 3/1994 Phillips
 5,368,599 A 11/1994 Hirsch
 5,443,197 A 8/1995 Malis et al.
 5,465,895 A 11/1995 Knodel
 5,470,010 A 11/1995 Rothfuss et al.
 5,522,797 A 6/1996 Grimm
 5,562,682 A 10/1996 Oberlin et al.
 5,582,616 A 12/1996 Bolduc et al.
 5,601,573 A 2/1997 Fogelberg
 5,680,981 A 10/1997 Mililli et al.
 5,692,665 A 12/1997 Lee
 5,704,534 A 1/1998 Huitema et al.

5,810,882 A 9/1998 Bolduc et al.
 5,817,054 A 10/1998 Grimm
 5,830,221 A 11/1998 Stein
 5,833,700 A 11/1998 Fogelberg et al.
 5,878,193 A 3/1999 Wang et al.
 5,921,997 A 7/1999 Fogelberg et al.
 6,132,368 A * 10/2000 Cooper A61B 34/76
 606/1
 6,319,258 B1 11/2001 Mcallen, III et al.
 6,457,625 B1 10/2002 Toermelaet et al.
 6,457,645 B1 10/2002 Gardner, Jr.
 6,551,333 B2 4/2003 Kuhns et al.
 6,626,916 B1 9/2003 Yeung et al.
 6,702,737 B2 3/2004 Hino et al.
 6,746,450 B1 6/2004 Wall et al.
 7,044,353 B2 5/2006 Mastri et al.
 7,485,124 B2 2/2009 Kuhns et al.
 7,524,320 B2 4/2009 Tierney et al.
 7,533,790 B1 5/2009 Knodel et al.
 7,547,312 B2 6/2009 Bauman et al.
 7,722,610 B2 5/2010 Viola et al.
 7,758,612 B2 7/2010 Shipp
 7,776,060 B2 8/2010 Mooradian et al.
 7,938,307 B2 5/2011 Bettuchi
 8,091,755 B2 1/2012 Kayan et al.
 8,123,766 B2 2/2012 Bauman et al.
 8,123,767 B2 2/2012 Bauman et al.
 8,167,895 B2 5/2012 D'Agostino et al.
 8,215,533 B2 7/2012 Viola et al.
 8,236,011 B2 * 8/2012 Harris A61B 90/94
 606/232
 8,241,320 B2 8/2012 Lyons
 8,418,908 B1 4/2013 Beardsley
 8,430,291 B2 4/2013 Heinrich
 8,464,925 B2 6/2013 Hull et al.
 8,474,679 B2 7/2013 Felix
 8,529,819 B2 9/2013 Ostapoff et al.
 8,579,920 B2 11/2013 Nering et al.
 8,657,176 B2 2/2014 Shelton, IV et al.
 8,679,137 B2 3/2014 Bauman et al.
 8,690,889 B2 4/2014 Colesanti et al.
 8,714,352 B2 * 5/2014 Farascioni A61B 17/068
 206/363
 8,728,098 B2 5/2014 Daniel et al.
 8,728,099 B2 5/2014 Cohn et al.
 8,740,919 B2 6/2014 Straehnz et al.
 8,789,739 B2 7/2014 Swensgard
 8,894,669 B2 * 11/2014 Nering A61B 17/10
 606/151
 8,920,439 B2 12/2014 Cardinale et al.
 8,967,445 B2 * 3/2015 Kostrzewski A61B 17/068
 227/176.1
 8,986,287 B2 3/2015 Park et al.
 9,005,243 B2 4/2015 Stopek
 9,010,609 B2 4/2015 Carter et al.
 9,010,612 B2 4/2015 Stevenson
 9,038,880 B1 5/2015 Donohoe
 9,055,945 B2 6/2015 Miksza et al.
 9,113,885 B2 8/2015 Casasanta, Jr. et al.
 9,119,617 B2 9/2015 Souls et al.
 9,119,629 B2 9/2015 Cardinale et al.
 9,125,662 B2 9/2015 Shelton, IV
 9,144,369 B2 9/2015 Ostrovsky et al.
 9,161,753 B2 10/2015 Prior
 9,186,144 B2 11/2015 Stevenson et al.
 D744,646 S 12/2015 Nering et al.
 9,198,561 B2 12/2015 Smith et al.
 9,204,783 B2 12/2015 Kappel et al.
 9,211,134 B2 12/2015 Stroup et al.
 9,289,225 B2 3/2016 Shelton, IV et al.
 9,351,751 B2 5/2016 Malkowski
 9,398,917 B2 7/2016 Whitfield et al.
 9,545,253 B2 1/2017 Worrell
 9,554,802 B2 1/2017 Williams
 9,561,038 B2 2/2017 Shelton, IV et al.
 9,675,395 B2 6/2017 Averous et al.
 9,775,618 B2 10/2017 Bettuchi
 9,782,200 B2 10/2017 Burbank
 D804,666 S 12/2017 Guo et al.

(56)

References Cited**U.S. PATENT DOCUMENTS**

10,117,649	B2	11/2018	Baxter, III et al.	
10,154,840	B2	12/2018	Viola	
10,285,704	B2	5/2019	Prior	
10,357,249	B2	7/2019	Carter	
10,390,827	B2	8/2019	Hodgkinson	
10,499,913	B2	12/2019	Vendely	
10,568,627	B2 *	2/2020	Guo	A61B 17/10
10,743,867	B2	8/2020	Cohn et al.	
10,918,381	B2	2/2021	Guo et al.	
12,089,837	B2	9/2024	Biedermann et al.	
2002/0019636	A1	2/2002	Ogilvie et al.	
2002/0068947	A1	6/2002	Kuhns et al.	
2002/0128669	A1	9/2002	Field et al.	
2003/0009195	A1	1/2003	Field et al.	
2005/0005996	A1	1/2005	Mizutani	
2005/0059996	A1	3/2005	Bauman et al.	
2005/0059997	A1	3/2005	Bauman et al.	
2005/0149064	A1	7/2005	Peterson et al.	
2005/0273108	A1	12/2005	Groiso	
2006/0122635	A1 *	6/2006	Naegeli	A61B 17/068 606/142
2006/0281332	A1	12/2006	Duinveld	
2007/0021737	A1	1/2007	Lee	
2007/0045379	A1	3/2007	Shelton, IV	
2007/0088390	A1	4/2007	Paz et al.	
2007/0142846	A1 *	6/2007	Catanese, III	A61F 2/0063 606/142
2007/0162030	A1	7/2007	Aranyi et al.	
2007/0239278	A1	10/2007	Heinz	
2007/0246505	A1	10/2007	Pace-Florida et al.	
2007/0287993	A1	12/2007	Hinman et al.	
2008/0065153	A1	3/2008	Allard et al.	
2008/0085296	A1	4/2008	Powell et al.	
2009/0088792	A1	4/2009	Hoell, Jr.	
2010/0001036	A1	1/2010	Marczyk et al.	
2010/0012703	A1	1/2010	Calabrese	
2010/0023022	A1 *	1/2010	Zeiner	A61B 17/0401 606/139
2010/0191282	A1	7/2010	Harris et al.	
2010/0270354	A1	10/2010	Rimer et al.	
2010/0292712	A1	11/2010	Nering et al.	
2011/0079627	A1	4/2011	Cardinale	
2011/0082472	A1 *	4/2011	Harris	A61B 17/0469 606/139
2011/0084110	A1	4/2011	Ueno et al.	
2011/0208169	A1	8/2011	Nash	
2012/0080495	A1 *	4/2012	Holcomb	A61B 17/0469 227/176.1
2012/0109186	A1	5/2012	Parrott et al.	
2012/0209254	A1	8/2012	Park et al.	
2012/0241497	A1	9/2012	Mandakolathur et al.	
2012/0241499	A1	9/2012	Baxter, III et al.	
2012/0248169	A1	10/2012	Widenhouse et al.	
2013/0018393	A1	1/2013	Bengtson	
2013/0023868	A1	1/2013	Worrell et al.	
2013/0123816	A1	5/2013	Hodgkinson et al.	
2013/0153638	A1	6/2013	Carter et al.	
2013/0218177	A1	8/2013	Miksza et al.	
2013/0301091	A1	11/2013	Martinez	
2013/0304091	A1	11/2013	Straehnz et al.	
2014/0005662	A1	1/2014	Shelton, IV	
2014/0005694	A1	1/2014	Shelton, IV et al.	
2014/0008412	A1 *	1/2014	Zemlok	A61B 17/07207 227/176.1
2014/0197224	A1	7/2014	Penna	
2014/0217148	A1	8/2014	Penna	
2014/0252065	A1 *	9/2014	Hessler	A61B 17/0682 227/176.1
2014/0257340	A1	9/2014	Ostrovsky et al.	
2014/0276965	A1	9/2014	Ranucci et al.	
2014/0276968	A1	9/2014	Miksza et al.	
2014/0379001	A1	12/2014	Cohn et al.	
2015/0001272	A1	1/2015	Sniffin et al.	
2015/0080919	A1	3/2015	Nering et al.	

2015/0265262	A1	9/2015	Dewaele et al.	
2016/0074028	A1	3/2016	Castro	
2016/0262738	A1	9/2016	Altman	
2016/0287256	A1	10/2016	Marczyk	
2018/0140299	A1	5/2018	Weaner et al.	
2018/0153547	A1	6/2018	Cohn et al.	
2018/0153550	A1	6/2018	Souls et al.	
2018/0153551	A1 *	6/2018	Guo	A61B 17/08

FOREIGN PATENT DOCUMENTS

CN	1644174	A	7/2005
CN	101040798	A	9/2007
CN	101273908	A	10/2008
CN	101507643	A	8/2009
CN	1907238	B	2/2012
CN	102458268	A	5/2012
CN	103892882	A	7/2014
CN	104337558	A	2/2015
CN	104812987	A	7/2015
CN	105188565	A	12/2015
CN	106061405	A	10/2016
DE	3704759	C1	3/1988
EP	0517488	A1	12/1992
EP	0593920	A1	4/1994
EP	0598529	A2	5/1994
EP	0598592	A2	5/1994
EP	2090254	A1	8/2009
EP	2361562	A1	8/2011
EP	1494596	B1	10/2011
EP	2604195	A1	6/2013
EP	2606838	A1	6/2013
EP	3120781	A2	1/2017
EP	3569162	A1	11/2019
JP	H0663054	A	3/1994
JP	2001523512	A	11/2001
JP	2009082704	A	4/2009
JP	2012526644	A	11/2012
WO	2009058671	A2	5/2009
WO	2012040593	A2	3/2012
WO	2013009993	A1	1/2013
WO	2013119817	A1	8/2013
WO	2013151858	A1	10/2013
WO	2014162440	A1	10/2014
WO	2016007626	A1	1/2016
WO	2016061291	A1	4/2016

OTHER PUBLICATIONS

International Preliminary Report on Patentability issued in the corresponding International Application No. PCT/US2017/061273, dated Jun. 11, 2019, 8 pages.

International Preliminary Report on Patentability issued in the corresponding International Application No. PCT/US2017/061275, dated Jun. 11, 2019, 6 pages.

International Preliminary Report on Patentability issued in the corresponding International Application No. PCT/US2017/061279, dated Jun. 11, 2019, 6 pages.

International Preliminary Report on Patentability issued in the corresponding International Application No. PCT/US2017/061282, dated Jun. 11, 2019, 6 pages.

International Search Report issued in the corresponding International Application No. PCT/US2017/061275, dated Apr. 25, 2018, 5 pages.

International Search Report issued in the corresponding International Application No. PCT/US2017/061264, dated Apr. 20, 2018, 6 pages.

International Search Report issued in the corresponding International Application No. PCT/US2017/061273, dated Apr. 26, 2018, 6 pages.

International Search Report issued in the corresponding International Application No. PCT/US2017/061279, dated Apr. 26, 2018, 5 pages.

International Search Report issued in the corresponding International Application No. PCT/US2017/061282, dated Feb. 26, 2018, 5 pages.

(56)

References Cited**OTHER PUBLICATIONS**

Office Action for related Japanese Application No. 2019-530419, dated Aug. 31, 2001, 5 pages.

Patent Examination Report for related Mexican Application No. MX/a/2019/006613, dated Nov. 30, 2022, 5 pages.

Patent Examination Report No. 1 for related Australian Application No. 2017370465, dated Sep. 21, 2022, 3 pages.

Patent Examination Report No. 1 for related Australian Application No. 2017370466, dated Jun. 27, 2022, 3 pages.

Patent Examination Report No. 1 for related Australian Application No. 2017372463, dated Jun. 17, 2022, 2 pages.

Patent Examination Report No. 1 for related Australian Application No. 2017372464, dated Jun. 25, 2022, 3 pages.

Patent Examination Report No. 1 for related Australian Application No. 2017372465, dated Apr. 28, 2022, 4 pages.

Translation of Notification of Reasons for Refusal for related Japanese Application No. 2019-530395, dated Sep. 7, 2021, 2 pages.

Translation of Notification of Reasons for Refusal for related Japanese Application No. 2019-530424, dated Aug. 31, 2021, 3 pages.

Translation of Search Report for related Chinese Application No. 201780075927.3, dated Jan. 17, 2022, 3 pages.

Translation of Search Report for related Chinese Application No. 201780075939.6, dated Jan. 19, 2022, 2 pages.

Translation of Search Report for related Chinese Application No. 201780075944.7, dated Jan. 18, 2022, 2 pages.

Translation of Search Report for related Chinese Application No. 201780076267.0, dated Jan. 14, 2022, 2 pages.

Translation of Search Report for related Chinese Application No. 201780076281.0, dated May 17, 2022, 3 pages.

Written Opinion of the International Searching Authority issued in the corresponding International Application No. PCT/US2017/061273, dated Apr. 26, 2018, 9 pages.

Written Opinion of the International Searching Authority issued in the corresponding International Application No. PCT/US2017/061275, dated Apr. 25, 2018, 7 pages.

Written Opinion of the International Searching Authority Issued in the corresponding International Application No. PCT/US2017/061279, dated Apr. 26, 2018, 7 pages.

Written Opinion of the International Searching Authority issued in the corresponding International Application No. PCT/US2017/061282, dated Feb. 26, 2018, 7 pages.

Written Opinion of the International Searching Authority issued in the corresponding International Application No. PCT/US2017/061284, dated Apr. 20, 2018, 5 pages.

Extended European Search Report received for European Application No. 17808674.0, mailed on Mar. 3, 2023, 4 pages.

Invitation to pay Additional Fees received for PCT Application No. PCT/US2017/061286, mailed on Mar. 26, 2018, 11 pages.

Extended European Search Report received for European Application No. 17812278.4, mailed on Feb. 27, 2023, 8 pages.

Extended European Search Report received for European Application No. 17812279.2, mailed on Feb. 21, 2023, 7 pages.

Ho et al., "Techniques for colorectal anastomosis", World Journal of Gastroenterology, vol. 16, No. 13, pp. 1610-1621, 2010.

International Search Report and Written Opinion received for PCT Application No. PCT/US2017/024467, mailed on Jun. 16, 2017, 16 pages.

International Search Report and Written Opinion received for PCT Application No. PCT/US2017/061286, mailed on Jun. 25, 2018, 16 pages.

Jönsson et al., "Breaking strength of small intestinal anastomoses", The American Journal of Surgery, vol. 145, No. 06, pp. 800-803, 1983.

* cited by examiner

FIG. 1A

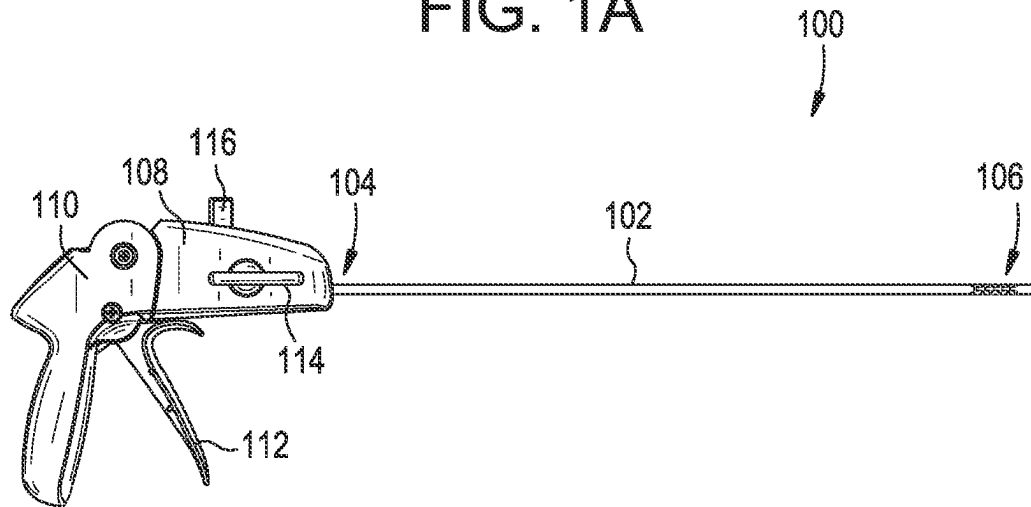


FIG. 1B

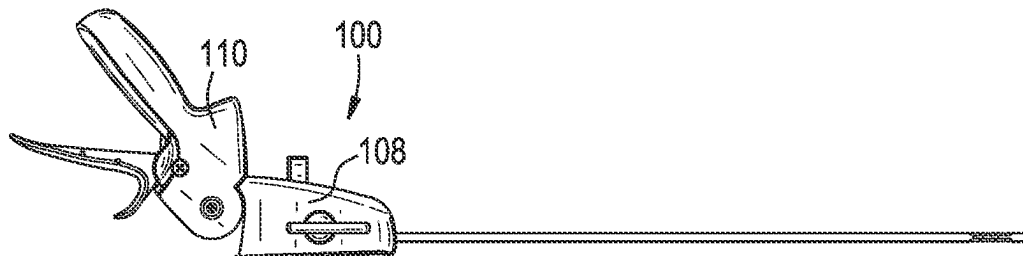


FIG. 2A

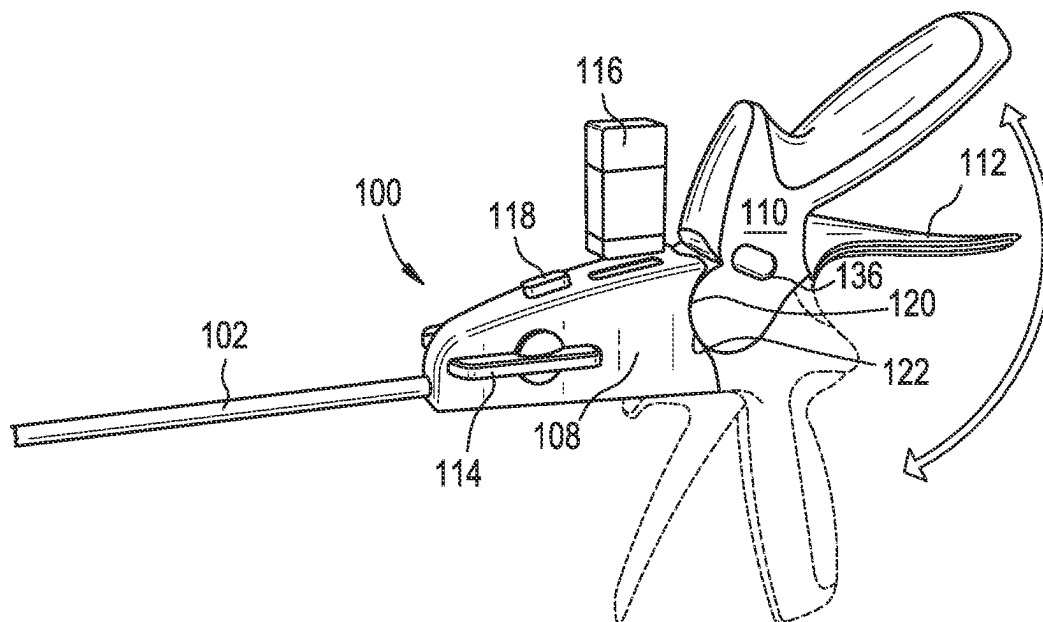


FIG. 2B

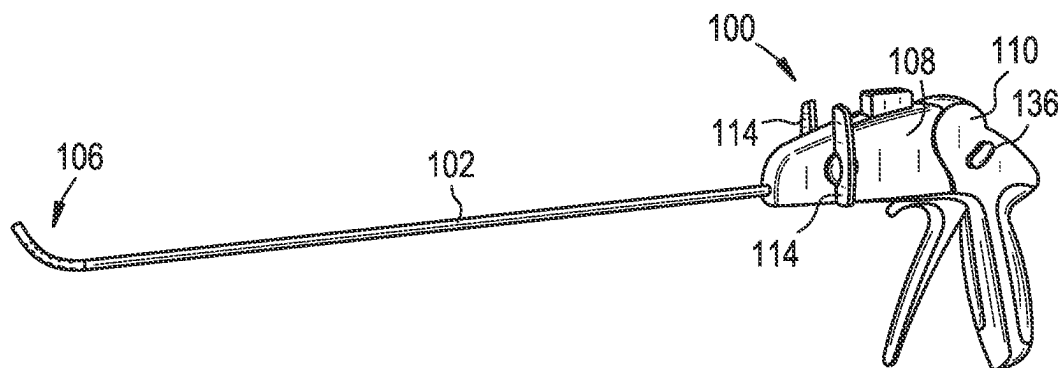


FIG. 3

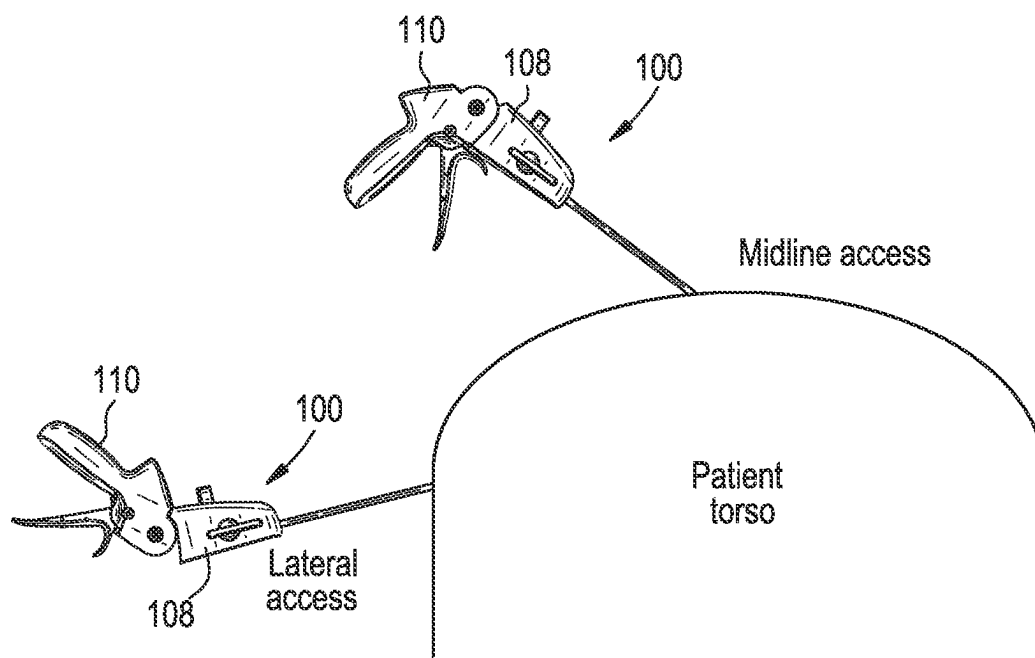


FIG. 4A

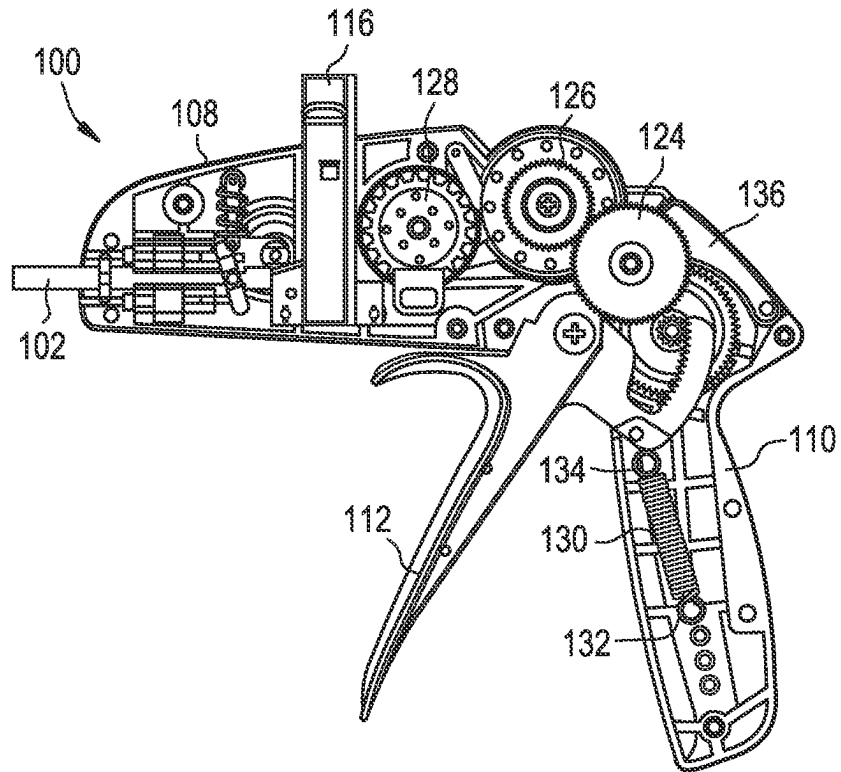


FIG. 4B

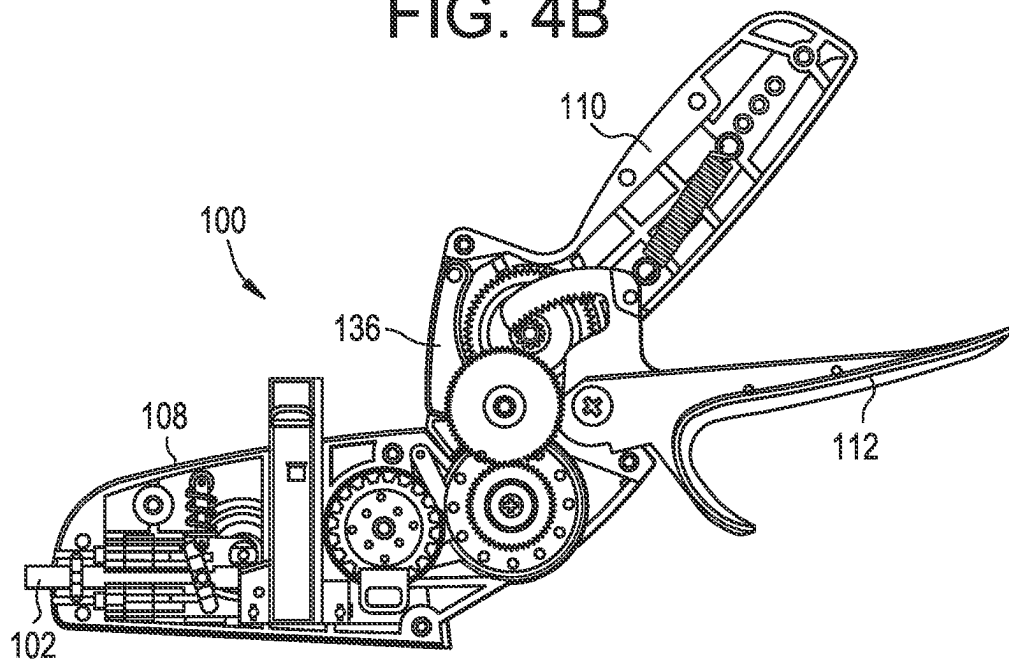


FIG. 5

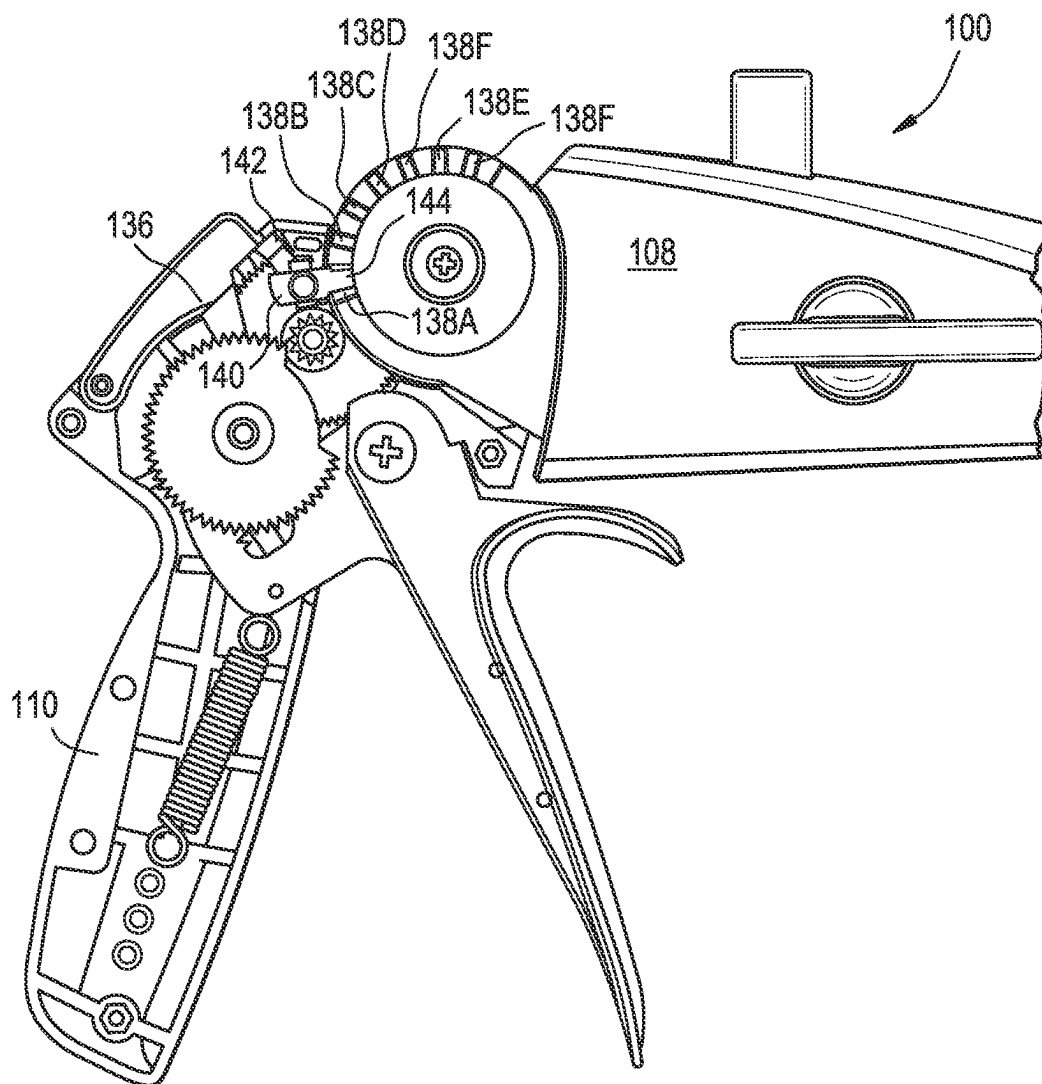


FIG. 6A

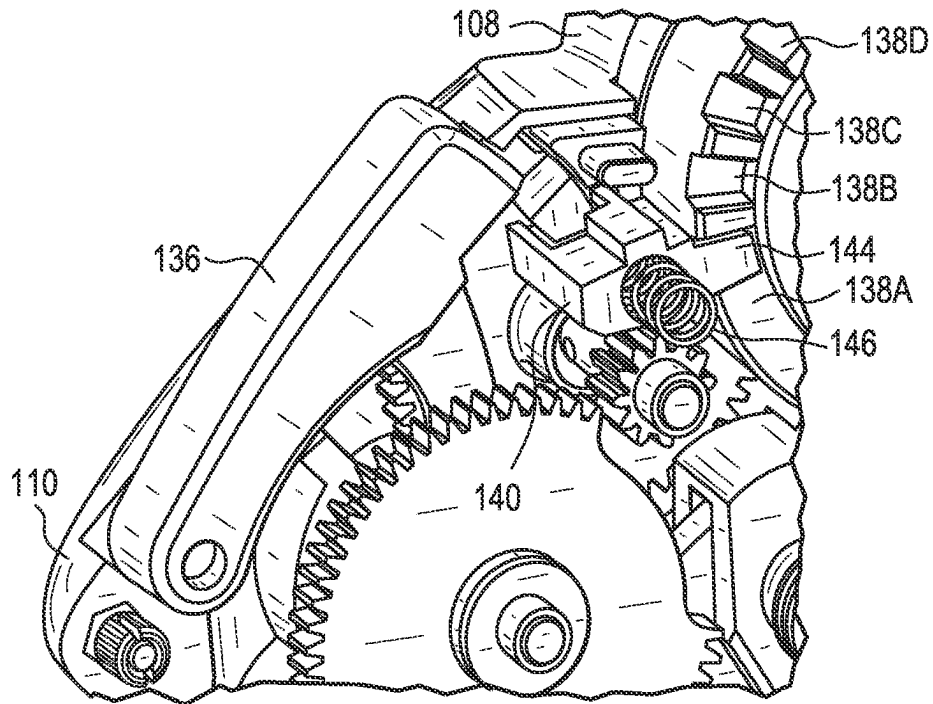


FIG. 6A-1

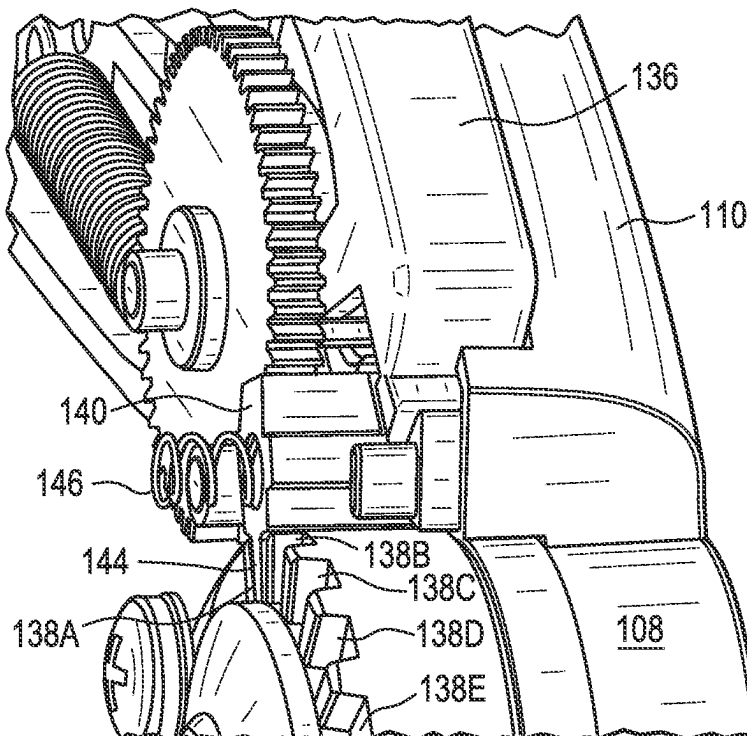


FIG. 6B

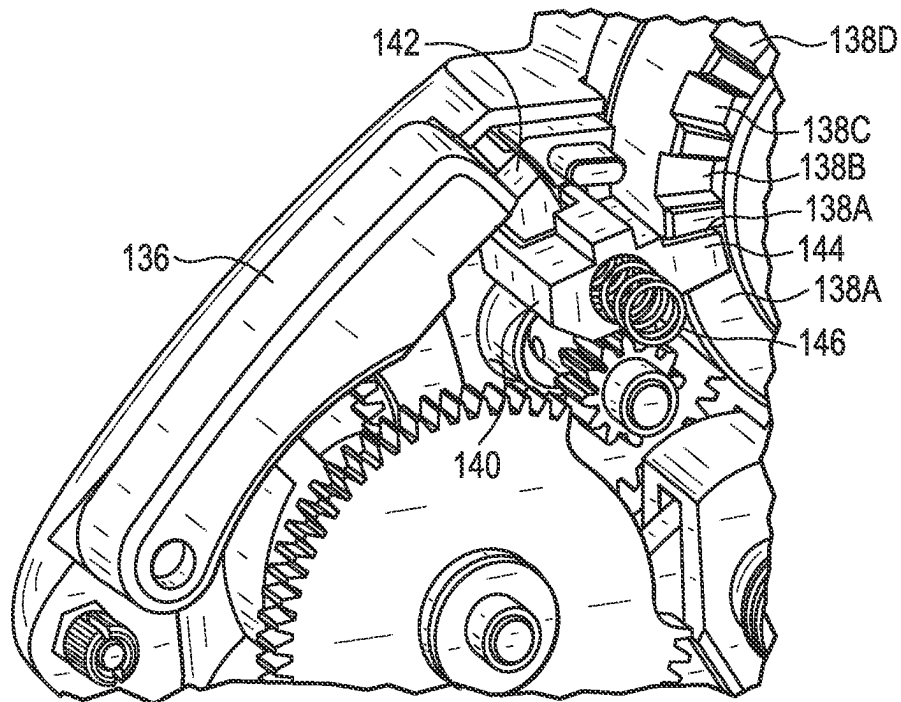


FIG. 6B-1

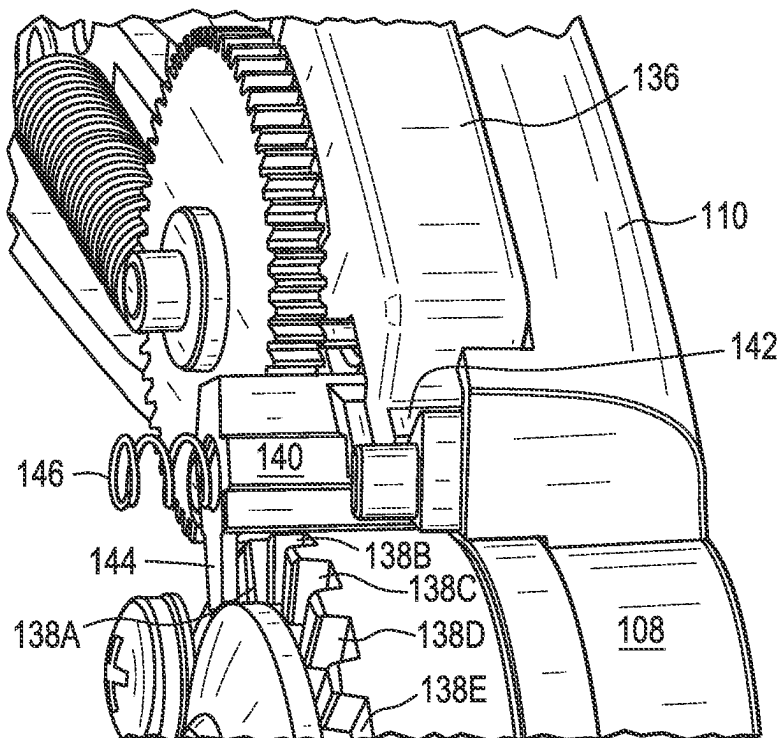


FIG. 7

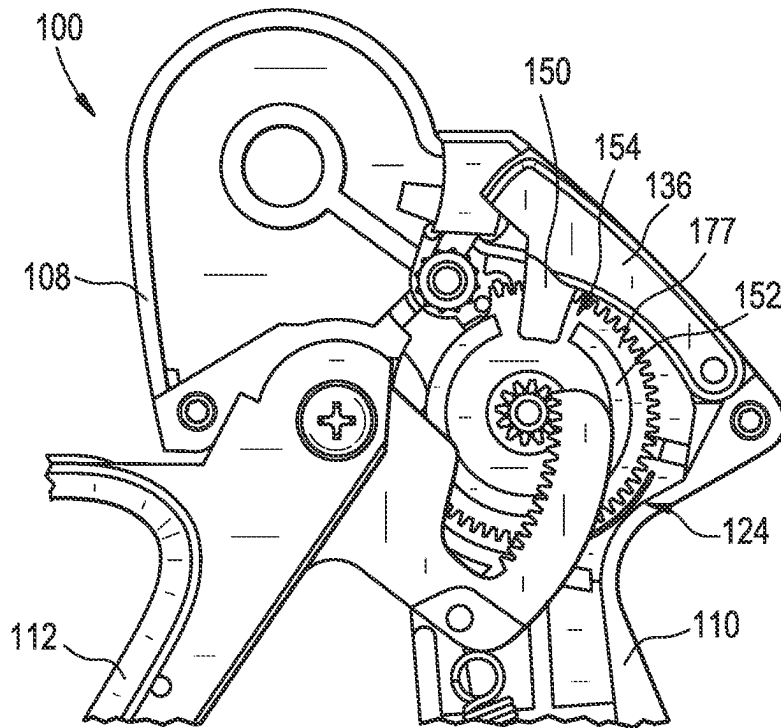


FIG. 8

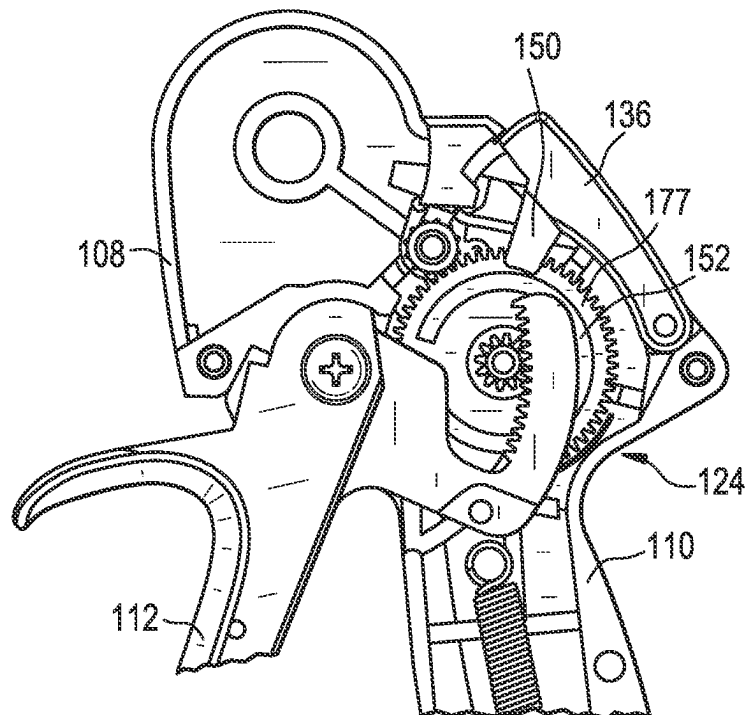


FIG. 9A

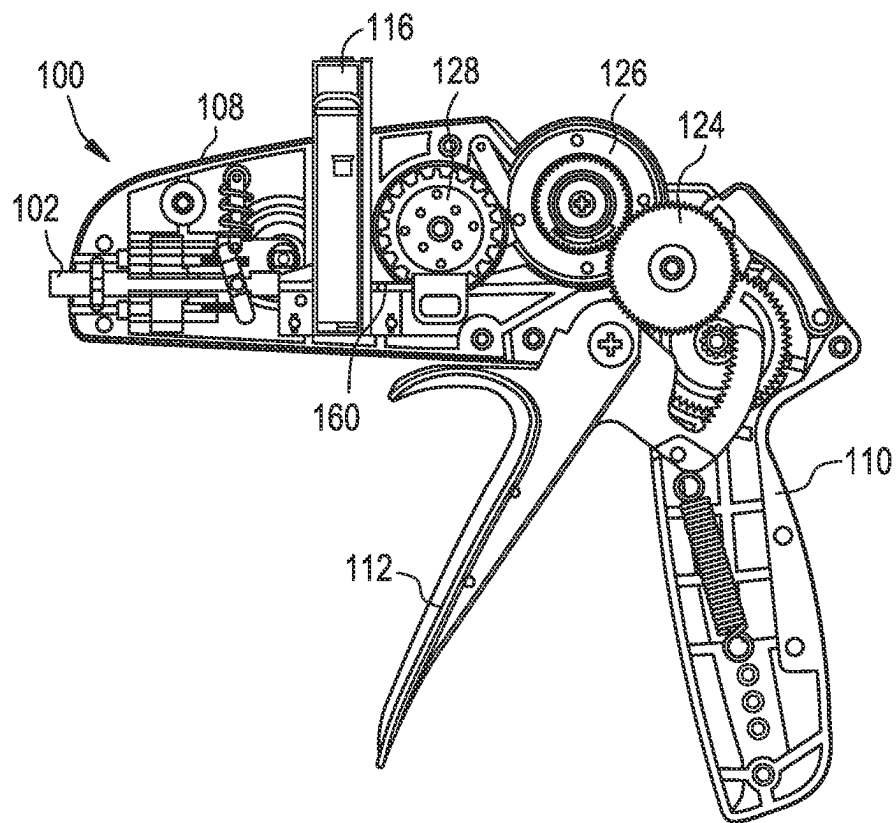


FIG. 9B

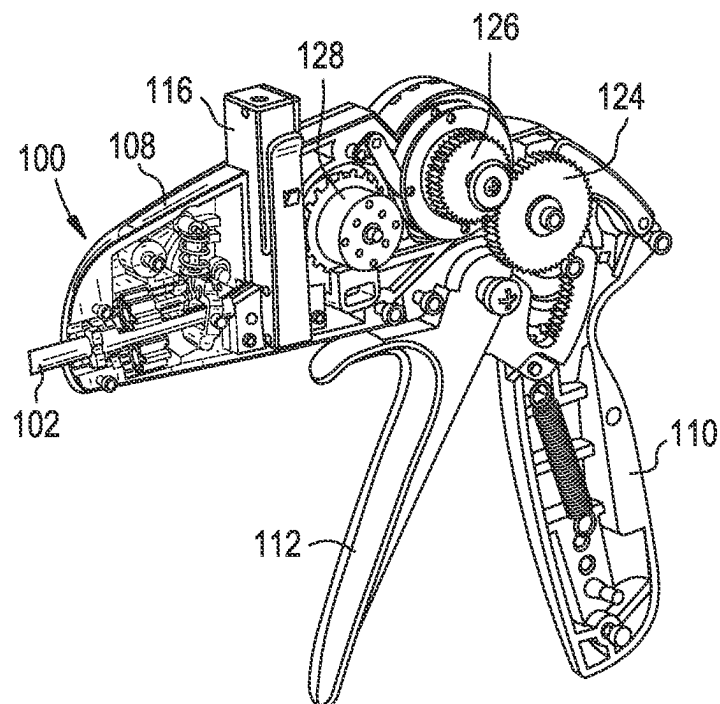


FIG. 10A

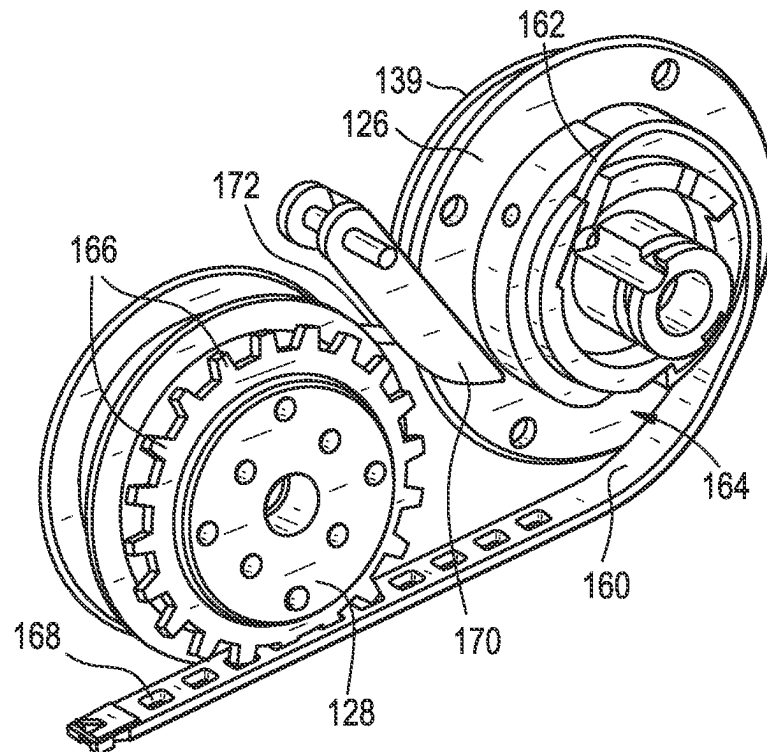


FIG. 10B

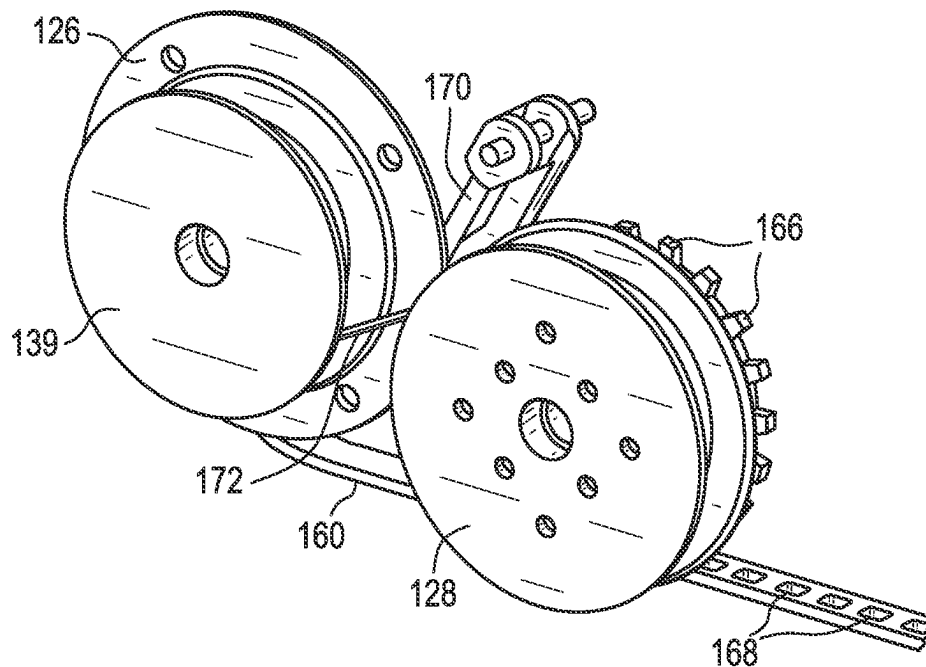


FIG. 11

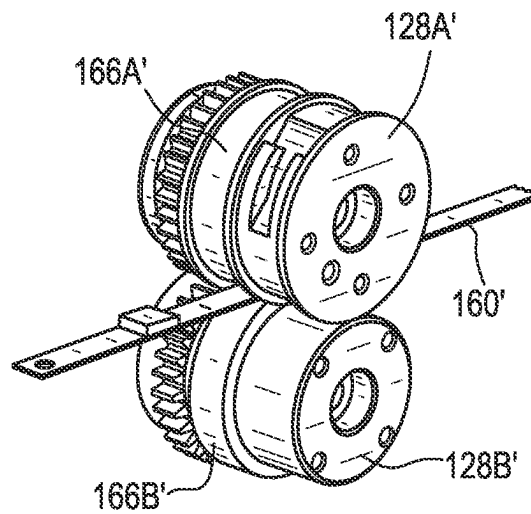


FIG. 12

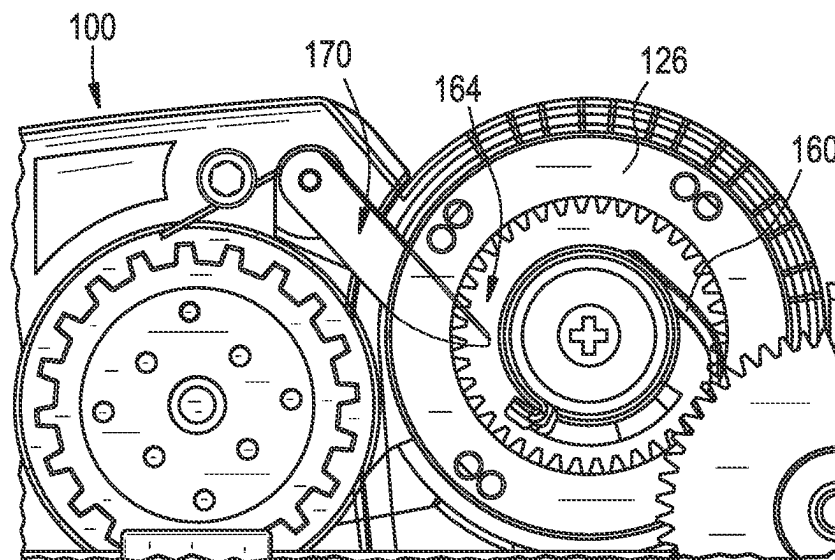


FIG. 13A

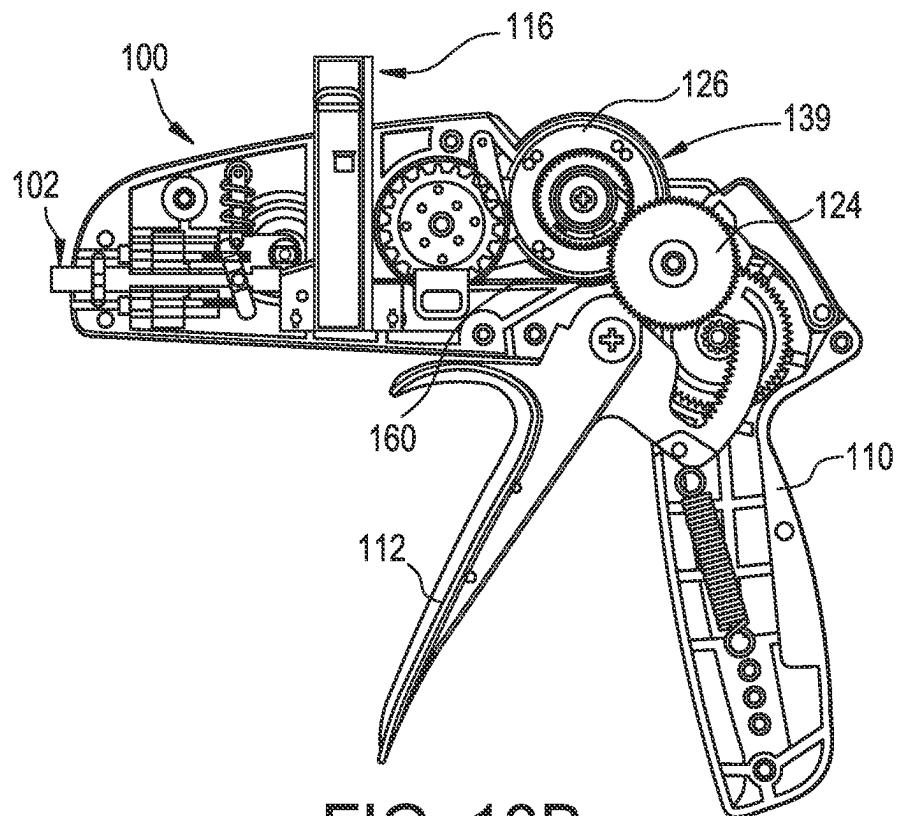


FIG. 13B

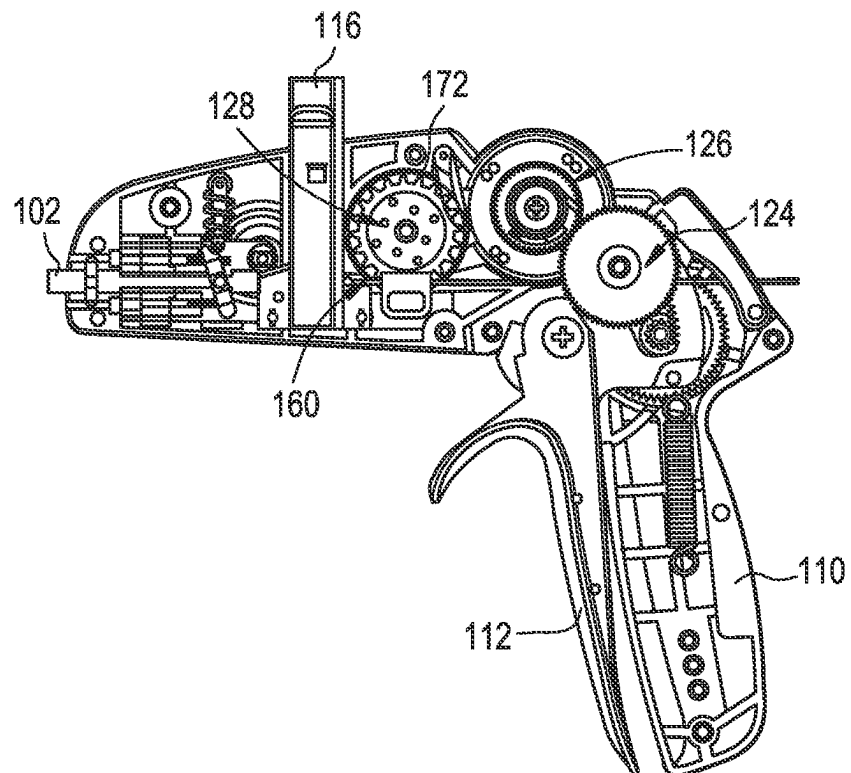


FIG. 14A

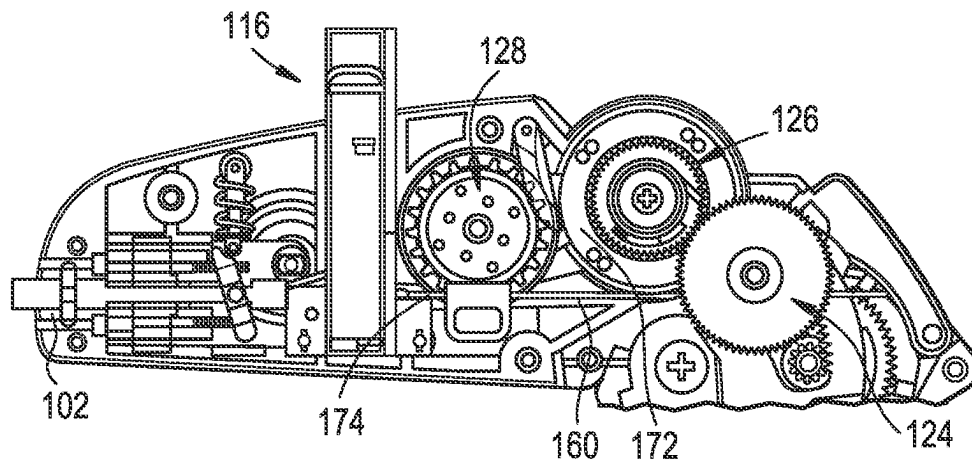


FIG. 14A-1

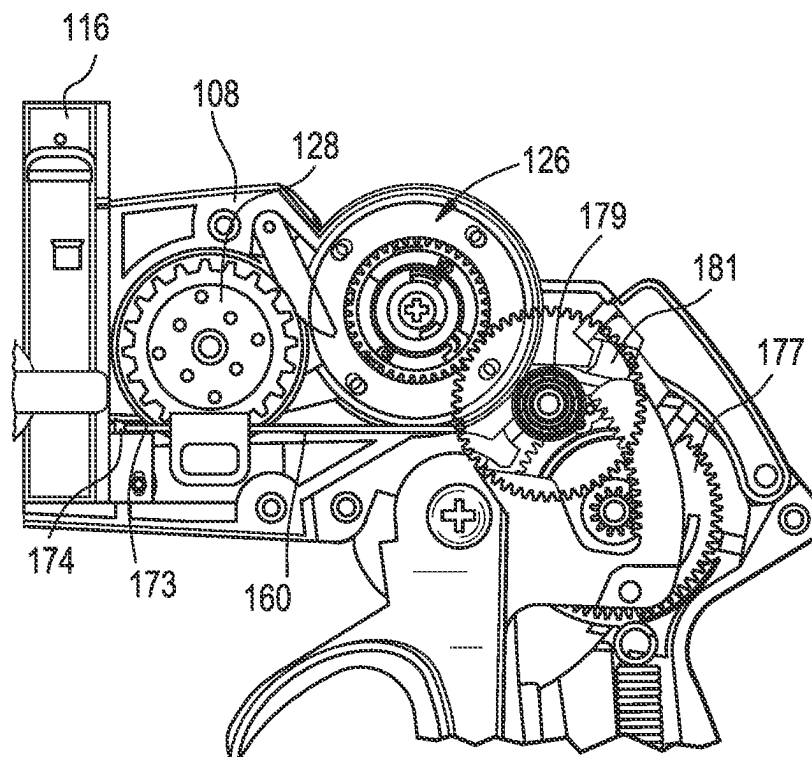


FIG. 14B

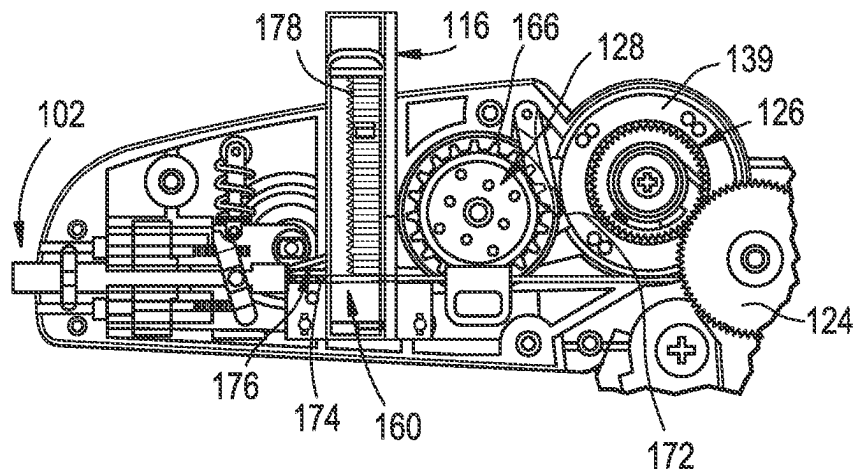


FIG. 14B-1

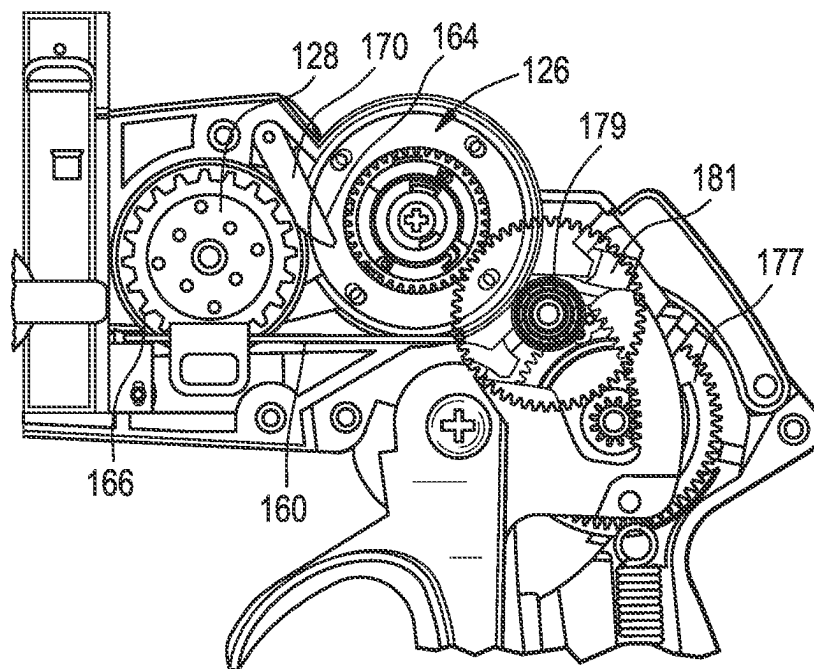


FIG. 15A

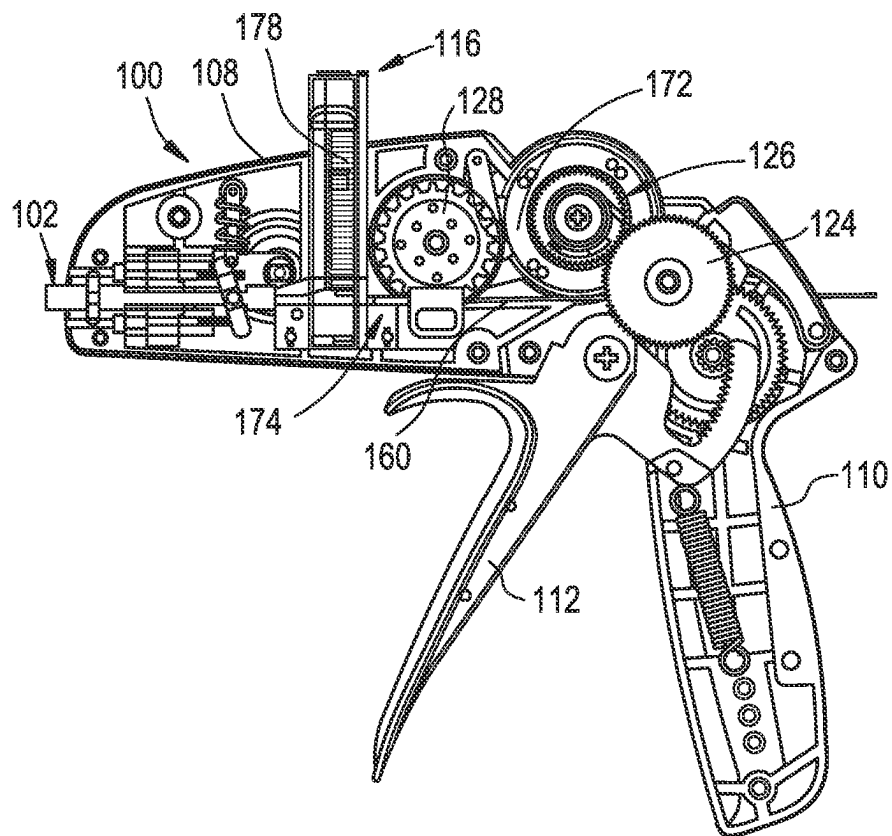


FIG. 15B

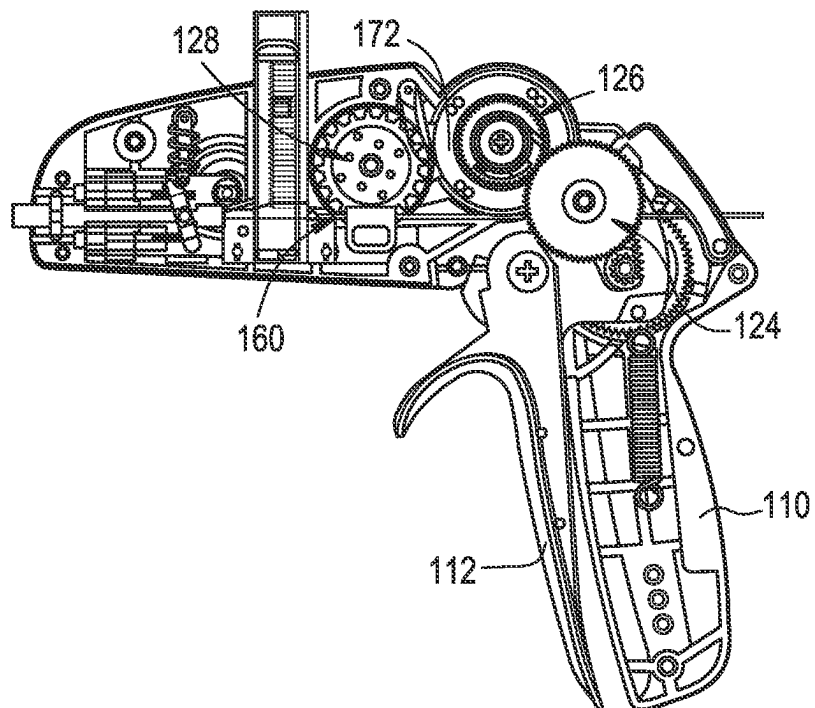


FIG. 16A

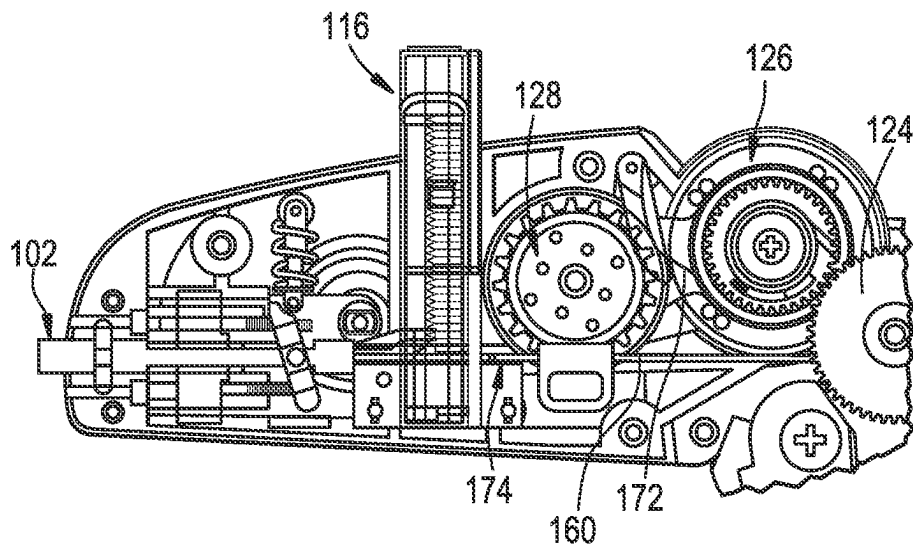


FIG. 16B

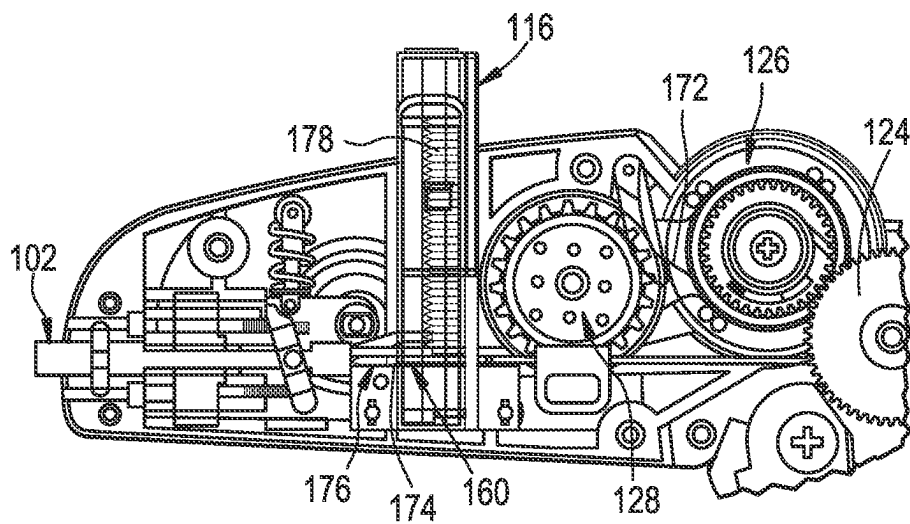


FIG. 16C

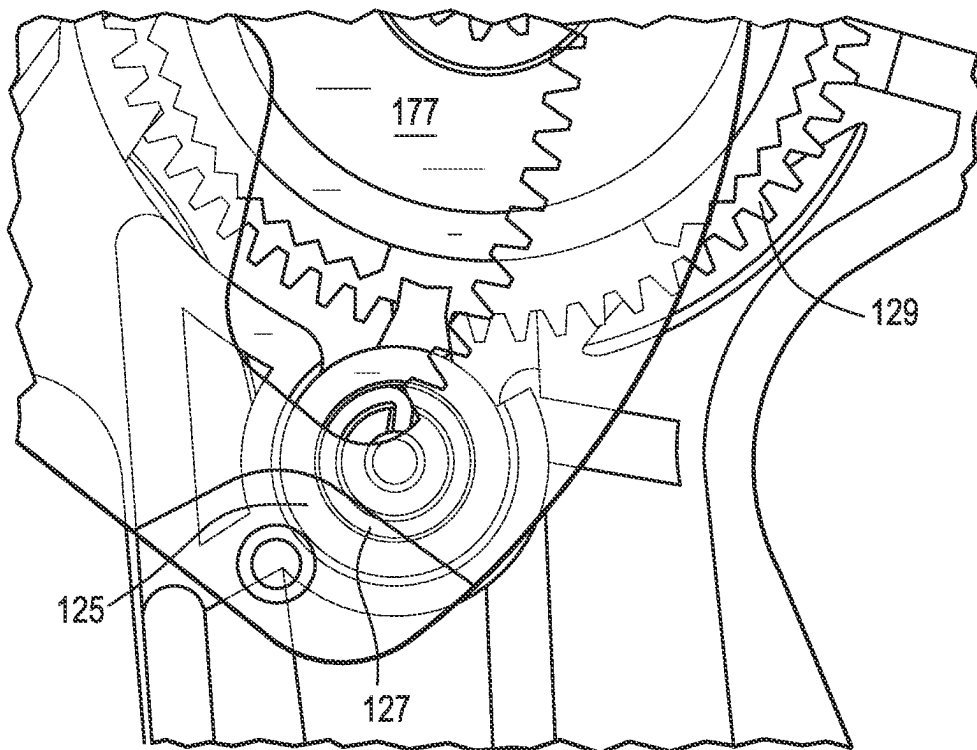


FIG. 16D

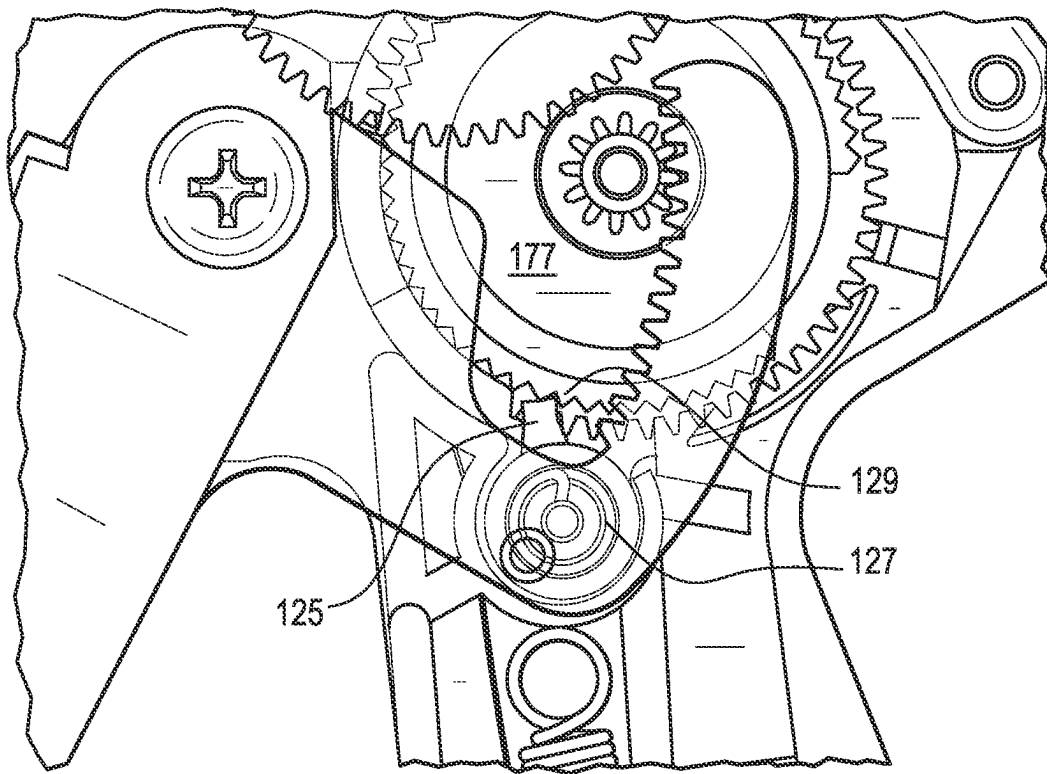


FIG. 17

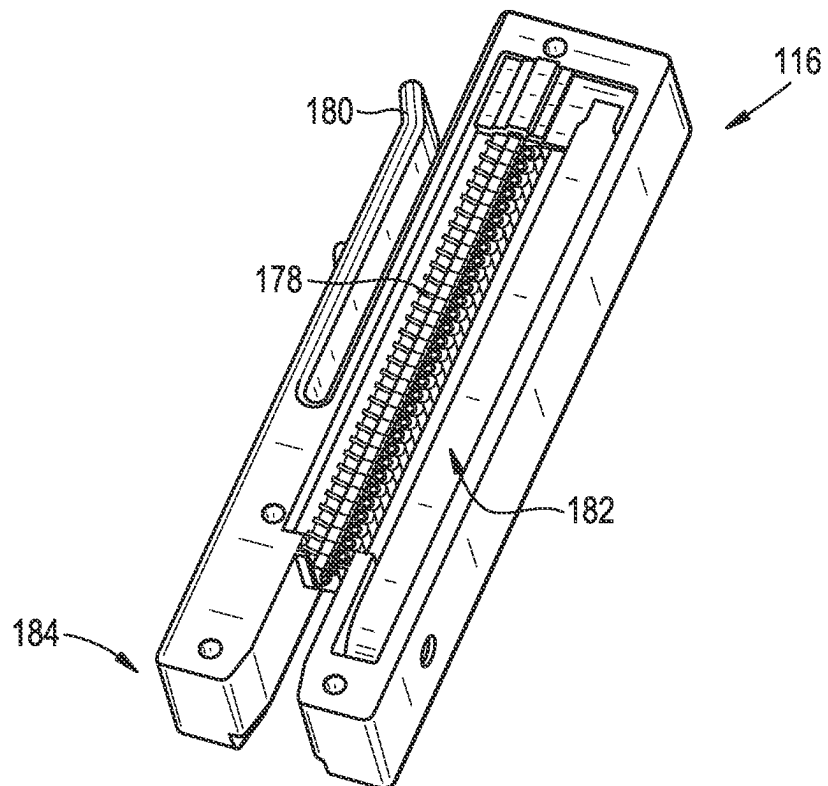


FIG. 18A

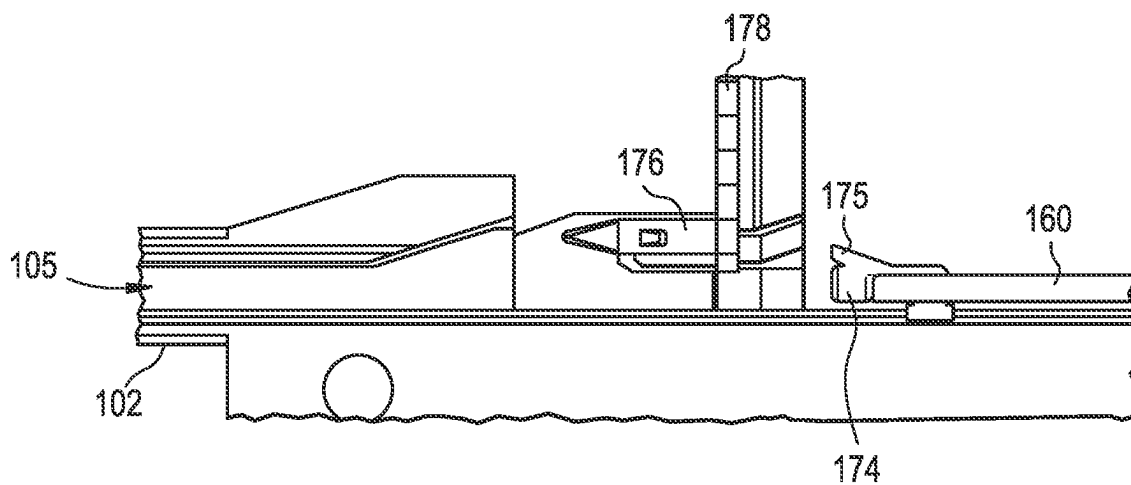


FIG. 18B

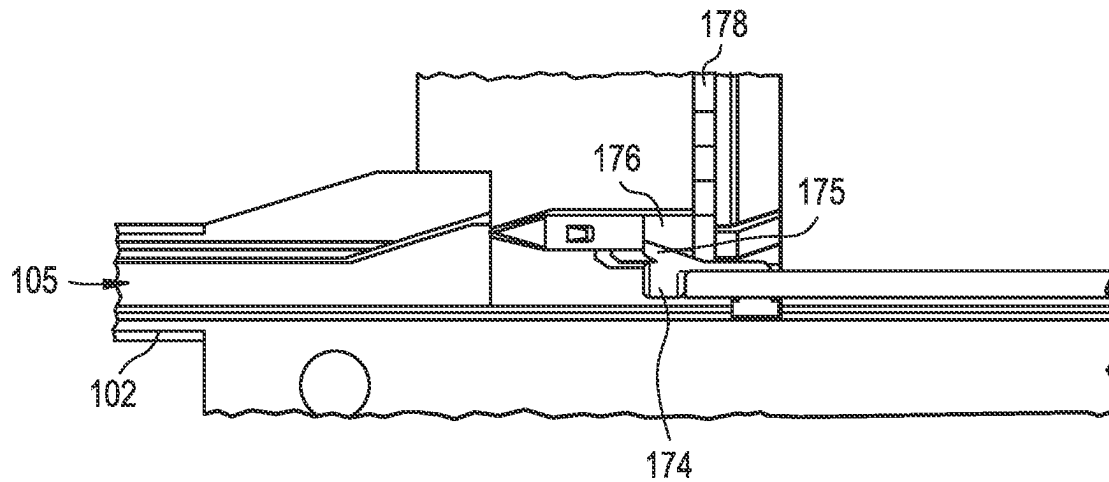


FIG. 18C

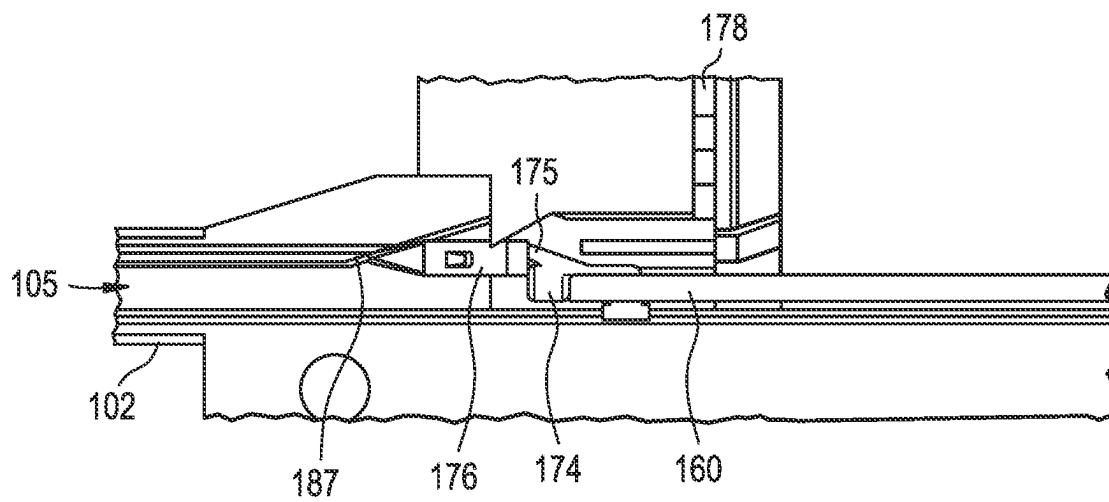


FIG. 18D

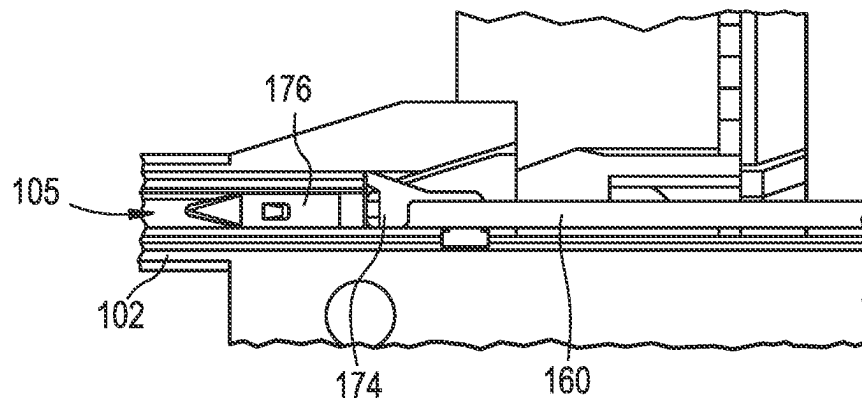


FIG. 19A

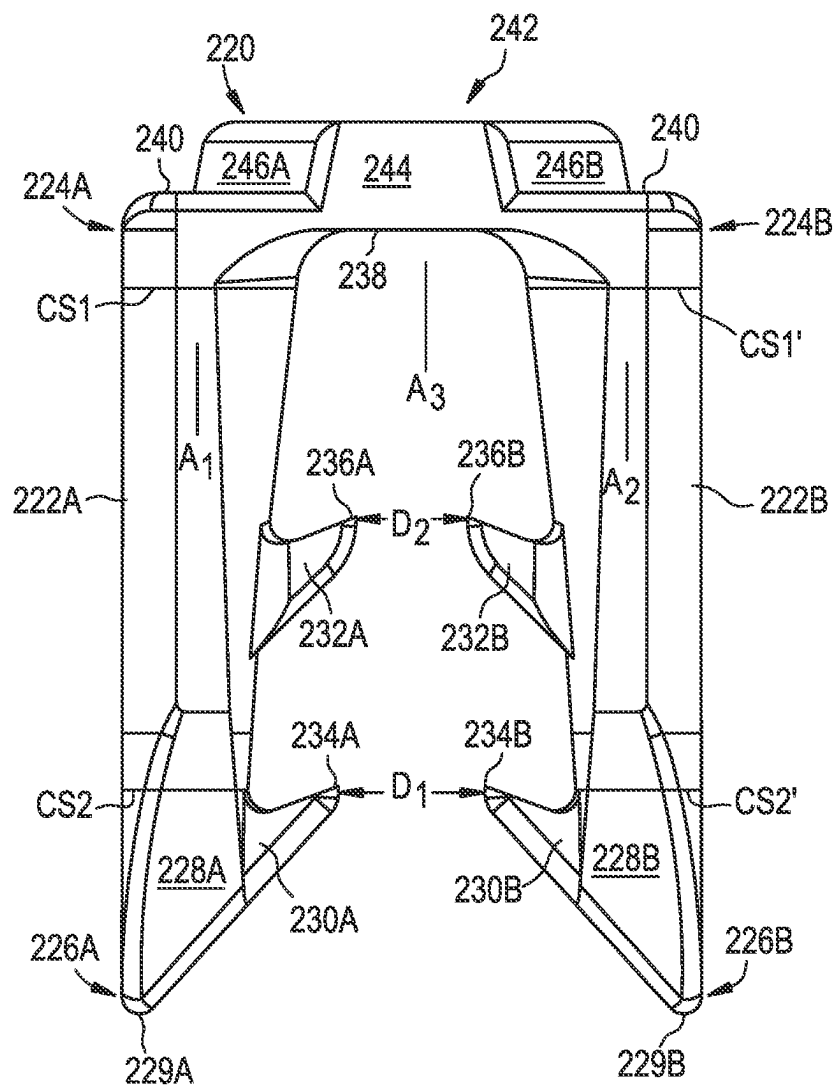


FIG. 19B

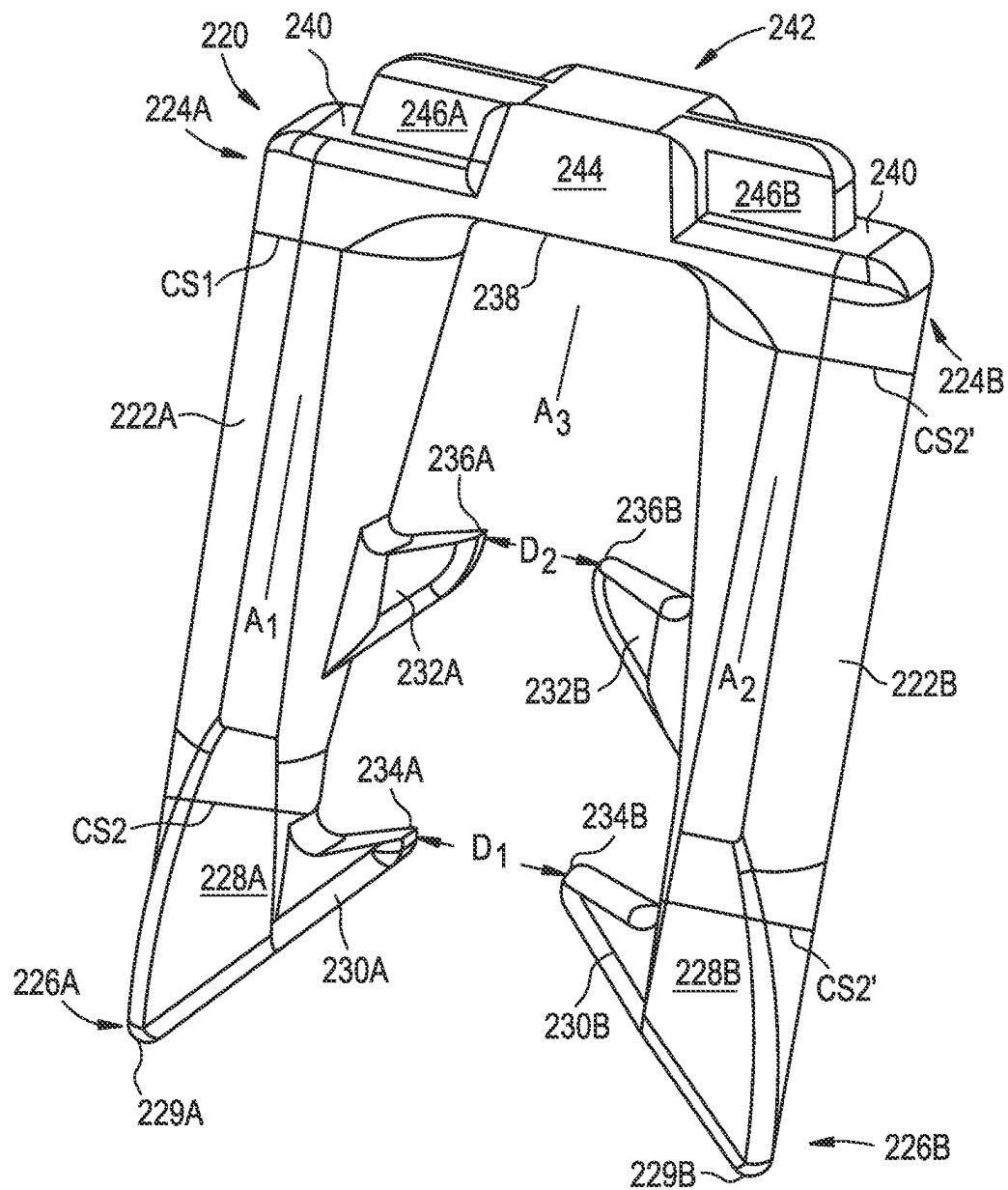


FIG. 19D

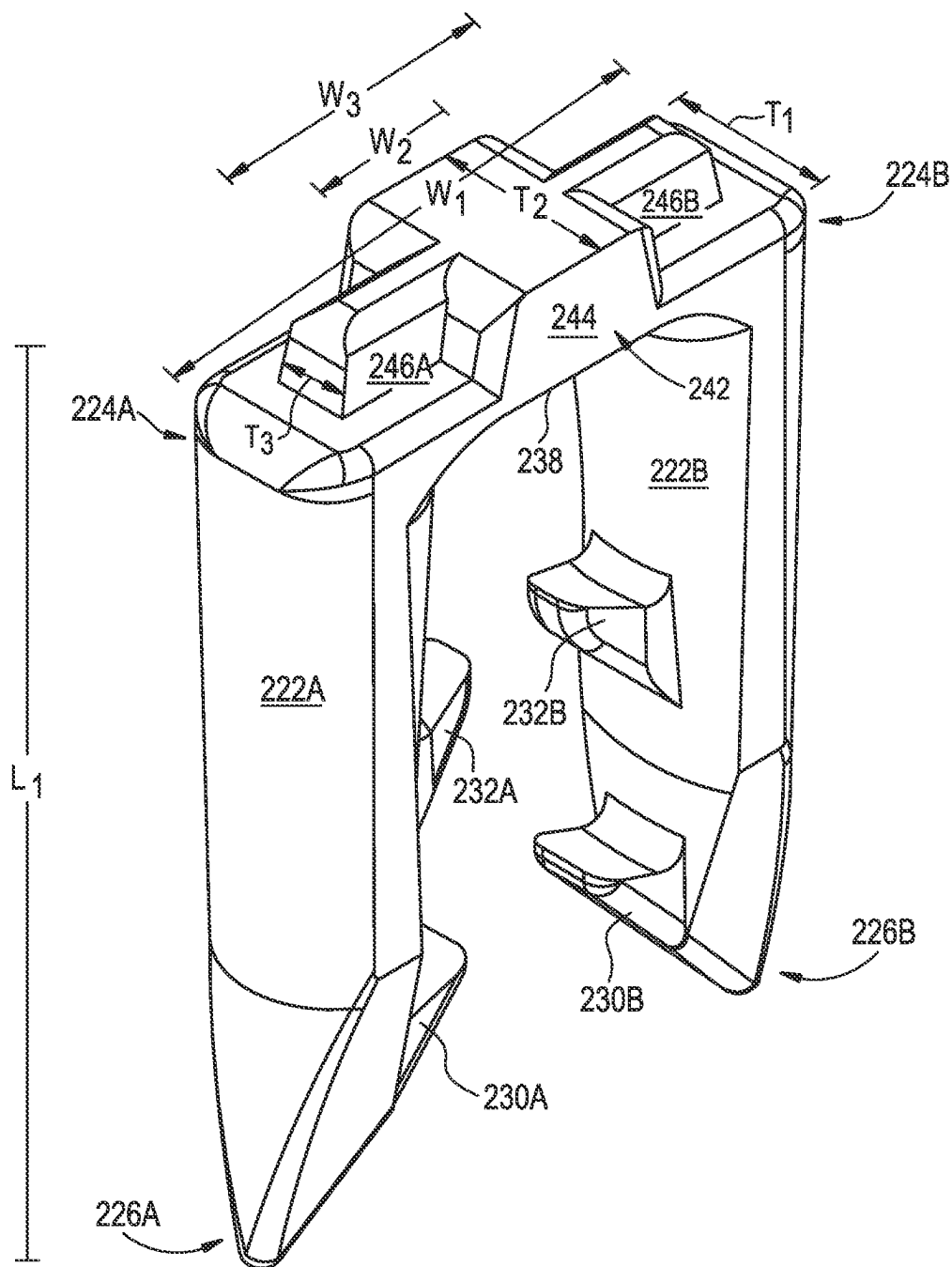


FIG. 19E

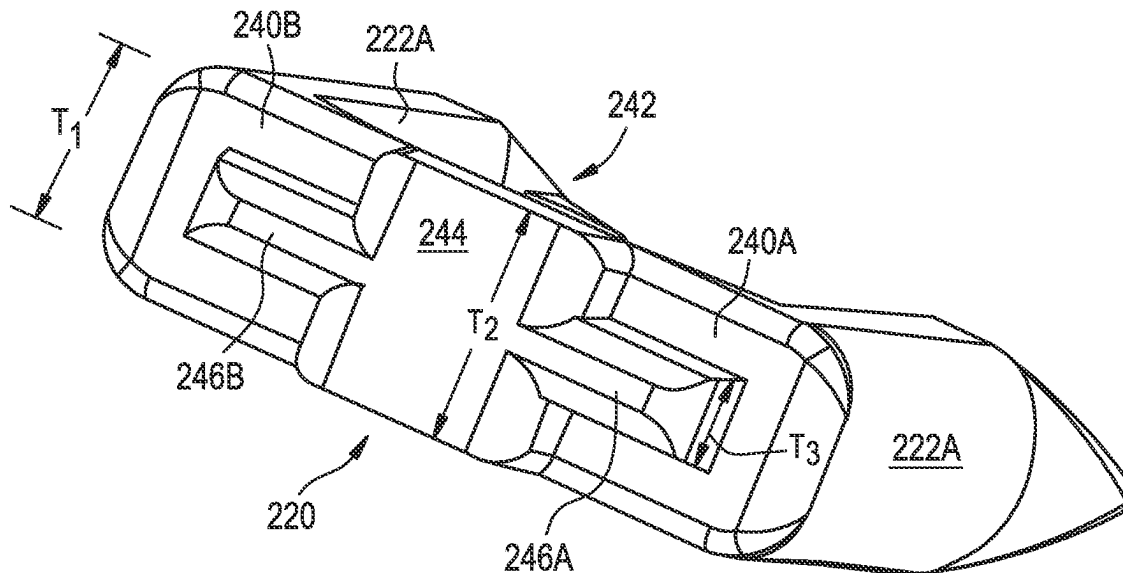


FIG. 19F

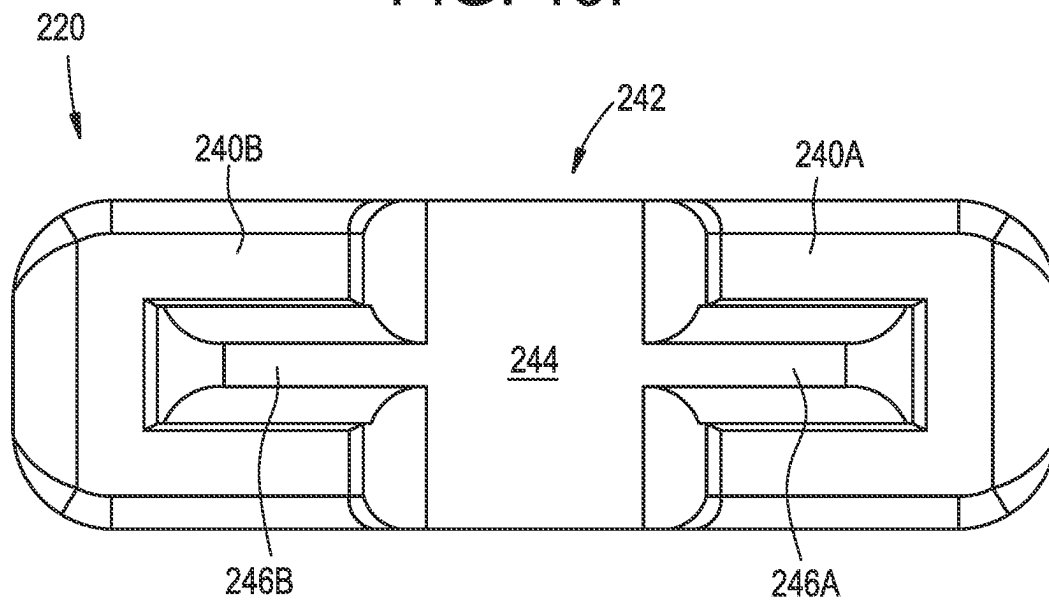


FIG. 19G

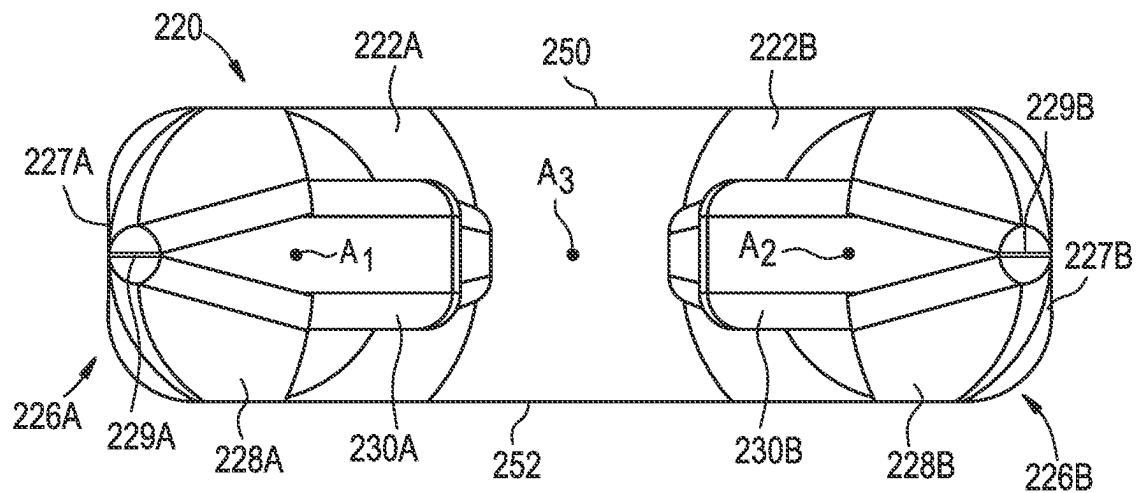


FIG. 19H

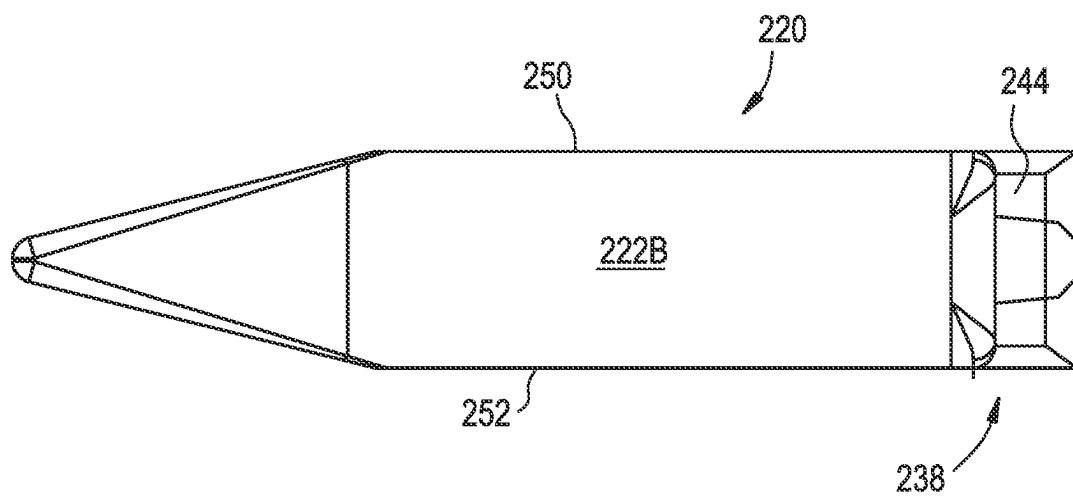


FIG. 20

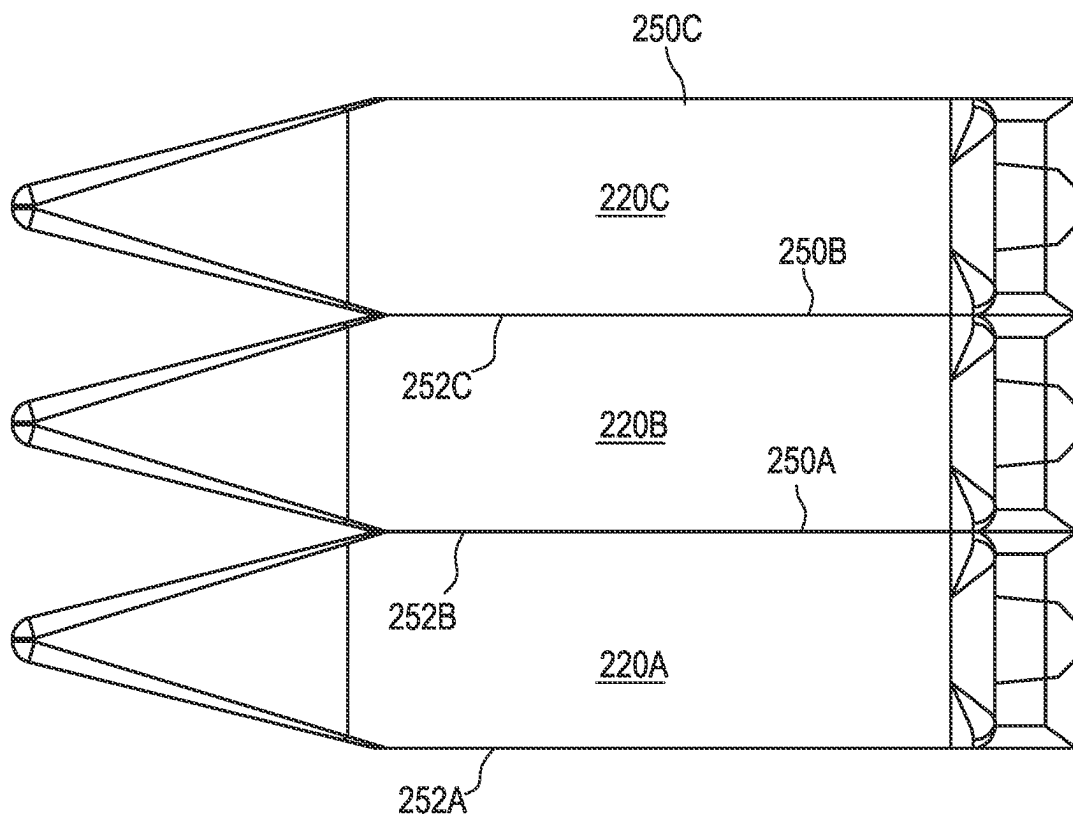


FIG. 21A

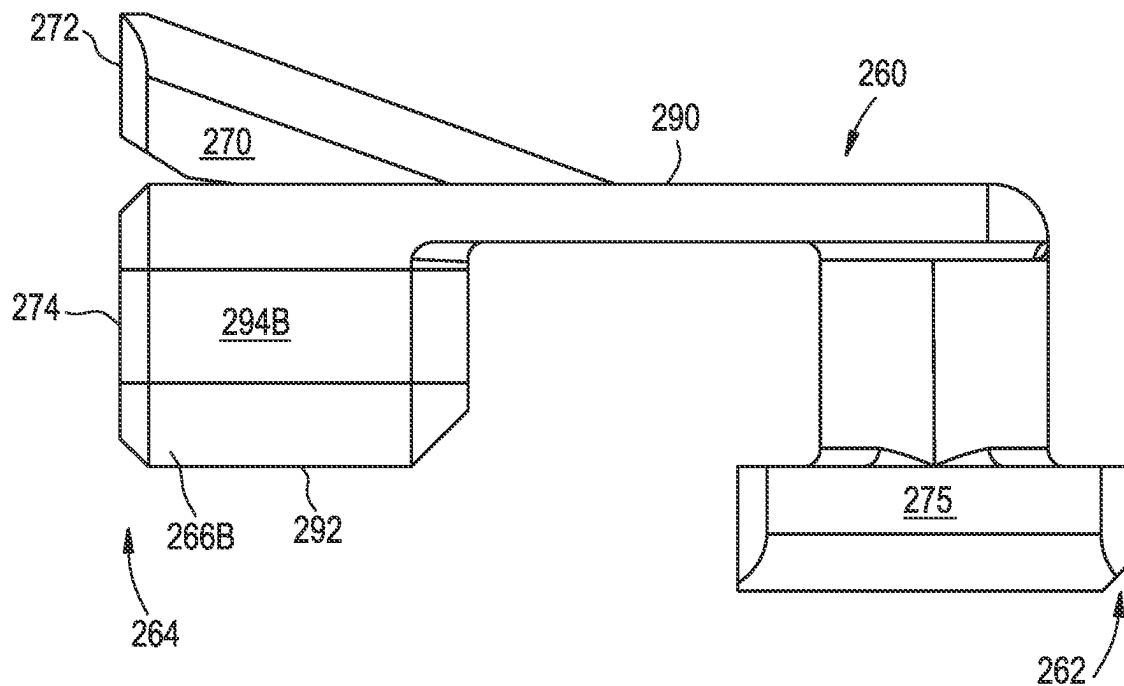


FIG. 21B

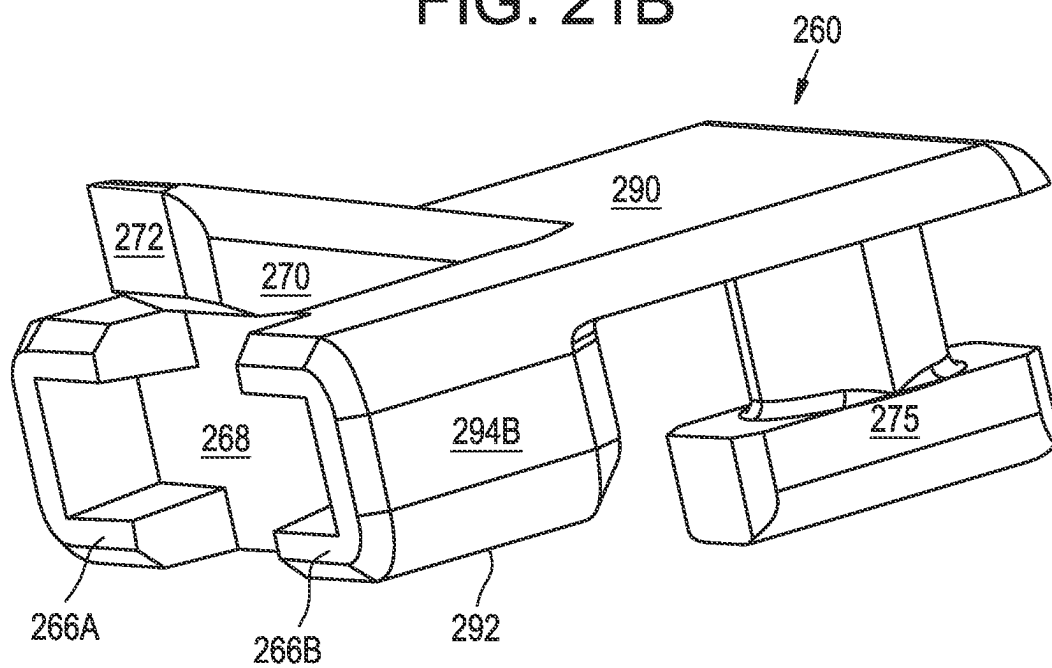


FIG. 21C

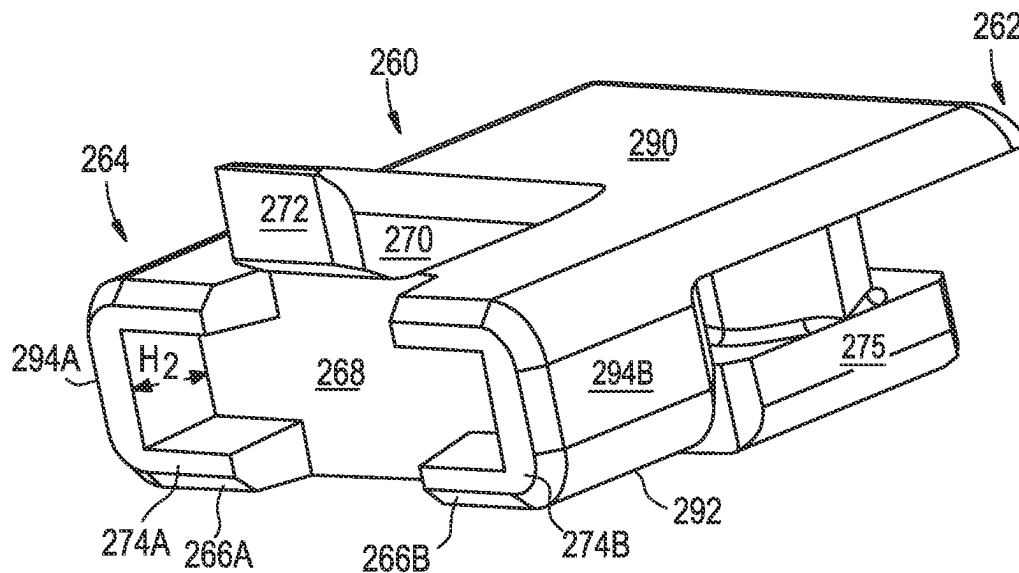


FIG. 21D

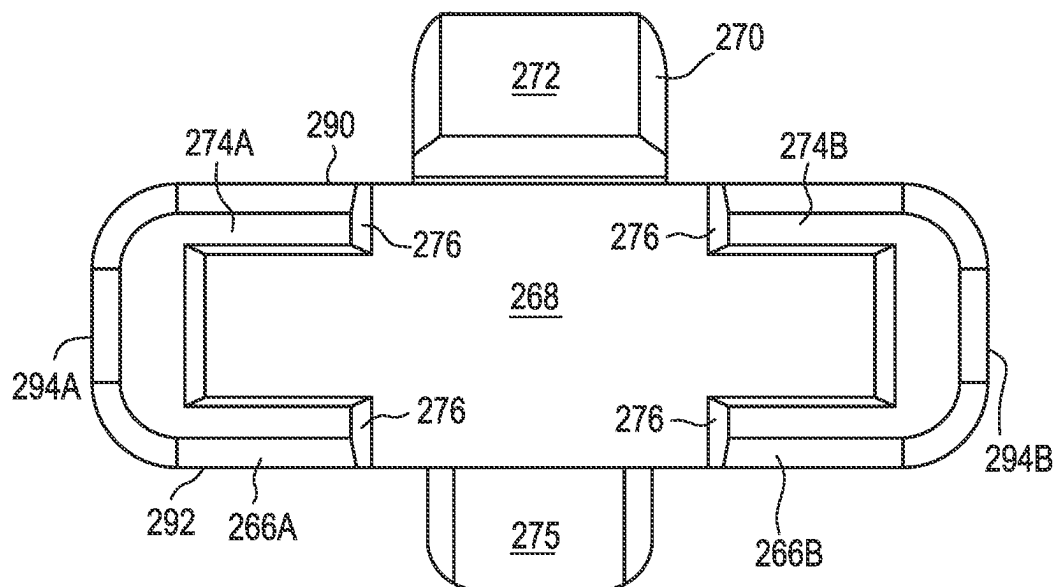


FIG. 22A

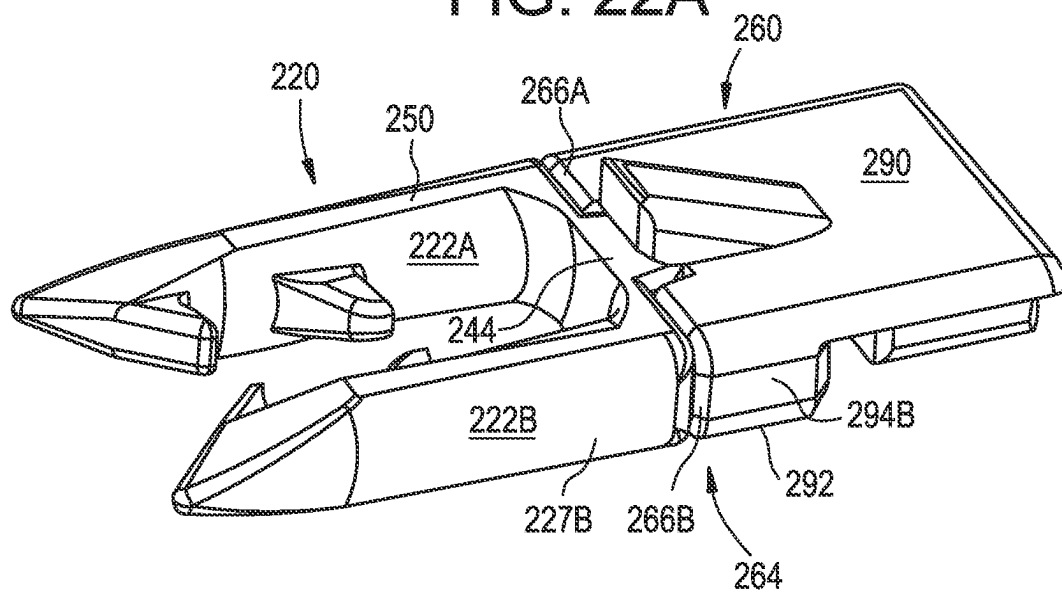


FIG. 22B

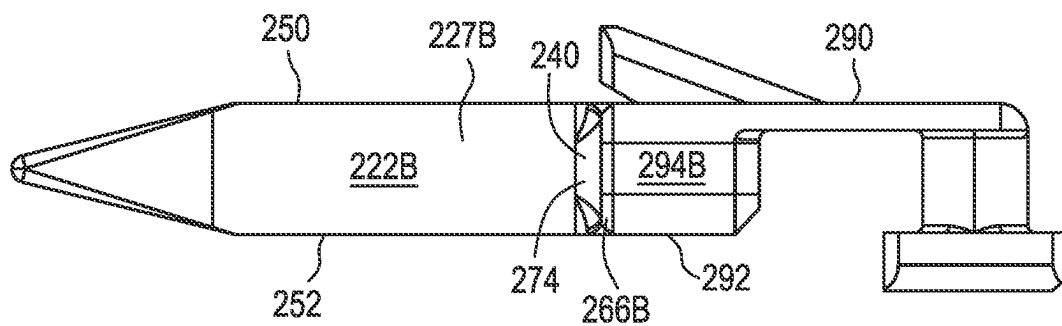


FIG. 22C

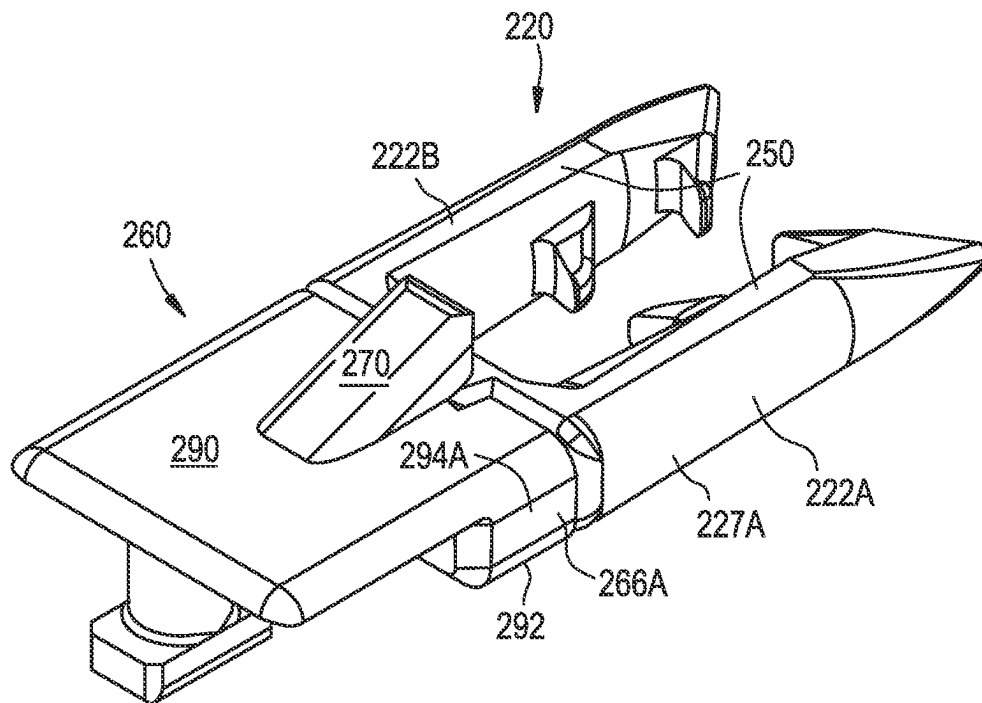


FIG. 22D

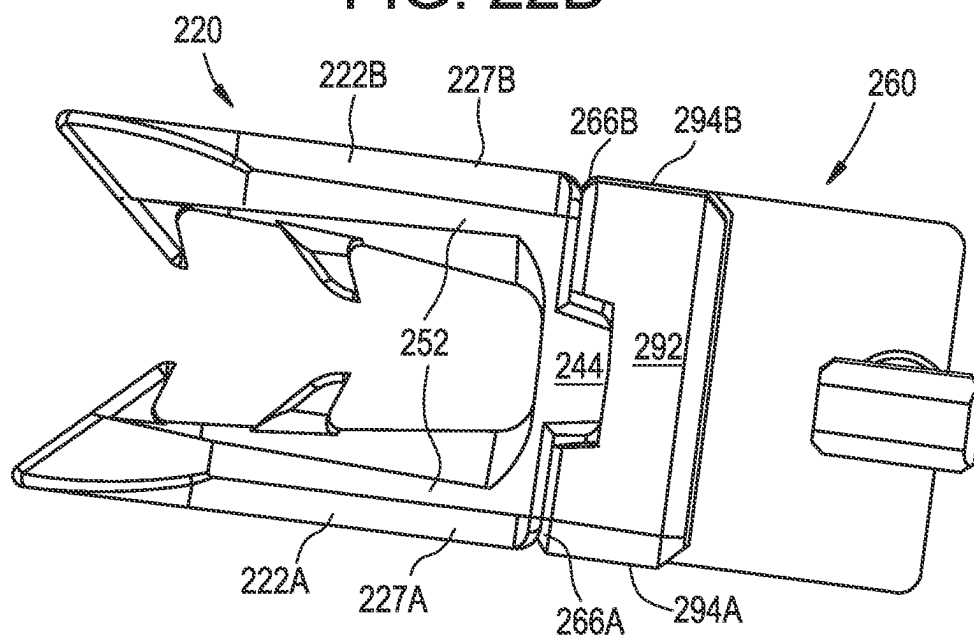


FIG. 23

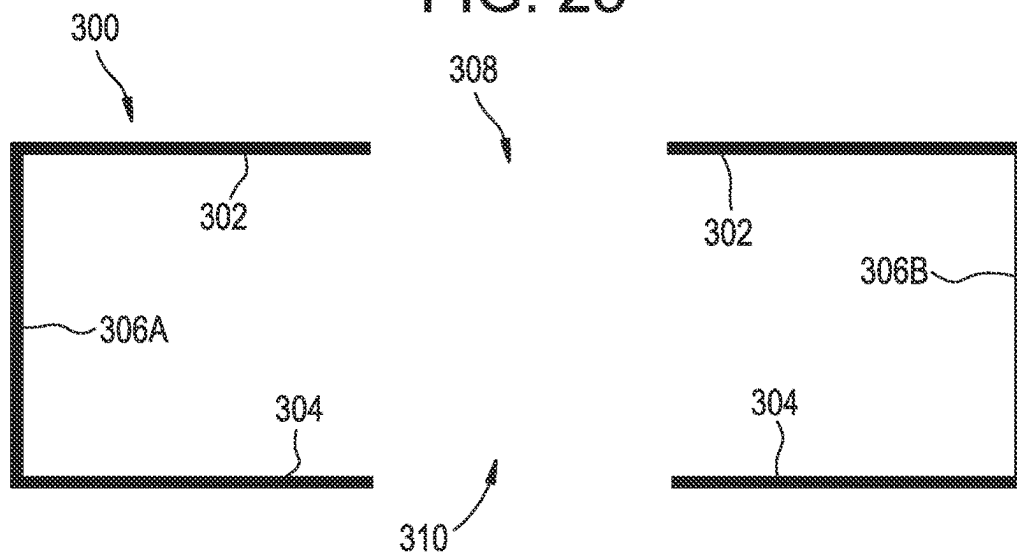


FIG. 24

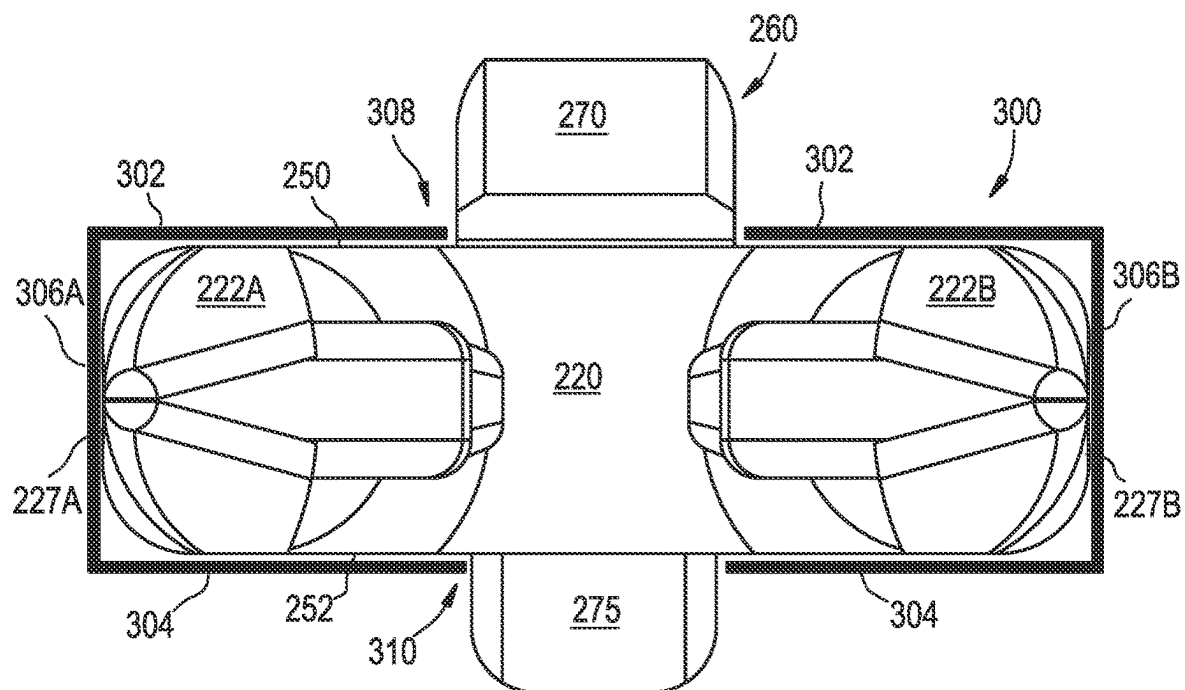


FIG. 25A

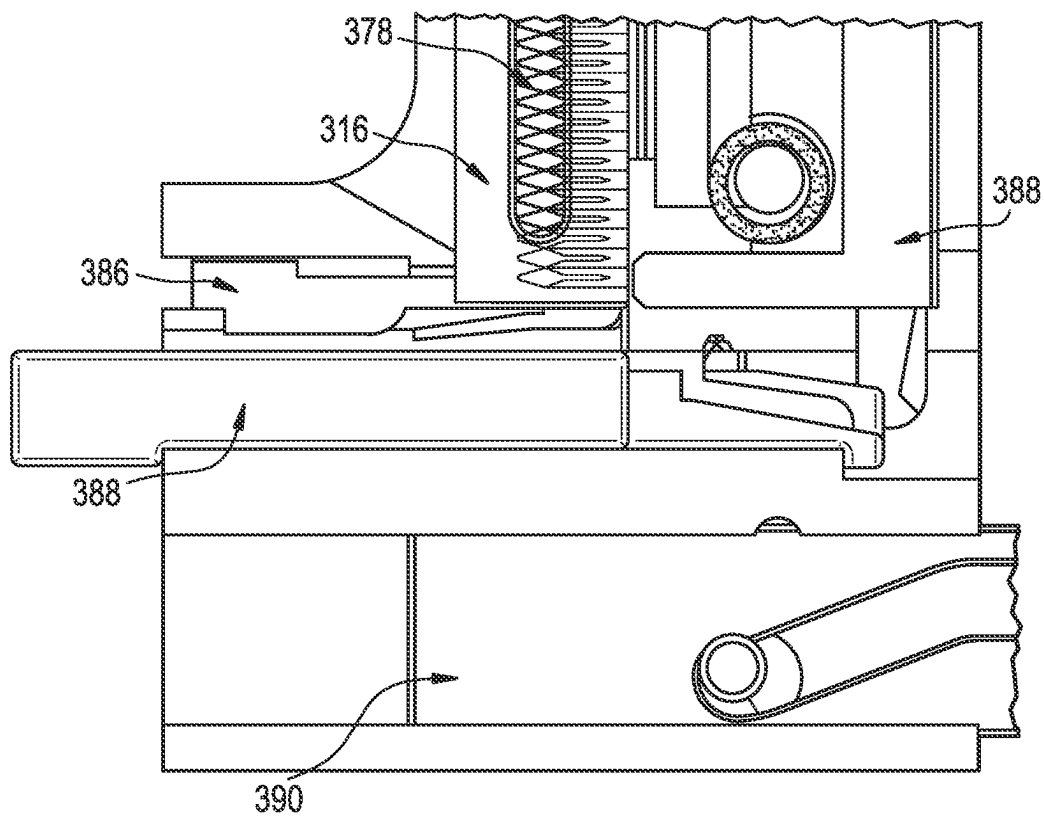


FIG. 25B

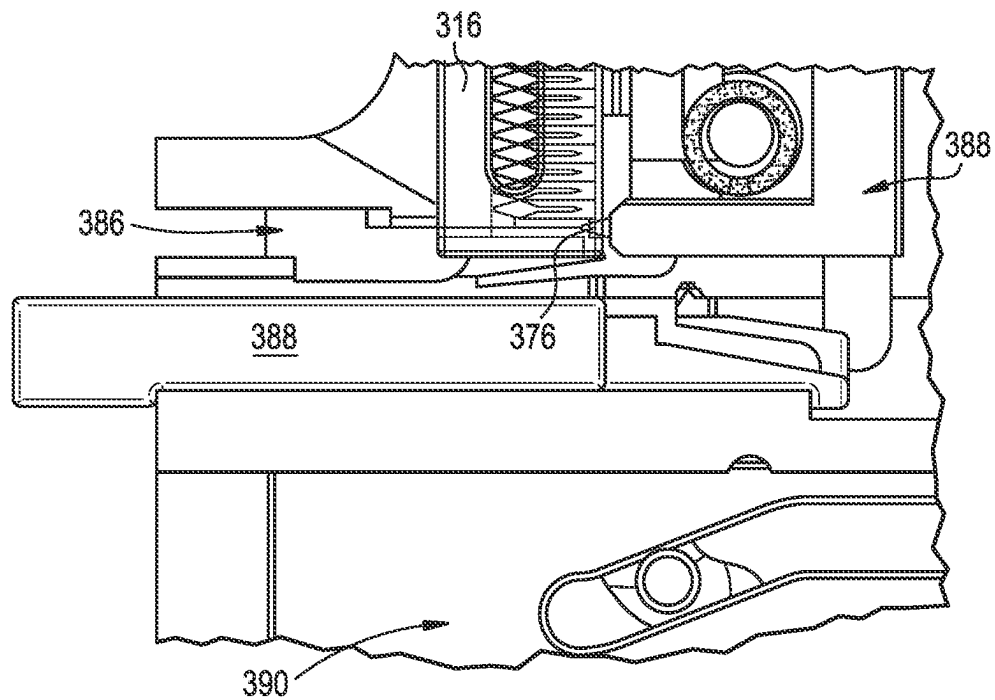


FIG. 25C

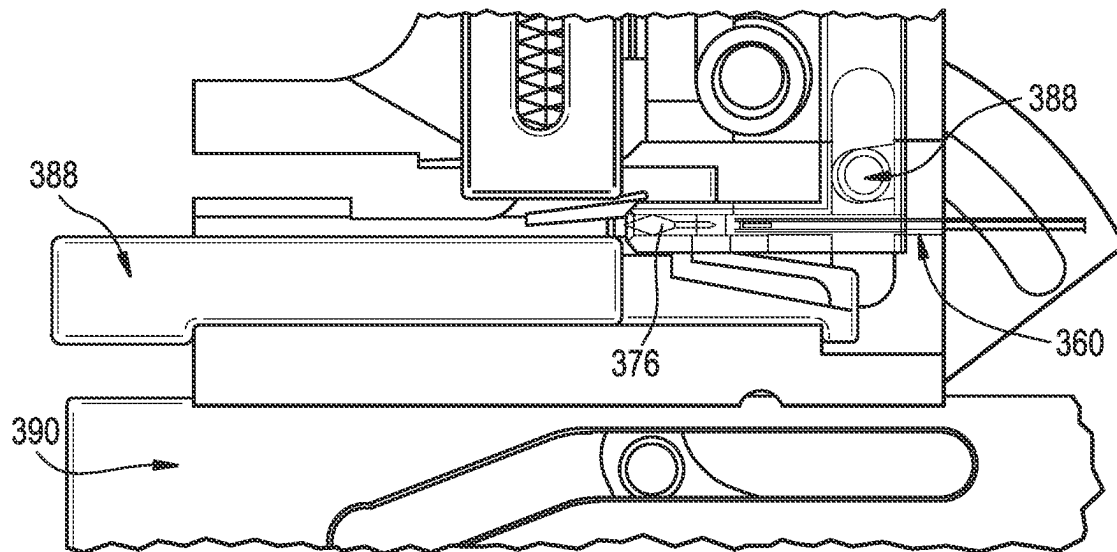


FIG. 25D

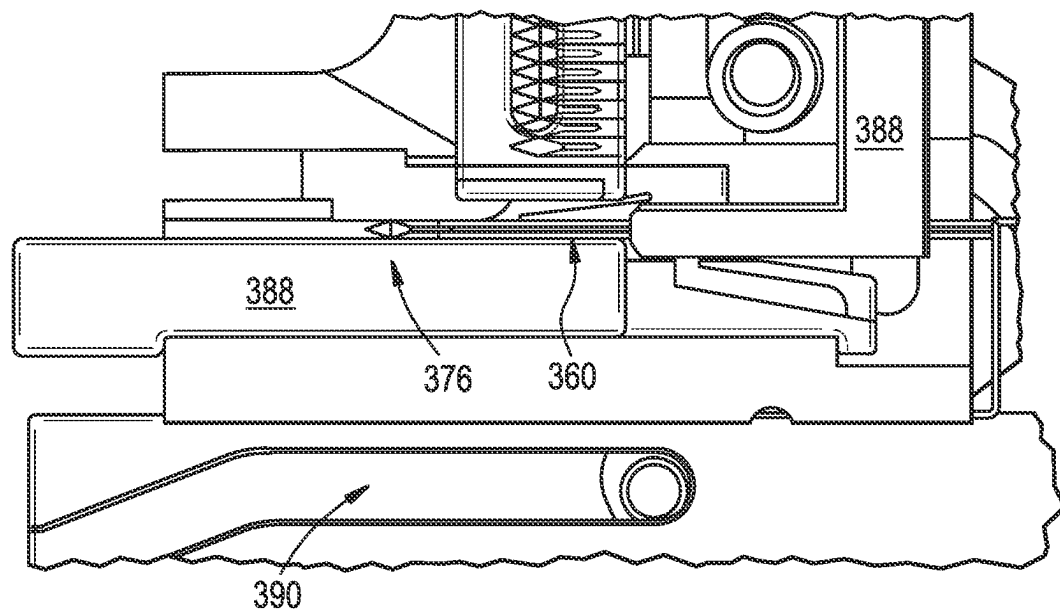


FIG. 26A

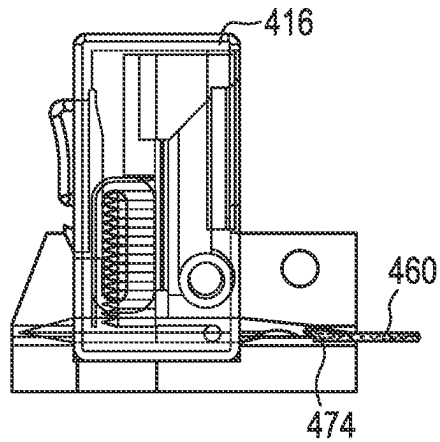


FIG. 26B

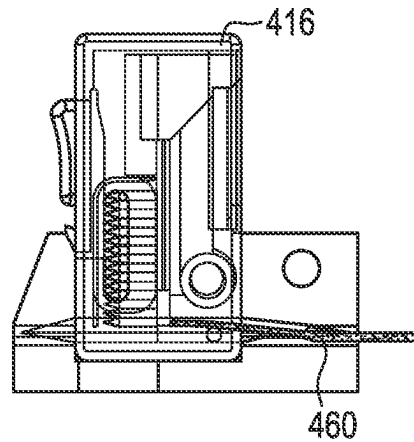


FIG. 26B-1

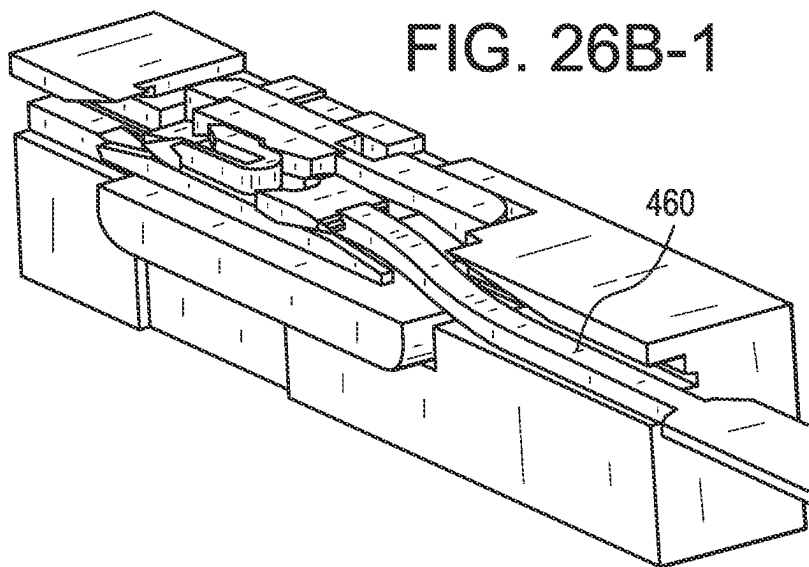


FIG. 26C

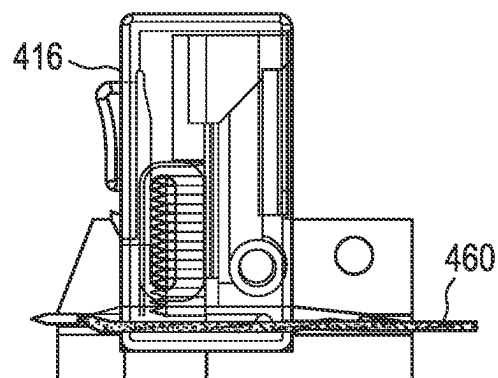


FIG. 27A

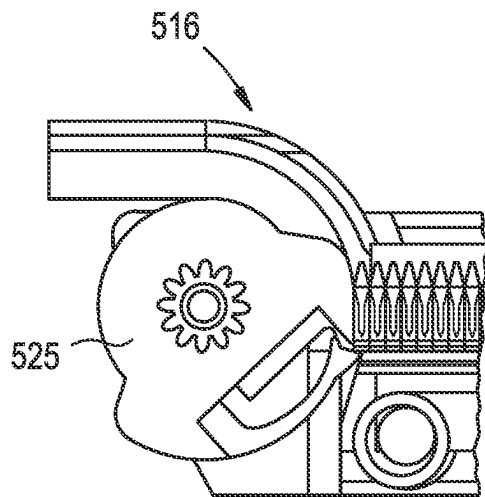


FIG. 27B

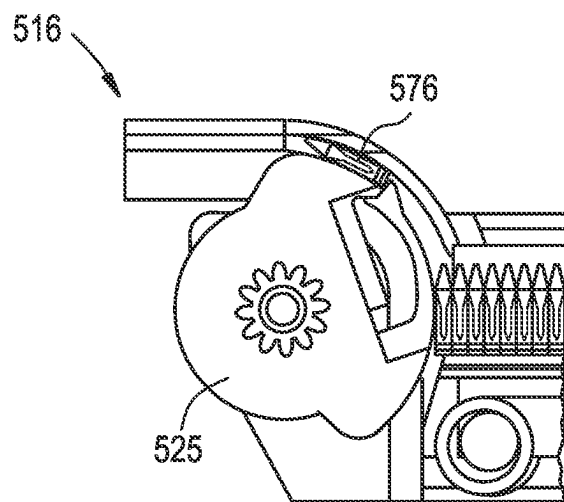


FIG. 27C

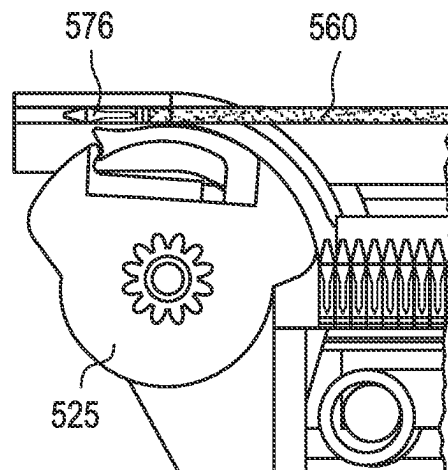


FIG. 28A

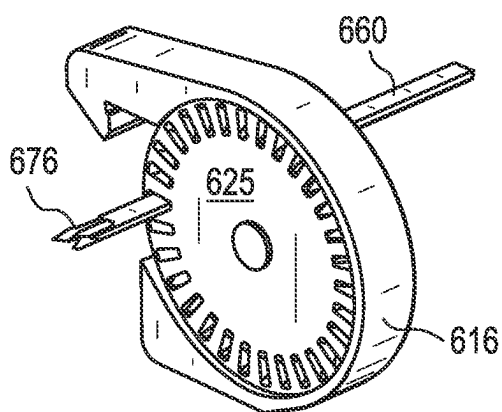


FIG. 28B

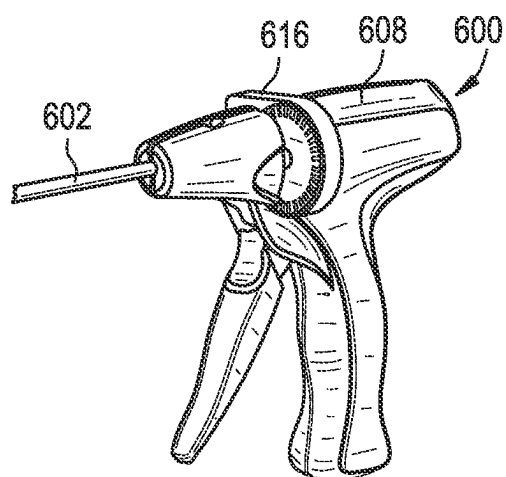


FIG. 29A

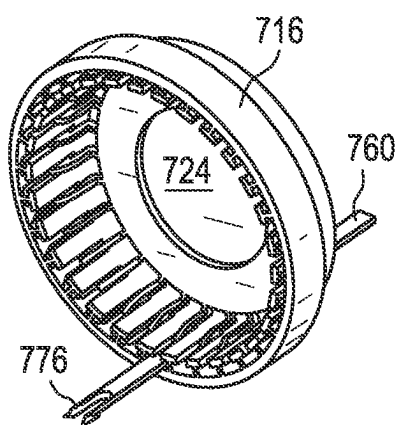


FIG. 29B

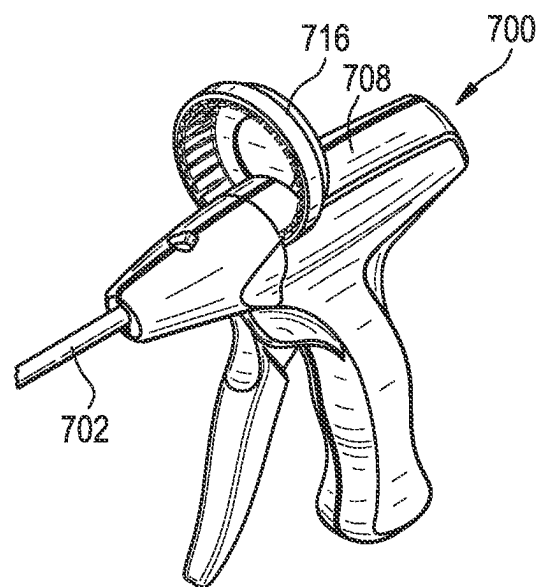


FIG. 30A

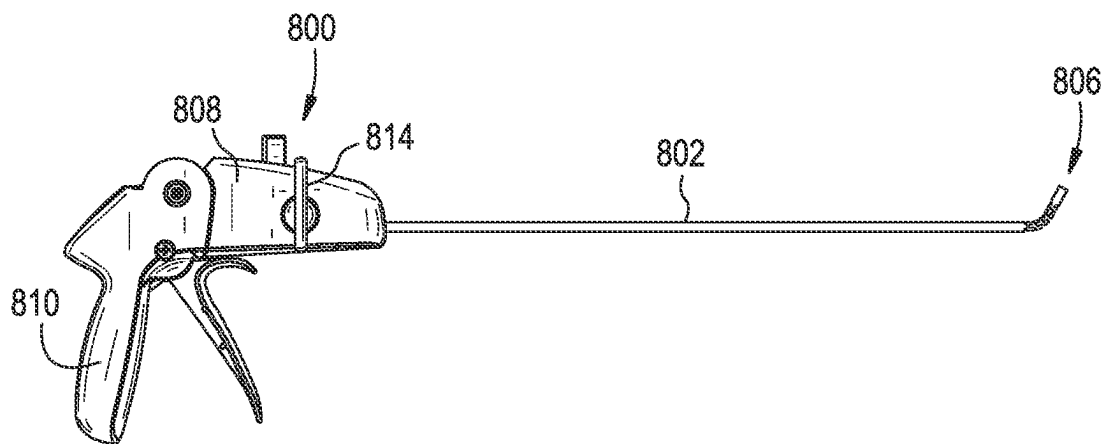


FIG. 30B

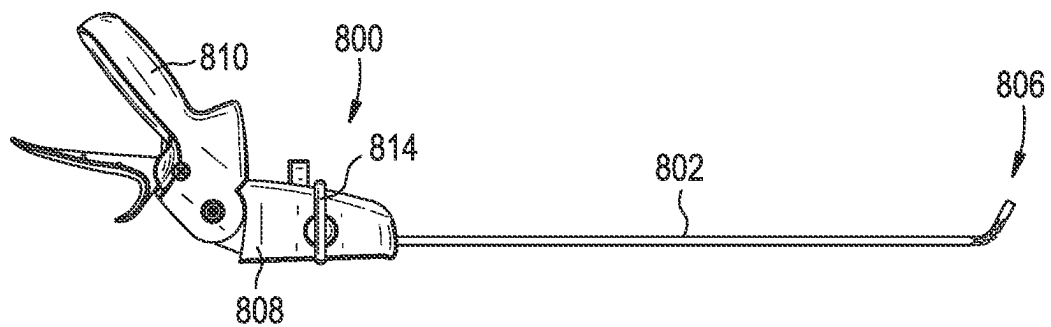


FIG. 31A

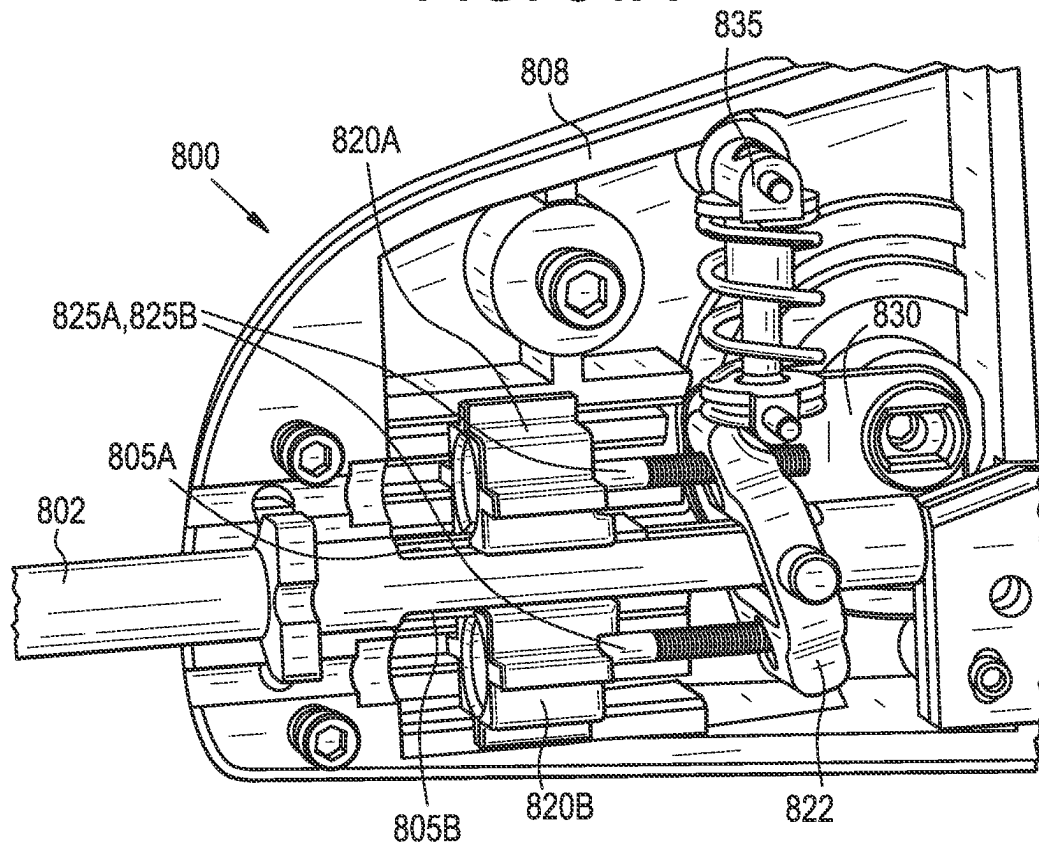


FIG. 31B

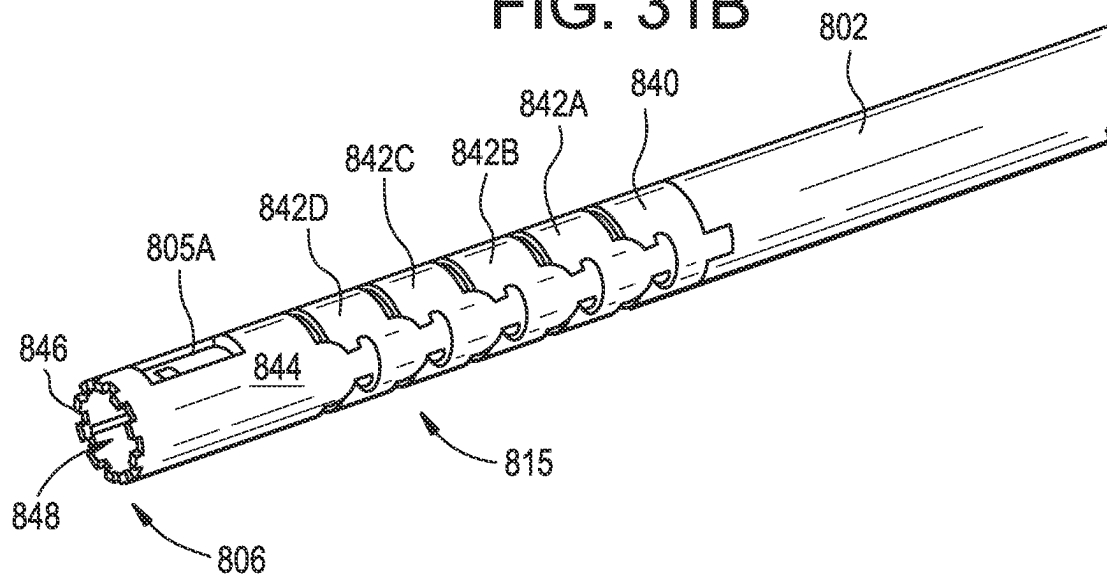


FIG. 32A

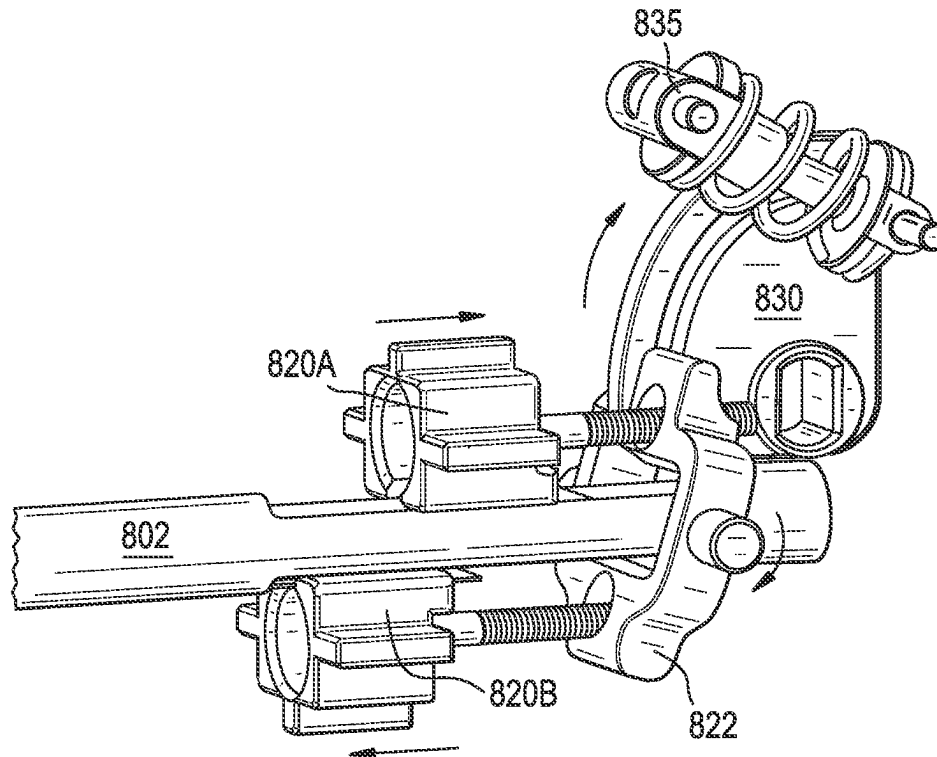


FIG. 32B

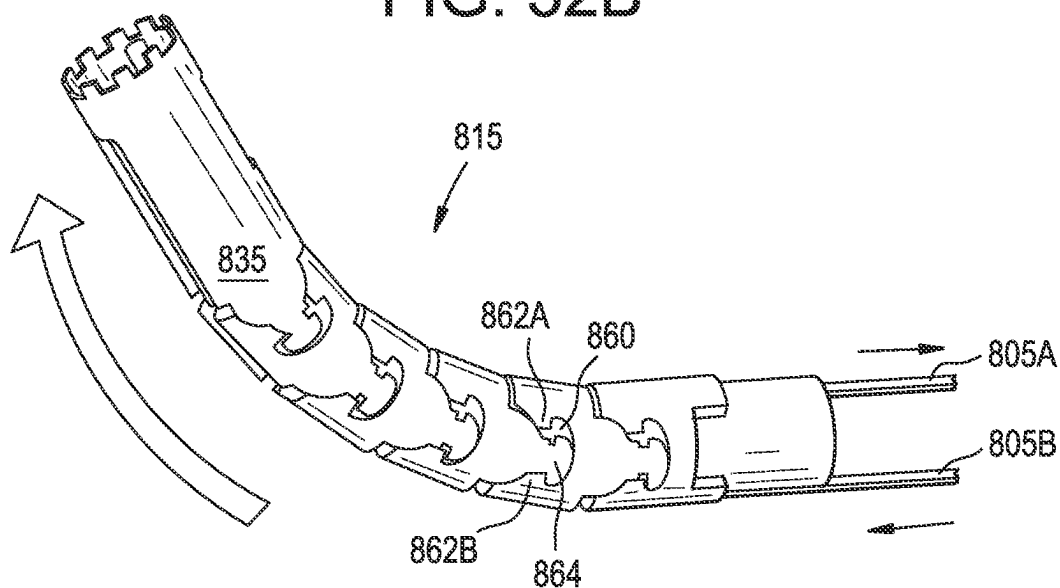


FIG. 33A

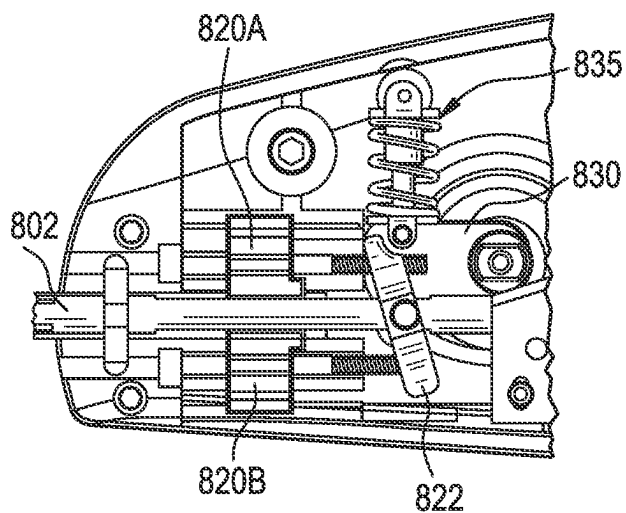


FIG. 34A

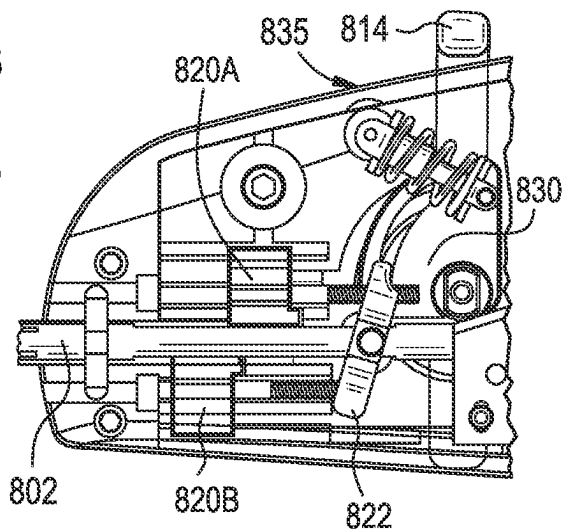


FIG. 33B

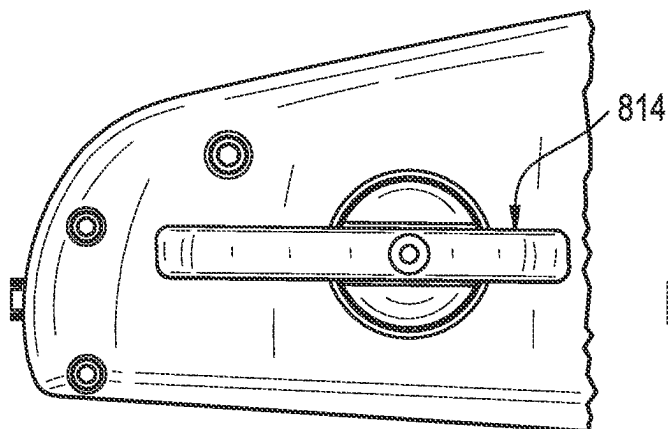


FIG. 34B

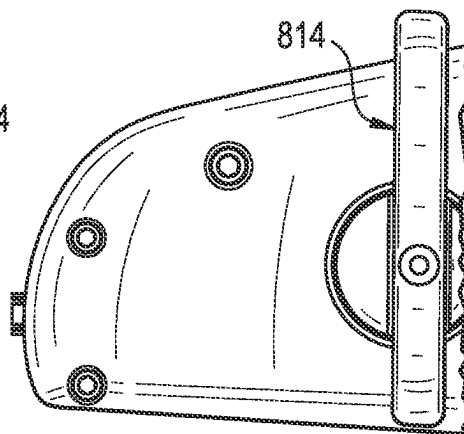


FIG. 33C

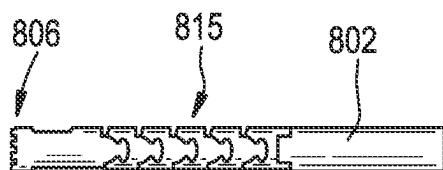


FIG. 34C



FIG. 35

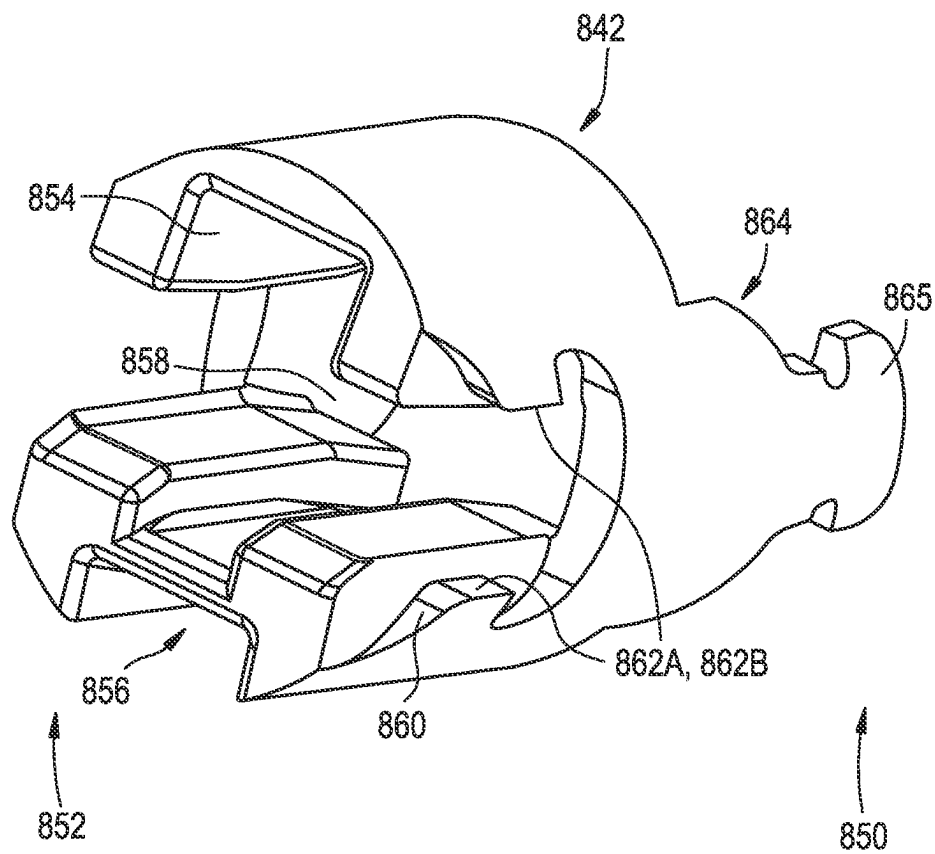


FIG. 36

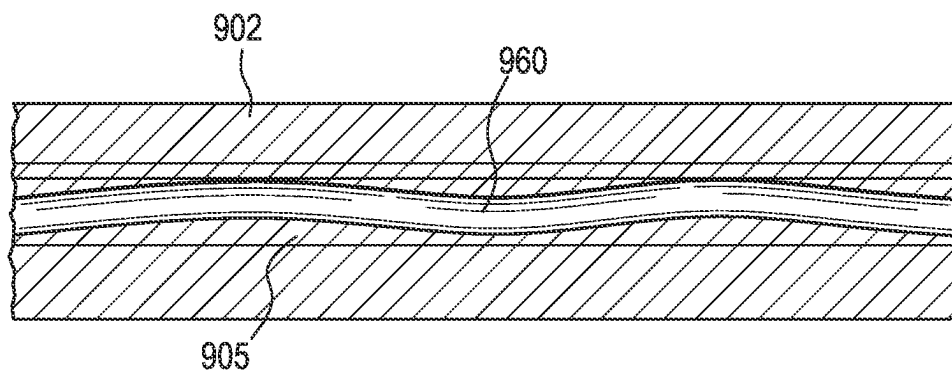


FIG. 37

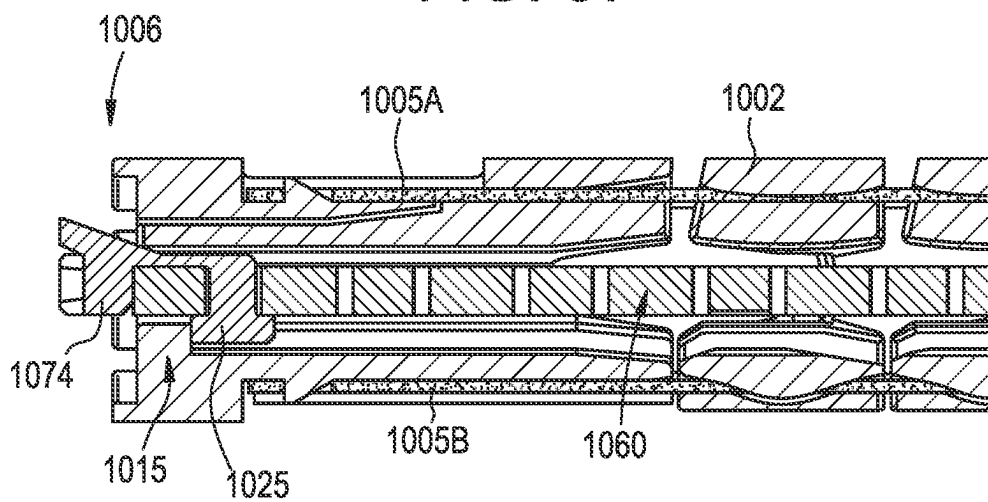


FIG. 38

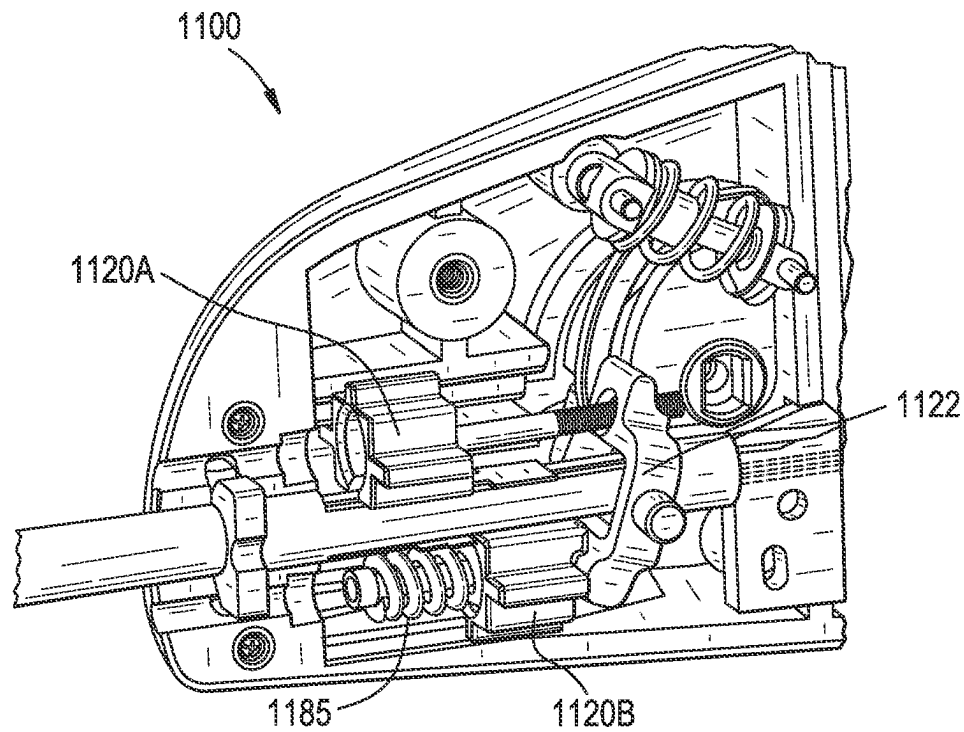


FIG. 39

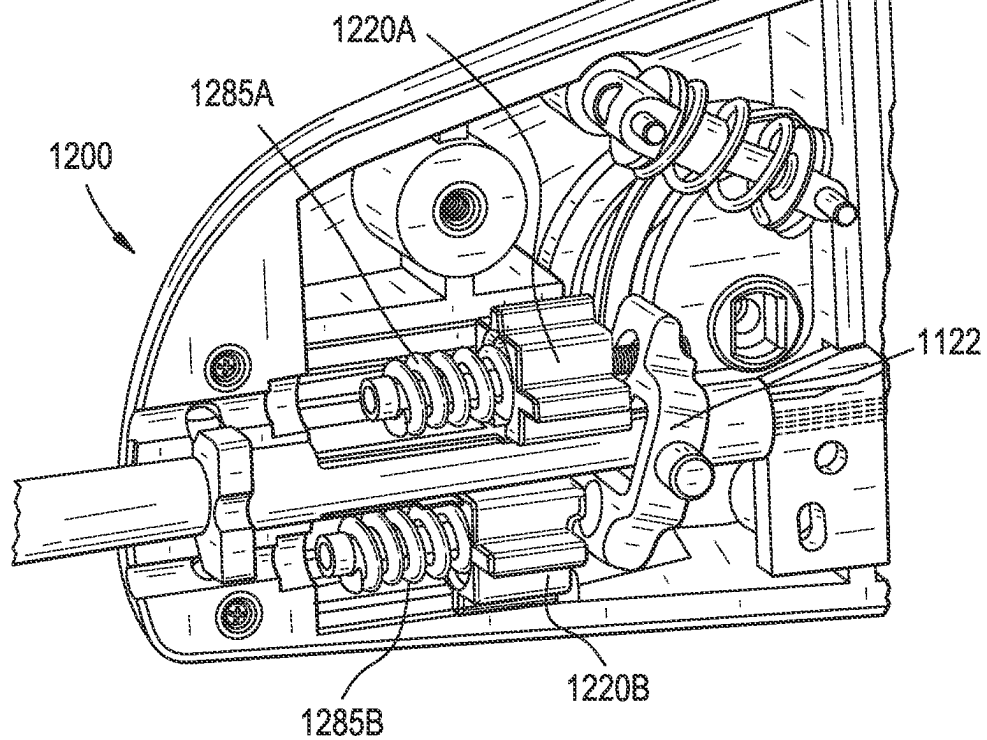


FIG. 40

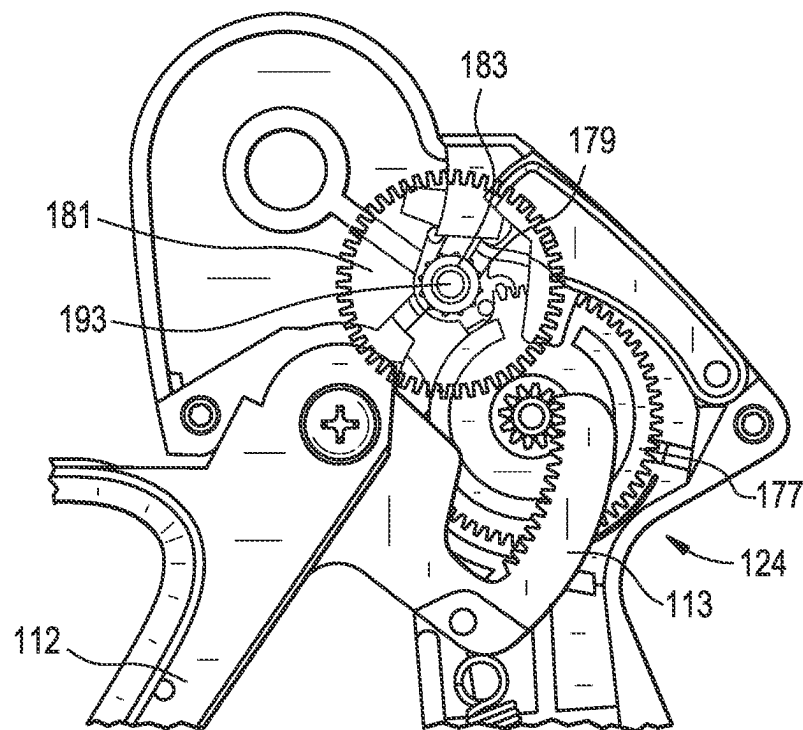


FIG. 41A

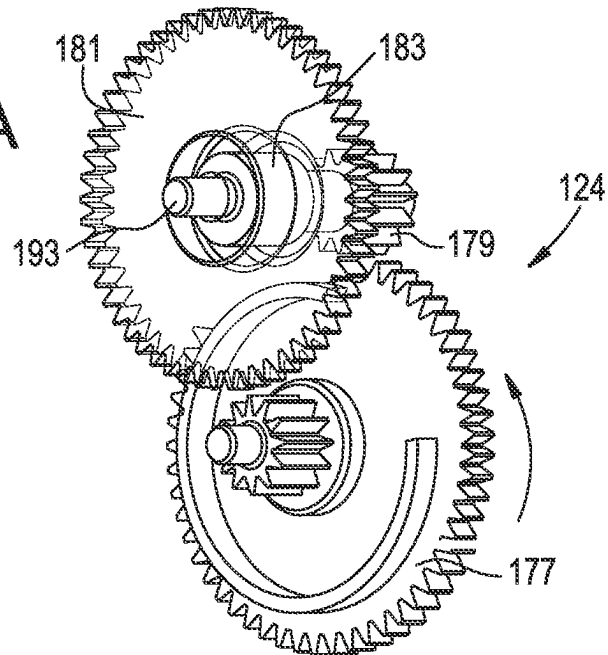


FIG. 41B

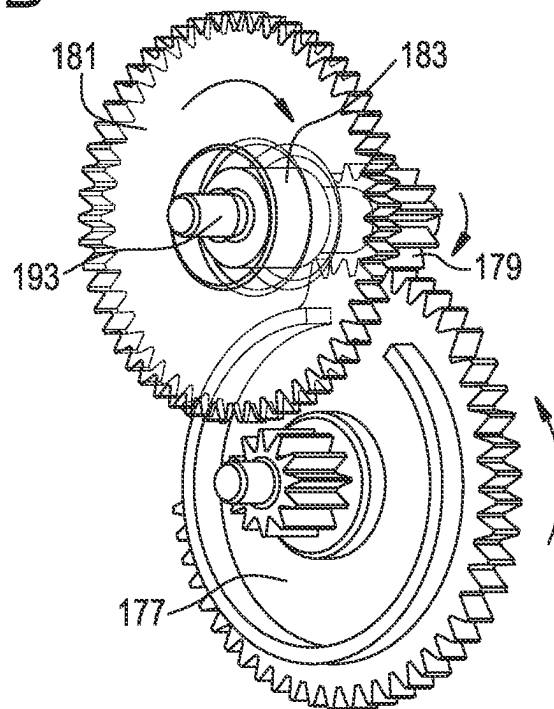


FIG. 41C

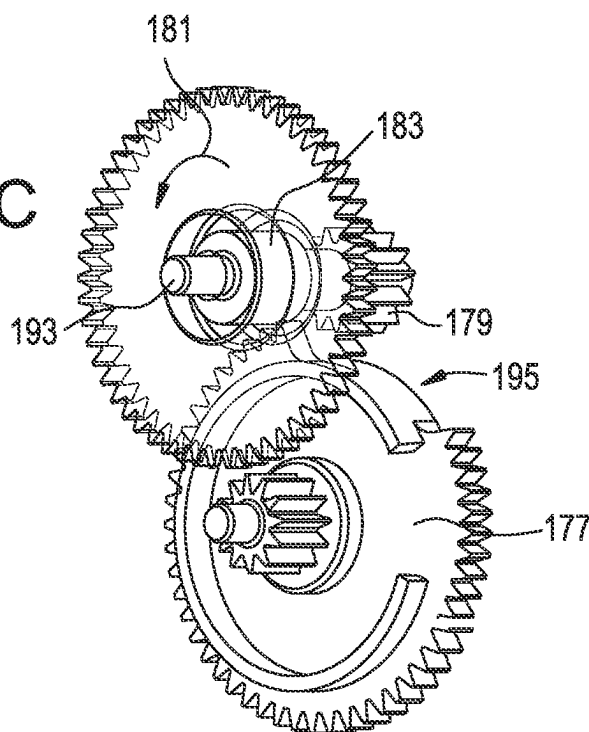


FIG. 41D

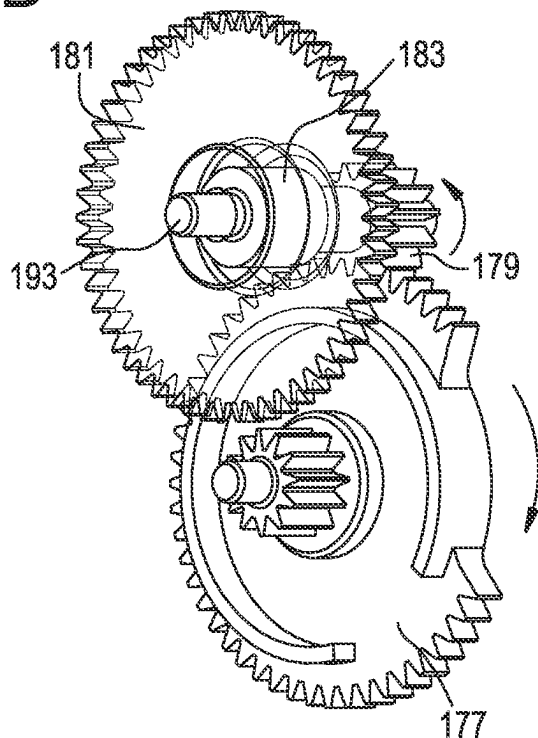


FIG. 42A

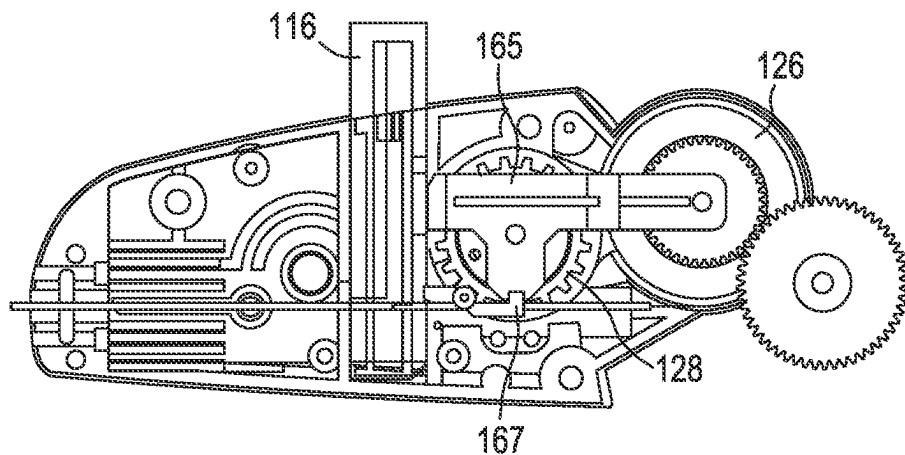


FIG. 42B

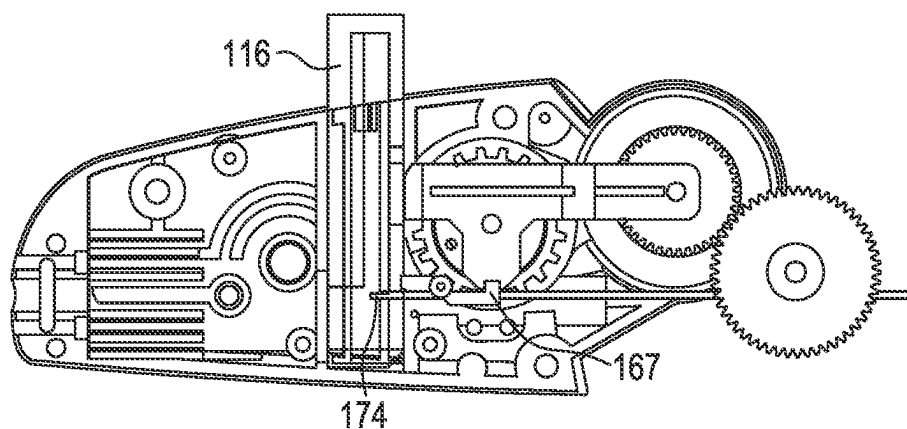


FIG. 42C

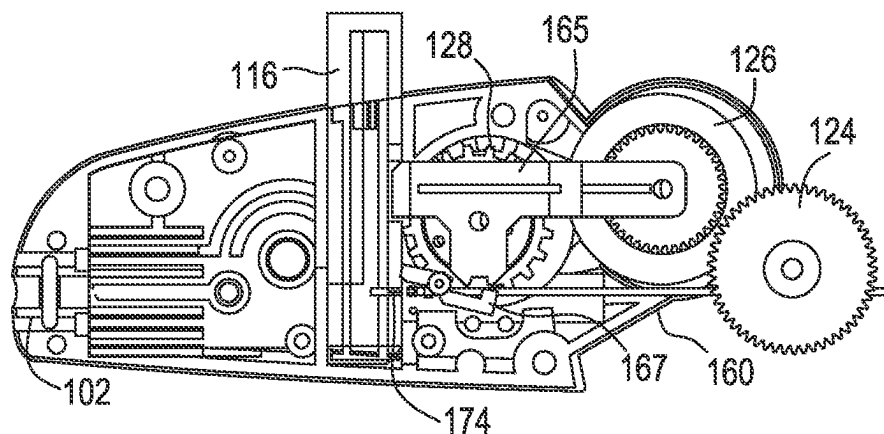


FIG. 42D

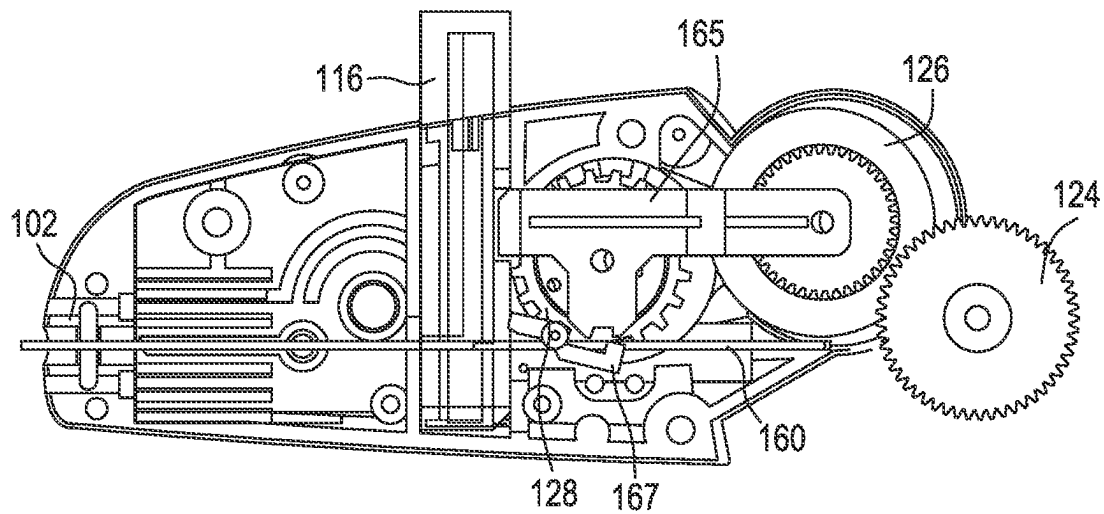


FIG. 42E

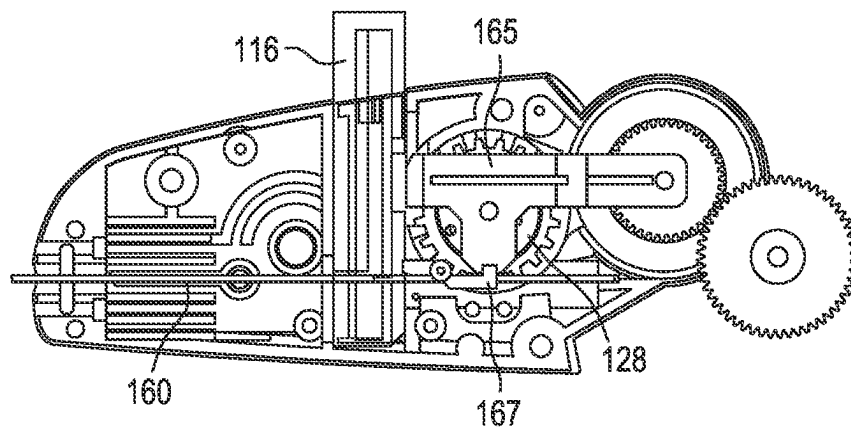


FIG. 43A

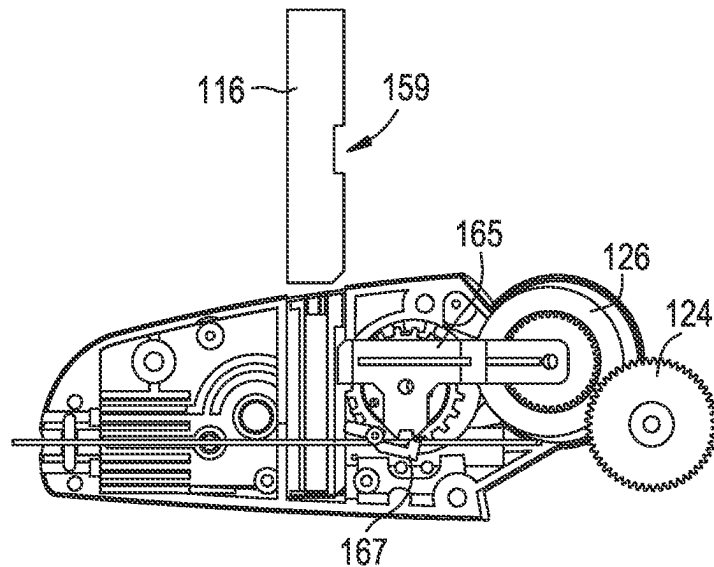


FIG. 43B

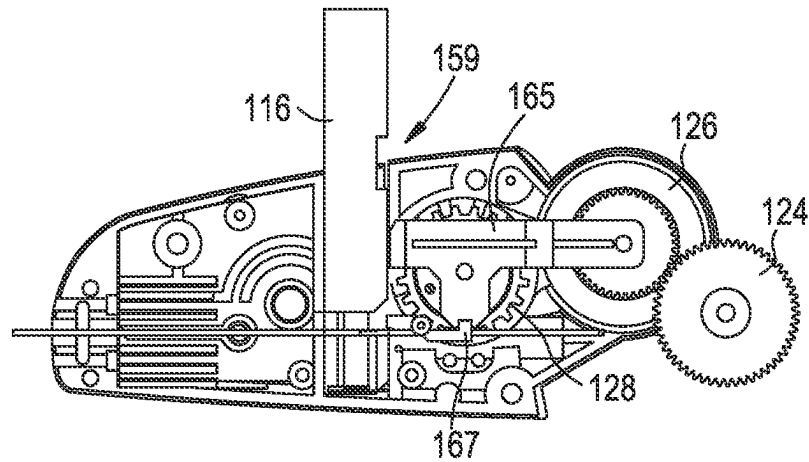


FIG. 43C

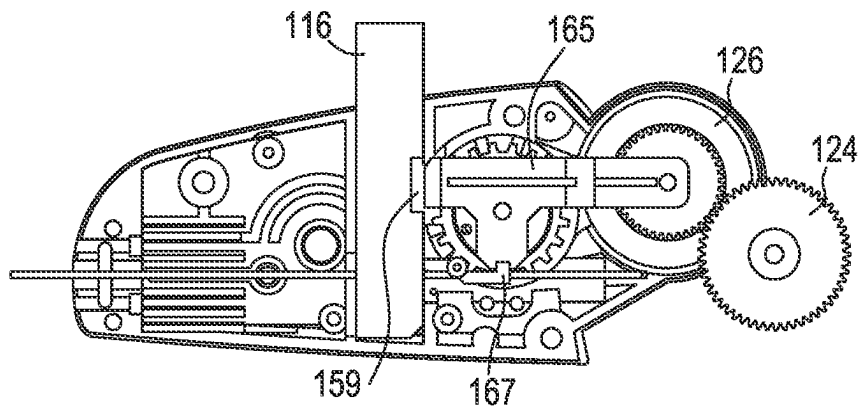


FIG. 44A

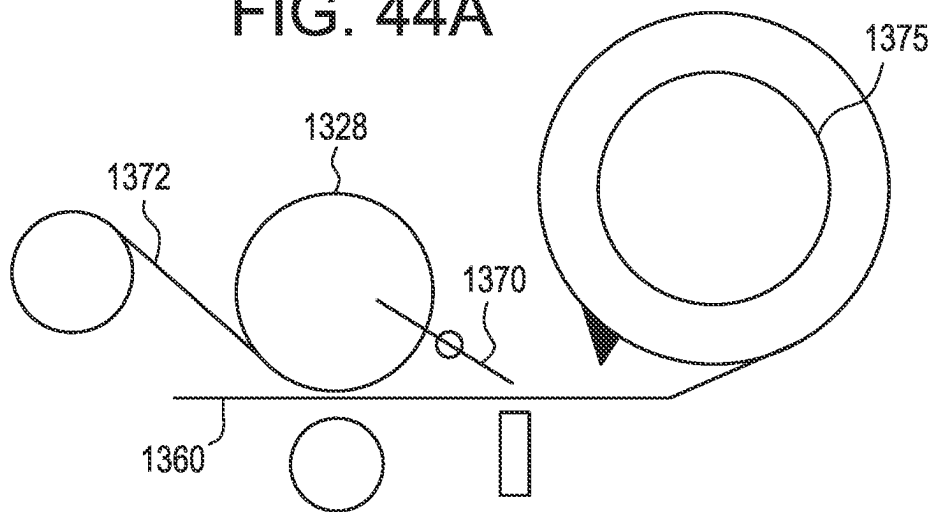


FIG. 44B

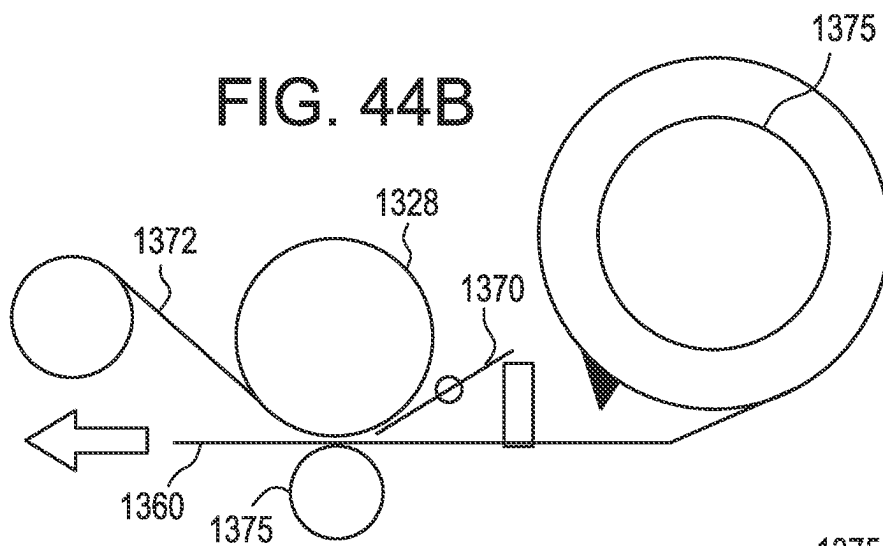


FIG. 44C

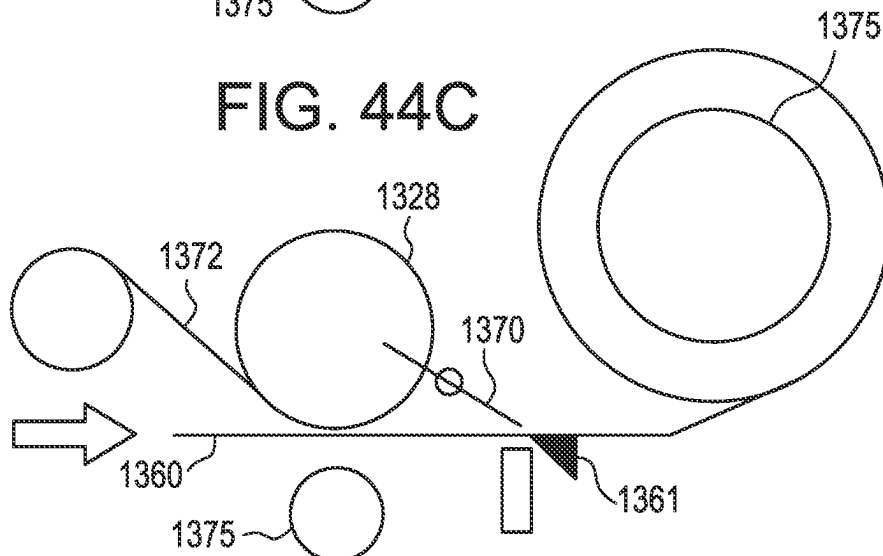


FIG. 45A

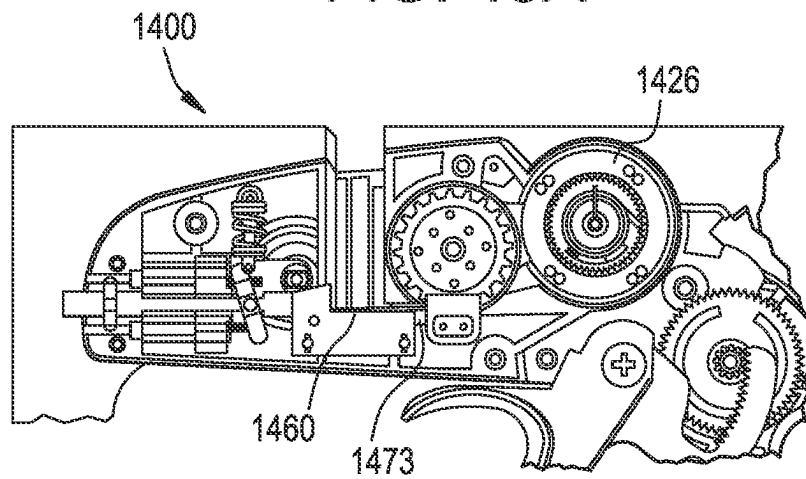


FIG. 45B

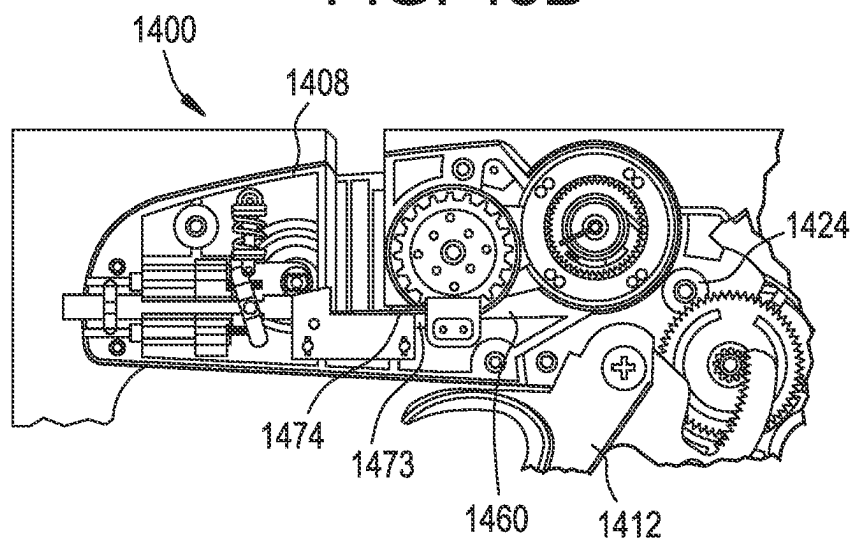


FIG. 46A

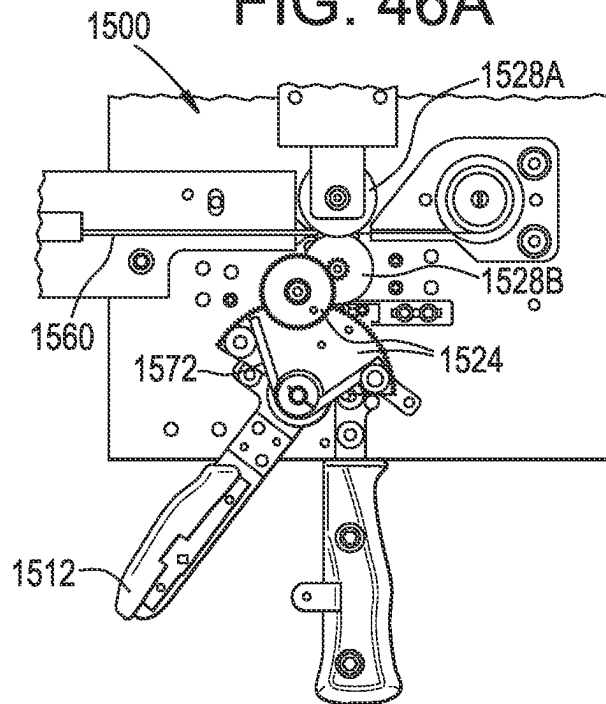


FIG. 46B

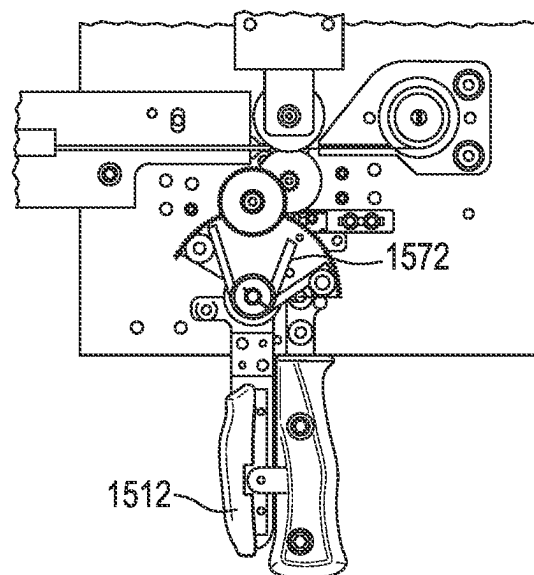


FIG. 46C

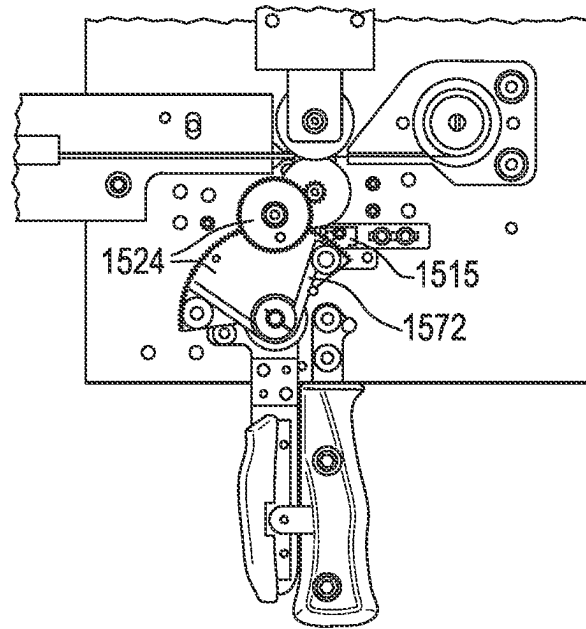


FIG. 46D

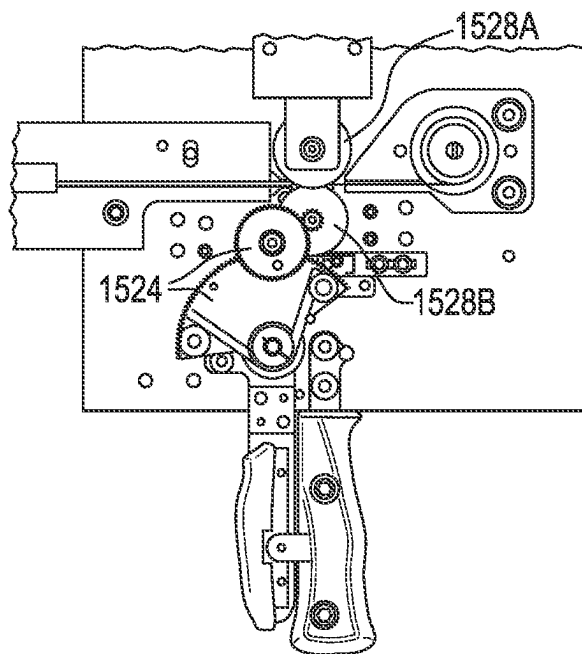
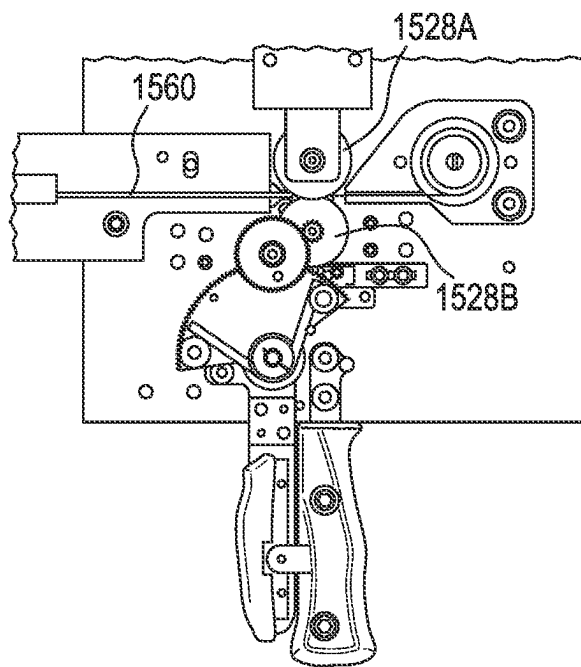


FIG. 46E



1

APPLICATOR INSTRUMENTS FOR DISPENSING SURGICAL FASTENERS, THE APPLICATOR INSTRUMENTS BEING CONTROLLED BY SURGICAL ROBOTS

CROSS REFERENCE TO RELATED APPLICATION

The present patent application is a continuation of U.S. patent application Ser. No. 15/494,012, filed on Apr. 21, 2017, now allowed, which claims benefit of U.S. Provisional Application Ser. No. 62/431,355, filed on Dec. 7, 2016, the disclosures of which are hereby incorporated by reference herein.

BACKGROUND OF THE INVENTION

Field of the Invention

The present patent application is generally related to medical devices, and is more specifically related to medical devices that dispense surgical fasteners for securing prosthetic devices to tissue.

Description of the Related Art

A hernia is a condition where a small loop of bowel or intestine protrudes through a weak place or defect within the abdominal muscle wall or groin of a patient. This condition commonly occurs in humans, particularly males. Hernias of this type may result from a congenital defect whereby the patient is born predisposed with this condition, prior abdominal surgery, or may be caused by straining or lifting heavy objects. Heavy lifting may be known to create a large amount of stress upon the abdominal wall and can cause a rupture or tearing at a weak point of the abdominal muscle to create the defect or opening. In any case, the patient may be left with an unsightly bulge of intestinal tissue protruding through the defect, which may result in pain, reduced lifting abilities, and in some cases, impaction of the bowel, or possibly other complications if the flow of blood is cut off to the protruding tissue.

A common solution to the above-described problem may be surgery. During a surgical procedure, the defect is accessed and carefully examined, either through an open incision or endoscopically through an access port such as a trocar. In either case, careful examination is required due to the network of vessels and nerves which exist in the area of a typical defect, which requires a surgeon to conduct a hernia repair with great skill and caution. Within this area can be found vascular structures such as gastric vessels, the external iliac vessels, and the inferior epigastric vessels, as well as reproductive vessels such as the vas deferens extending through the inguinal floor.

Once the surgeon is familiar with the anatomy of a patient, the surgeon carefully places the viscera back into the patient's abdomen through the defect. Repairing the defect can involve closure of the defect with sutures or fasteners but generally involves placing a surgical prosthetic such as a mesh patch over the defect, and attaching the mesh patch to the abdominal wall or inguinal floor with conventional suture or with surgical fasteners. The mesh patch acts as a barrier and prevents expulsion of bowel through the defect. Suturing of the mesh patch to the inguinal floor can be well suited to open procedures but can be much more difficult and time consuming with endoscopic procedures. With the adoption of endoscopic surgery, endoscopic surgical instruments

2

that apply surgical fasteners can be used. However, the tissue of the inguinal floor may offer special challenges to the surgeon when a needle or fastener is used to penetrate structures such as Cooper's ligament.

At present, there are a variety of surgical instruments and fasteners available for the surgeon to use in an endoscopic or open procedure to attach the mesh patch to the abdominal wall or inguinal floor. One of the earliest types of endoscopic surgical instruments used is a surgical stapler. A plurality or stack of these unformed staples may be generally contained within a stapling cartridge in a serial fashion, and may be sequentially advanced or fed within the instrument by a spring mechanism. A secondary valving or feeding mechanism may be employed to separate the distal-most staple from the stack, to hold the remainder of the spring loaded stack, and may be used to feed the distal-most staples into the staple forming mechanism. Feeding mechanisms of this type are found in U.S. Pat. No. 5,470,010 to Rothfuss et al., and in U.S. Pat. No. 5,582,616, also to Rothfuss et al.

Another hernia mesh attachment instrument uses a helical wire fastener that resembles a small section of spring. Multiple helical wire fasteners may be stored serially within the 5 mm shaft, and may be corkscrewed or rotated into tissue. A load spring may be used to bias or feed the plurality of helical fasteners distally within the shaft. A protrusion extends into the shaft to possibly prevent the ejection of the stack of fasteners by the load spring and may permit passage of a rotating fastener. Instruments and fasteners of these types are found in U.S. Pat. No. 5,582,616 to Bolduc et al., U.S. Pat. No. 5,810,882 to Bolduc et al., and in U.S. Pat. No. 5,830,221 to Stein et al.

Whereas the above surgical instruments may be used for hernia fastening applications, they use a spring mechanism to feed a plurality of fasteners through the surgical instrument. Spring mechanisms typically use a long soft coil spring to push a stack of fasteners through a guide or track within the shaft of the surgical instrument. These types of feeding mechanisms may be generally simple and reliable, but may require an additional secondary valving mechanism or protrusion to separate and feed one fastener from the stack.

Other surgical fasteners may be used for hernia mesh attachment but utilize either a reloadable single shot instrument or a rotary magazine that holds a small number of fasteners. These types of surgical fastening instruments can be found in U.S. Pat. Nos. 5,203,864 and 5,290,297, both to Edward Phillips. These instruments have not gained acceptance by the surgical community, possibly due to their single shot capabilities and the large size of the rotary magazine, which can restrict such an instrument to an open procedure.

Whereas all the above surgical instruments may be used for hernia fastening applications, they either use a spring mechanism to feed the plurality of fasteners through the surgical instrument, or a rotary magazine in lieu of a feeding mechanism. Other types of surgical fasteners may be available, such as surgical clips, and they can utilize feeding mechanisms that do not require the use of a spring to feed the clips distally. A reciprocating feeding mechanism is described in U.S. Pat. Nos. 5,601,573; 5,833,700; and 5,921,997 to Fogelberg et al. The Fogelberg et al. references teach a clip applier with a feeding mechanism that utilizes a reciprocating feed bar to feed a serial stack of clips. A feeder shoe may operably engage with and move with the distally moving feed bar and may slidably engage with the proximally moving feed bar. Thus, the feeder shoe may index or push the stack of clips distally with the distally moving feed bar and remains stationary relative to the proximally moving

3

feed bar. A valving mechanism may be also required to separate the distal-most clip from the stack and to hold the stack stationary as the distal-most clip may be applied onto a vessel. Whereas the Fogelberg et al. references teach a reciprocating feeding mechanism with a single reciprocating member, they do not teach the use of the clip applier in the attachment of hernia mesh, nor do they teach the individual driving or feeding of each clip by a moving member.

U.S. Pat. No. 3,740,994 to DeCarlo Jr. discloses a reciprocating feeding mechanism that indexes a plurality of staples or clips, and readies them for discharge by reciprocating one of a pair of opposing leaf spring assemblies. The staples reside serially within a guide rail with a fixed leaf spring assembly extending into the plane of the guide rail. A reciprocating leaf spring assembly may extend inwardly towards the fixed leaf spring assembly. As the reciprocating leaf spring assembly moves distally, each of individual leaf springs of the assembly may engage a staple and move it distally. The distally moving staples deflect the local individual leaf springs of the fixed leaf spring assembly, and the deflected leaf springs may return to the un-deflected position after passage of the staple. As the moving leaf spring assembly moves proximally, the leaf springs of the fixed leaf spring assembly hold the staples stationary and prevent proximal movement thereof. A secondary guide rail and valving mechanism may be provided to separate a single staple from the stack for forming and can hold the stack of staples stationary as the single clip is formed.

Additionally, similar feeding mechanisms are disclosed in U.S. Pat. No. 4,478,220 to DiGiovanni et al. and U.S. Pat. No. 4,471,780 to Menges et al. Both of these related patents teach a reciprocating feeding mechanism that uses one fixed member and one reciprocating member to feed or index a plurality of clips distally. Angled flexible fingers may be hingedly attached to the reciprocating member and operatively engage the clips when moving distally, and slidably engage with the clips when moving proximally. The angled flexible fingers within the fixed member deflect out of the way when the clips move distally and spring up to stop proximal movement of the clip after the clip has passed. A secondary valving mechanism is also disclosed.

Commonly assigned U.S. Pat. No. 7,485,124, the disclosure of which is hereby incorporated by reference herein, teaches a device for delivering a plurality of individual surgical fasteners. In one embodiment, the delivery device includes a drive mechanism having distal and proximal ends. The drive mechanism has a moving member and a fixed opposing member, whereby the moving member is moveable proximally and distally with respect to the delivery device. The moving member has a sharpened distal end for piercing tissue. The device includes at least one surgical fastener located between the first and the second members. Each of the at least one surgical fasteners has a proximal end and a distal end. The device also has an actuator having at least two sequential positions. A first position for moving the moving member distally and piercing tissue, and a second position for moving the moving member proximally, thereby deploying the distal end of the fastener.

Tacks for fixing meshes used laparoscopically have generally been made of metal, such as stainless steel, nitinol, or titanium. The metal tacks were necessary to provide for sufficient holding strength, penetration of various prosthetic meshes, and for ease of manufacture. Until recently, there were no absorbable tacks available on the market, and surgeons could only use absorbable sutures in order to provide a fixation means that did not permanently stay in the body. However, using sutures is exceedingly difficult for

4

laparoscopic procedure, and so they are generally not used unless the repair is done in an open fashion. With surgical trends leading to more minimally invasive techniques with minimum foreign body accumulation, an absorbable tack with minimum profile that can be applied laparoscopically is needed.

Commonly assigned U.S. Pat. No. 8,920,439, the disclosure of which is hereby incorporated by reference herein, discloses an applicator instrument for dispensing surgical fasteners having an elongated shaft with a proximal shaft section and a distal shaft section. The applicator instrument has an articulation controller coupled with the distal shaft section for selectively changing the angle between the distal shaft section and the proximal shaft section. The articulation controller has at least one flexible linkage extending through the shaft and has a proximal end connected with an actuator and a distal end connected with the distal shaft section. The actuator is mounted on a housing for sliding between proximal and distal ends of the housing for moving the at least one flexible linkage in proximal and distal directions. Surgical fasteners are disposed within elongated shaft for being dispensed one at a time from the distal end of the elongated shaft.

In spite of the above advances, intra-operative conditions during laparoscopic surgery remain challenging for the surgeon. There is a need for flexibility, both with respect to surgeon ergonomics and fastener options. Regarding ergonomics, there remains a need for applicator instruments for dispensing surgical fasteners that have improved ergonomics, that enable ipsilateral (same side) mesh tensioning, and that provide maneuverability both inside and outside of a body cavity. There also remains a need for applicator instruments for dispensing surgical fasteners that have an optimal distal shaft strength when the shaft is articulated, and that provide pre-defined articulation angles for simplifying the device complexity and the user experience. There also remains a need for applicator instruments that have improved ergonomics for accommodating a diverse range of trocar placements, including both midline and lateral trocar placements. In addition, there is a need for tailored fastener solutions to accommodate the diverse needs of patients. Moreover, related to this need, there is a need to reduce procedural costs and increase flexibility during surgical procedures.

SUMMARY OF THE INVENTION

In one embodiment, a method of dispensing surgical fasteners preferably includes providing an applicator instrument having a housing with a cartridge opening, an elongated shaft extending from the housing, a firing system disposed within the housing for dispensing surgical fasteners, and an actuator for activating the firing system. The method desirably includes forming a surgical opening in tissue, inserting a distal end of the elongated shaft into the surgical opening, inserting a cartridge containing a plurality of first surgical fasteners into the cartridge opening, and engaging the actuator for dispensing one of the first surgical fasteners from the distal end of the elongated shaft. In one embodiment, while maintaining the distal end of the elongated shaft inside the surgical opening, the cartridge is removed from the cartridge opening, a second cartridge containing a plurality of second surgical fasteners is inserted into the cartridge opening, and the actuator is engaged for dispensing one of the second surgical fasteners from the distal end of the elongated shaft.

5

In one embodiment, the first surgical fasteners in the first cartridge are different than the second surgical fasteners in the second cartridge. In one embodiment, the first surgical fasteners have different properties than the second surgical fasteners. In one embodiment the different properties may be one or more of the material composition, absorption time, size, length, width, leg length, barb length, number of barbs on a leg, curved legs, straight legs, color, opacity, and cartridge quantity.

In one embodiment, a method preferably includes positioning a prosthetic device at a surgical site, dispensing one of the first surgical fasteners for securing the prosthetic device to the tissue at the surgical site, and dispensing one of the second surgical fasteners for securing the prosthetic device to the tissue at the surgical site, whereby the distal end of the elongated shaft remains inside the surgical opening between the dispensing one of the first surgical fasteners step and the dispensing one of the second surgical fasteners step.

In one embodiment, a method desirably includes abutting the distal end of the elongated shaft against the prosthetic device, and during the abutting step, engaging the actuator for dispensing one of the first and second surgical fasteners for securing the prosthetic device to the tissue at the surgical site. In one embodiment, the prosthetic device may be a surgical implant such as a surgical mesh or part of a hernia repair kit.

In one embodiment, engaging the actuator activates the firing system for commencing a firing cycle during which one of the first and second surgical fasteners is removed from one of the first and second cartridges and the removed surgical fastener is driven distally from the housing and through the elongated shaft for being dispensed from the distal end of the elongated shaft.

In one embodiment, the applicator instrument has a snap-fit connection for securing the first and second cartridges to the housing. In one embodiment, the applicator instrument may include a cartridge release that is engageable for releasing the first and second cartridges from the snap-fit connection. In one embodiment, the snap-fit connection is located inside the housing, and the cartridge release includes a cartridge release button located on the housing.

In one embodiment, a method of dispensing surgical fasteners desirably includes providing an applicator instrument having a housing with a cartridge opening, an elongated shaft extending from the housing, a handle coupled with a proximal end of the housing, a trigger coupled with the handle, a firing system including an energy storing assembly disposed within the housing, and a gear train disposed within the handle and coupled with the trigger.

In one embodiment, a method includes inserting a first cartridge containing first surgical fasteners into the cartridge opening, and squeezing the trigger for storing energy in the energy storing assembly of the firing system and then releasing the stored energy for removing one of the first surgical fasteners from the cartridge, advancing the removed first surgical fastener from the housing to the elongated shaft, and dispensing the removed first surgical fastener from the distal end of the elongated shaft.

In one embodiment, after dispensing the removed first surgical fastener, the first cartridge is removed from the cartridge opening and a second cartridge containing second surgical fasteners is inserted into the cartridge opening. In one embodiment, after inserting the second cartridge, the trigger is squeezed for storing energy in the energy storing assembly of the firing system and then the stored energy is released for removing one of the second surgical fasteners

6

from the cartridge, advancing the removed second surgical fastener from the housing to the elongated shaft, and dispensing the removed second surgical fastener from the distal end of the elongated shaft.

In one embodiment, a method desirably includes forming a surgical opening in tissue, inserting the distal end of the elongated shaft into the surgical opening, and maintaining the distal end of the elongated shaft inside the surgical opening between the step of dispensing the removed first surgical fastener and the step of dispensing the removed second surgical fastener.

In one embodiment, a method preferably includes positioning a prosthetic device at a surgical site, and dispensing one of the first and second surgical fasteners for securing the prosthetic device to the tissue at the surgical site.

In one embodiment, a method may include abutting the distal end of the elongated shaft against a first section of the prosthetic device, squeezing the trigger for dispensing one of the first surgical fasteners for securing the prosthetic device to the tissue at the surgical site, abutting the distal end of the elongated shaft against a second section of the prosthetic device, and squeezing the trigger for dispensing one of the second surgical fasteners for securing the prosthetic device to the tissue at the surgical site. In one embodiment, the cartridges are changed and the first and second surgical fasteners are dispensed while maintaining the distal end of the elongated shaft inside the surgical opening.

In one embodiment, the applicator instrument may include an articulation lever provided on the housing that is coupled with the elongated shaft for moving the distal end of the elongated shaft between a straight configuration and an articulated configuration. In one embodiment, a method includes, with the elongated shaft in the straight configuration, squeezing the trigger for dispensing one of the first and second surgical fasteners, and engaging the articulation lever for moving the distal end of the elongated shaft into the articulated configuration and again squeezing the trigger for dispensing another one of the first and second surgical fasteners from the distal end of the articulated elongated shaft.

In one embodiment, a method preferably includes providing a handle reconfiguration actuator having a first position for locking the handle in place relative to the housing and a second position for enabling the handle to rotate relative to the housing between a pistol configuration and an in-line configuration. In one embodiment, a method may include moving the handle into the in-line configuration and squeezing the trigger for dispensing a surgical fastener from the distal end of the elongated shaft, and moving the handle into the pistol configuration and again squeezing the trigger for dispensing another surgical fastener from the distal end of the elongated shaft.

In one embodiment, after dispensing the removed second surgical fastener, the second cartridge is removed from the cartridge opening and a third cartridge containing third surgical fasteners is inserted into the cartridge opening. The third surgical fasteners preferably have different properties than the second surgical fasteners, which have different properties than the first surgical fasteners. In one embodiment, after inserting the third cartridge, a method preferably includes squeezing the trigger for storing energy in the energy storing assembly of the firing system and then releasing the stored energy for removing one of the third surgical fasteners from the cartridge, advancing the removed third surgical fastener from the housing to the elongated

shaft, and dispensing the removed third surgical fastener from the distal end of the elongated shaft.

In one embodiment, a method of dispensing surgical fasteners desirably includes providing an applicator instrument having a housing with a cartridge opening, an elongated shaft extending from the housing, a firing system disposed in the housing for dispensing surgical fasteners, and an actuator for activating the firing system. In one embodiment, a method may include forming a surgical opening in tissue, positioning a prosthetic device at a surgical site, inserting a distal end of the elongated shaft into the surgical opening and abutting the distal end of the elongated shaft against the prosthetic device, inserting a first cartridge containing a plurality of first surgical fasteners into the cartridge opening, and engaging the actuator for activating the firing system and dispensing one of the first surgical fasteners from the distal end of the elongated shaft for securing the prosthetic device to the tissue at the surgical site. In one embodiment, a method desirably includes, while maintaining the distal end of the elongated shaft inside the surgical opening, removing the first cartridge from the cartridge opening, inserting a second cartridge containing a plurality of second surgical fasteners into the cartridge opening, and engaging the actuator for activating the firing system for dispensing one of the second surgical fasteners from the distal end of the elongated shaft for securing the prosthetic device to the tissue at the surgical site.

In one embodiment, the first cartridge may include a cartridge body having an upper end and a lower end, whereby the lower end of the cartridge body is insertable into the cartridge opening, the first surgical fasteners being arrayed in a stack that extends between the upper and lower ends of the cartridge body, and a constant force spring that normally urges the stack of first surgical fasteners toward the lower end of the cartridge body for being fed into the firing system.

In one embodiment, the cartridge body may have a catch for forming a snap-fit connection with the housing for securing the first cartridge in the cartridge opening, and the applicator instrument desirably includes a cartridge release button that is engageable for releasing the cartridge body from the snap-fit connection with the housing.

In one embodiment, an applicator instrument for dispensing surgical fasteners during surgical procedures has a reconfigurable handle that may be moved between a pistol configuration, an in-line configuration, and an inverted pistol. The applicator instrument having a reconfigurable handle is particularly useful during laparoscopic procedures such as hernia repair procedures where ergonomics and instrument maneuverability are critical.

In one embodiment, the applicator instrument includes a distal housing assembly, a proximal handle assembly, a pivoting connection between the distal housing assembly and the proximal handle assembly, and a locking element for securing the distal housing assembly and the proximal handle assembly at a plurality of angular positions relative to each other.

In one embodiment, the locking element includes a button located on the proximal handle assembly that may be engaged for enabling the proximal handle assembly to be pivoted about the distal housing assembly. The proximal handle assembly is reconfigurable so that it may be placed in a plurality of positions relative to the housing assembly, including a pistol configuration, an in-line configuration, or an inverted pistol configuration. In one embodiment, the proximal handle assembly can be adjusted through a range of angles between 90 and 180 degrees.

In one embodiment, the applicator instrument desirably includes a gear train that is used to actuate a firing system for dispensing surgical fasteners. In one embodiment, the gear train preferably includes a first portion of a gear train located in the proximal handle assembly, which is configured to engage a second portion of a gear train located in the distal housing assembly in order to actuate the firing system. In one embodiment, at least one gear in the distal housing assembly is concentric with the axis of rotation of the reconfiguration pivot.

In one embodiment, the proximal handle assembly has a trigger that may be squeezed for activating the gear train. In one embodiment, during reconfiguration of the proximal handle assembly relative to the distal housing assembly, the gear train in the proximal handle assembly is disengaged from the gear/gear train in the distal housing assembly to allow for the reconfiguration while not affecting the stroke of the trigger/gear train.

In one embodiment, when the trigger has been squeezed to commence a firing cycle, the reconfiguration button is blocked to prevent reconfiguration of the proximal handle assembly relative to the distal housing assembly during the firing cycle.

In one embodiment, when the reconfiguration button is depressed for changing the angle of the proximal handle assembly relative to the distal housing assembly, the trigger or gear train is blocked to prevent firing of the applicator instrument prior to completing the reconfiguration of the proximal handle assembly.

Although the present invention is not limited by any particular theory of operation, it is believed that providing applicator instruments having reconfigurable handles will improve the ergonomics of surgical procedures and improve the maneuverability of the instruments. For example, a pistol configuration may be preferred for Totally Extra-Peritoneal (TEP) Inguinal repair procedures because the trocars are typically placed near the patient's midline and the surgeon is typically postured to hold instruments above the patient. In contrast, either a pistol or in-line configuration may be preferred for Trans-Abdominal Pre-Peritoneal (TAPP) inguinal and ventral repairs. For both of these repairs, the trocars are typically placed near the patient's side (i.e., lateral placement) and the surgeon will be working across the patient's body. On the contra-lateral side, either a pistol or in-line configuration may be advantageous. However, on the ipsilateral side, an in-line position provides the benefit of allowing the surgeon to maintain a neutral wrist position while leveraging the device to provide preload to the distal end just before firing.

Thus, in one embodiment, a single applicator instrument having a reconfigurable handle may be used for midline and lateral trocar placements, providing versatility and improved ergonomics.

In one embodiment, a cartridge contains a plurality of surgical fasteners that are stacked atop one another within the cartridge and urged toward a lower end of the cartridge by a spring. In one embodiment, an applicator may be used with different cartridges having different types of surgical fasteners. In one embodiment, an applicator instrument may have a cartridge receiving port that is located at the proximal end of the applicator instrument that is adapted to receive the different cartridges. In one embodiment, with an elongated shaft of the applicator instrument remaining inside a patient, the different cartridges may be exchanged between firing cycles so that a first type of surgical fastener may be fired during a first firing cycle and a second type of surgical fastener may be fired during a second firing cycle. The

ability to change cartridges without removing the distal end of the applicator instrument from the patient preferably enhances efficiency, safety and maintains sterile conditions.

In one embodiment, there is no reconfigurable handle. Instead, the distal housing assembly is docked directly to the arm of a surgical robot. The surgical robot then controls the articulating and firing functions through a standard interface on the robotic arm. The user can still change the cartridge and attach new cartridges to the housing assembly. In this manner, the instrument can be re-loaded or used to deliver a variety of surgical fasteners without changing the applicator instrument attached to the robotic arm.

In one embodiment, the proximal handle assembly has a reconfiguration button having a reconfiguration slider coupled therewith that is configured to engage reconfiguration notches located on the distal housing assembly. In one embodiment, when the reconfiguration button is depressed, the reconfiguration slider is moved away from engagement with one of the reconfiguration slots so that the proximal handle assembly may be pivoted relative to the distal housing assembly. When the reconfiguration button is released, a slider spring normally urges the slider to return to a locked position. In one embodiment, during reconfiguration of the handle, the gear train between the handle and the housing is decoupled.

In one embodiment, when the reconfiguration button is depressed, the deployed reconfiguration button blocks activation of the trigger or commencement of a firing cycle. Thus, in one embodiment, the applicator instrument may not be fired as the position of the handle is being reconfigured.

In one embodiment, when the trigger is squeezed for moving the gear train or commencing a firing cycle, the deployed trigger prevents a handle reconfiguration actuator (e.g., a depressible element or button) from being moved (e.g., depressed). Thus, in one embodiment, the position of the handle may not be reconfigured as the applicator instrument is being fired or after the commencement of a firing cycle.

In one embodiment, an applicator instrument for dispensing surgical fasteners engages a single surgical fastener from a location at or near the proximal end of the instrument and advances the surgical fastener to the distal end of the instrument. In one embodiment, a driving element, such as a spring, provides a pre-determined force, resulting in a consistent delivery of the surgical fastener. Over the course of the firing, this force accelerates the fastener, increasing its velocity and momentum allowing it to penetrate various meshes and tissues.

In one embodiment, the applicator instrument for delivering surgical fasteners preferably includes an elongated member, such as an elongated shaft, having a proximal end and a distal end, a surgical fastener (e.g., a tissue fastener or surgical staple) located adjacent the proximal end of the elongated member, and a mechanism for transporting the surgical fastener from the proximal end of the elongated member to the distal end of the elongated member and into tissue.

In one embodiment, the applicator instrument preferably includes a cartridge holding one or more surgical fasteners. In one embodiment, a plurality of surgical fasteners are stacked atop or adjacent one another within a spring-loaded cartridge. In one embodiment, the cartridge may be positioned adjacent the proximal end of the elongated member.

In one embodiment, the applicator instrument preferably includes an element for engaging a single surgical fastener held by the cartridge so that the singulated surgical fastener may be engaged by a firing system and/or advanced toward

the distal end of the elongated member. In one embodiment, the single surgical fastener may be stripped from the bottom of a stack of surgical fasteners.

In one embodiment, the applicator instrument desirably has a flexible member having a distal end for transporting the surgical fastener from the proximal end of the elongated member to the distal end of the elongated member. In one embodiment, the flexible member may be made of plastic, metal, other suitable materials, and/or combinations thereof. In one embodiment, the flexible member may be planar in cross section, or curved in cross section for enhancing the column strength of the flexible member.

In one embodiment, the flexible member has a distal end that pushes the surgical fastener in a distal direction. In one embodiment, the flexible member preferably has a proximal end that may be retracted and stored in a coiled manner (e.g., on a storage reel). In one embodiment, the flexible member may have features for engaging with a drive wheel. These features may be holes, pockets, or protrusions. In one embodiment, the one or more drive wheels may have surfaces designed to frictionally engage with the flexible member.

In one embodiment, the applicator instrument desirably has a drive wheel (e.g., cogged wheel or friction wheel) that engages a section of the flexible member that is proximal to the distal end of the flexible member. In one embodiment, the drive wheel may be driven by a constant torque spring, a torsion spring, an electrically powered motor, mechanically, electrically, electro-mechanically, and/or pneumatically, or a combination of the above. In one embodiment, the drive wheel may be driven by an external element, for example, by rotary motion from the arm of a robotic surgery system or by compressed air. In one embodiment, the drive wheel may be driven by a stored energy system such as a pre-wound spring.

In one embodiment, the flexible member may be directly connected to a constant torque spring and wound onto the same reel that is coupled with the constant torque spring, therefore not requiring a drive wheel component. In one embodiment, the flexible member and the constant torque spring may be layered together on the same coil.

In one embodiment, a spring such as a power spring or a constant torque spring may be connected to the proximal end of the flexible member to aid in retracting the flexible member from an extended position to a retracted position, and to provide tension to prevent billowing of the flexible member away from the reel during operation. In one embodiment, billowing is preferably minimized to reduce drag or losses in the system. In one embodiment, the constant torque spring or torsion spring desirably stores energy in response to squeezing a trigger or actuator coupled with the handle of the applicator instrument.

In one embodiment, the applicator instrument may have a positive stop coupled with or that contacts the flexible member or the drive wheel to limit or control distal movement of the flexible member. In one embodiment, a positive stop may be located at the distal end of the elongated member or in the housing portion of the applicator instrument, or both. A distal stop provides the benefit of precisely controlling the expulsion distance that the surgical fastener extends from the distal end of the device. A stop in the housing end of the device can engage directly with the flexible member, storage member, or drive wheel. If engaged with the drive wheel, it can provide the benefit of reducing compressive loads on the flexible member when left in the ready to fire position with the flexible member extended. Alternatively, a stop in the housing end of the device may

11

engage with the storage reel or flexible member to prevent over-rotation of the storage reel and subsequent damage to the proximal end of the flexible member. For either proximal stop, it is critical to delay the engagement of the stop until after the surgical fastener has sufficient stroke to embed into tissue. This also provides time for the length of the flexible member to compress, dampening the impact at the proximal stop.

In one embodiment, the flexible member is capable of elastically compressing and buckling within the constraints of a guide member, which preferably limits the force or stroke that may be applied to the surgical fastener.

In one embodiment, a cartridge may have tissue fasteners stacked at any angle within a magazine (e.g. horizontal, vertical or any angle in between), relative to the orientation of the elongated shaft.

In one embodiment, surgical fasteners may be singulated or stripped from the cartridge utilizing the flexible member. In one embodiment, a distal end of the flexible member includes a protruding portion (e.g., a solid or compressible fin) that pushes/strips a single surgical fastener from the cartridge to singulate the surgical fastener and move it into a cannula for delivery. In one embodiment, an insertion tool or insertion guide is affixed to the distal end of the flexible member. In one embodiment, the insertion tool is a feature of the flexible member, such that the flexible member and the insertion tool are a single component. The insertion tool may include the fin or a stripper ramp that engages a surgical fastener for stripping the surgical fastener from the bottom of a surgical fastener stack.

In one embodiment, a dual path arrangement is utilized for advancing a surgical fastener toward a distal end of the elongated member. In one embodiment, the dual path arrangement includes a first path in which the distal end of the flexible member strips off a single tissue fastener, and positions the tissue fastener in a staging position on a second path. A section of the flexible member proximal to the distal end of the flexible member drops down to the second path for subsequent deployment of the tissue fastener.

In one embodiment, a rotary motion element may be used to strip a single surgical fastener from a cartridge and place the surgical fastener in a staging position for being engaged by a distal end of the flexible member.

In one embodiment, an applicator instrument for dispensing surgical fasteners preferably includes a user actuated trigger that drives a gear train to rotate a storage reel, which, in turn, retracts a flexible member from a distal end of a cannula onto the storage reel. In one embodiment, the storage reel for the flexible member is a proximal storage reel located at the proximal end of the distal housing assembly of the instrument.

In one embodiment, retraction of the flexible member also rotates a drive wheel through a cogged interface or friction interface between the flexible member and the drive wheel. The rotation of the drive wheel winds a constant torque spring from a spool onto the drive wheel for storing energy in the constant torque spring. When the flexible member reaches a pre-determined retracted position behind a surgical fastener cartridge located in the housing or the handle, the gear train disengages and the portion of the constant torque spring that was wound onto the drive wheel unwinds back to the spool, which, in turn, rotates the drive wheel. The rotation of the drive wheel pulls the flexible member off the storage reel and drives the flexible member distally through the cogged interface and the flexible member strips a tissue

12

fastener from the cartridge and pushes the tissue fastener to the distal end of the cannula (e.g., elongated shaft) and into tissue.

In one embodiment, the applicator instrument may have a drive train that is directly connected to the drive wheel to wind the constant torque spring from the spool onto the drive wheel. In this embodiment, it is preferable to utilize a power spring or other means to wind and manage the flexible member as it is urged proximal onto the storage reel.

In one embodiment, the flexible member and surgical fastener may be continually accelerated as the surgical fastener is pushed down the length of the elongated shaft. This acceleration increases the velocity and momentum of the fastener, flexible member, and drive wheel system. A minimum velocity and inertia are necessary to allow the fastener to pierce meshes and abdominal wall tissue.

In one embodiment, the flexible member is adapted to push the tissue fastener via geometry that engages with the tissue fastener or via one or more flat contact surfaces.

In one method of dispensing a surgical fastener, similar to a "bow and arrow" method, the flexible member may be assembled at the distal end of the elongated member and, during the firing stroke, be drawn back to the handle to engage and deliver the tissue fastener. In this embodiment, the flat flexible member remains in a flat configuration during storage, reducing the likelihood that the flexible member will take a permanent set. In one method of dispensing a surgical fastener, referred to as a "coiled snake" method, the flexible member may start in the handle and during the firing stroke, experience a force to deliver the tissue fastener to the distal end of the elongated member, and then return back to the handle.

In one embodiment, the flexible member may be pulled from and return to a reel located in the handle. In one embodiment, the reel may have a power spring or the flexible member may be self-coiling to help retract the flexible member and to provide tension to prevent billowing of the flexible member away from the reel during operation. In one embodiment, the drive system may be spring powered, electrically powered, air powered, hydraulically powered, etc.

The flexible member may be fed through a straight cannula, a curved cannula, or through a cannula with an articulating end.

In one embodiment, an applicator instrument having an articulating distal end may have a bi-stable configuration that utilizes an over-center spring to bias the system toward no articulation of the elongated shaft or full articulation of the elongated shaft. The bi-stable configuration prevents the distal end of the elongate shaft from existing in an intermediate state where the articulation is not as stable.

In one embodiment, the applicator instrument has a user interface (e.g. an articulation control lever) that is in a first position (e.g., horizontal) when the shaft is not articulated and a second position (e.g., vertical) when the shaft is fully articulated, thereby mirroring the configuration of the articulated end of the shaft. The user interface for controlling articulation preferably provides the user with visual feedback regarding the configuration of the articulated end of the shaft when the shaft is inserted into a trocar and is not visible. Although the present invention is not limited by any particular theory of operation, it is believed that the bi-stable configuration provides a user experience that guides the user toward one of the two articulation positions and does not allow the user to leave the articulation of the shaft in an intermediate, state. It is also considered that a bi-stable

13

system will reduce the mental task load of the surgeon and simplify the user experience by simplifying the articulation control to a simple toggle.

In one embodiment, an applicator instrument maintains the integrity of a quantity of surgical fasteners during handling and allows the surgical fasteners to be removed one at a time for implantation by the instrument. In one embodiment, a protruding portion of a flexible member pushes a single tissue fastener from a cartridge to singulate the surgical fastener and move it into a cannula for delivery from a distal end of an elongated shaft. In one embodiment, surgical fasteners are stacked in the cartridge and a constant force spring is used to move the fasteners toward a staging location in the cartridge. In one embodiment, the surgical fasteners are desirably restricted and only allowed to move in a direction toward the staging location in the cartridge. The staging location in the cartridge preferably allows a single surgical fastener in the staging location to move in a direction parallel to the flexible member and then transition into the cannula.

In one embodiment, the distal end of the flexible member includes an insertion tool having a protruding portion shaped like a “shark fin” that is configured to engage with the back of a surgical fastener in the staging location of the cartridge and push the surgical fastener into the cannula for delivery into tissue. In one embodiment, the channel through the cannula has a relief groove for the fin.

In one embodiment, a cartridge containing surgical fasteners may have one or more orientation indicators that direct a user as to the correct orientation for inserting the cartridge into the applicator instrument. In one embodiment, the cartridge may have Poke-a-Yoke features so that the cartridge cannot be inserted incorrectly. In one embodiment, the cartridge may have a color indicator that indicates the type of surgical fasteners loaded into the cartridge. In one embodiment, the color indicator may be printed onto a label placed on the cartridge or the color indicator may be the color of the material used to make the cartridge. In one embodiment, the cartridge may have a snap feature that provides tactile feedback that the cartridge is fully/properly inserted into the applicator instrument. In one embodiment, the applicator instrument may have a cartridge release that may be engaged for locking or inserting the cartridge into the housing of the applicator instrument or unlocking and removing the cartridge from the housing of the applicator instrument. In one embodiment, the cartridge release may be on the cartridge itself.

In one embodiment, an applicator instrument has a cartridge system that utilizes a single linear path to strip a surgical fastener from the cartridge and place the stripped surgical fastener into a proper location for being delivered from a distal end of the instrument.

In one embodiment, prior to firing the applicator instrument, surgical fasteners are stacked in a cartridge. A pusher is staged distal to the surgical fastener, an elevator is aligned with the lower end of the cartridge, and a slide is in a proximal-most position. During a first stage of a firing cycle, as a user squeezes a trigger, the pusher pushes a single surgical fastener out of the cartridge and into the elevator, and the slide moves distally. During a second stage of the firing cycle, as the users fully squeezes the trigger, the pusher stops moving once the surgical fastener is fully loaded into the elevator, and the elevator moves down into alignment with the path of the distal end of the flexible member, on top of a lower guide. During a third stage of a firing cycle, the slide is in a distal-most position, the elevator

14

is all the way down, and the flexible member moves along the lower guide for delivering the surgical fastener down the elongated shaft.

In one embodiment, an applicator instrument for dispensing surgical fasteners has a cartridge system whereby the flexible member utilizes a first path to strip a surgical fastener from the cartridge and advance the surgical fastener to the distal end of the elongated shaft, and utilizes a second, different path for returning the flexible member to a location that is proximal to the cartridge (e.g., a “Racetrack” path).

In one embodiment, an applicator instrument may have a cartridge system that utilizes a rotary motion member to strip/singulate a surgical fastener and place the surgical fastener into a proper location for being engaged by a distal end of the flexible member. As a result of using a rotary motion element, surgical fasteners may be stacked inside the cartridge in any orientation relative to the path of the flexible member. In one embodiment, the rotary motion may rotate a surgical fastener through any angle necessary to bring it in-line with the distal end of the flexible member (e.g., 90 degree rotation).

In one embodiment, a cartridge system that stores surgical fasteners in a rotary drum may be side loaded onto an applicator instrument or may be top loaded onto the applicator instrument.

In one embodiment, an applicator instrument for dispensing surgical fasteners has an elongated shaft that may be articulated for moving between a straight configuration and an articulated, curved, or angled configuration. In one embodiment, the articulating shaft provides a high level of cannula rigidity in both the articulated configuration and the straight configuration, especially where high axial/lateral forces or pressure is exerted on the distal tip of the cannula during tack/strap application in hernia fixation or whenever a counter pressure on the cannula tip is applied (e.g., during mesh manipulation with the distal end of the instrument).

In one embodiment, an applicator instrument for dispensing surgical fasteners has an articulating shaft. The articulating instrument preferably includes an elongated shaft having a proximal end, a distal end, and a longitudinal axis extending between the proximal and distal ends. The applicator instrument preferably has a handle attached to the proximal end of the shaft, a cam assembly attached to the handle, at least one segmented member on the distal end of the elongated shaft, and at least one band having a proximal end and a distal end. In one embodiment, the distal end of the band is attached to the at least one segmented member at the distal end of the elongated shaft, and the proximal end of the band is adjustably attached to the cam assembly such that rotation of the cam assembly results in movement of the band along the longitudinal axis of the shaft resulting in articulation of the segmented member.

In one embodiment, the proximal end of the band is attached to a yoke, which, in turn, is attached to the cam assembly. In one embodiment, the yoke can pivot around a central axis in response to movement of the cam assembly.

In one embodiment, an articulation system preferably includes two bands, each having one end attached to the at least one segmented member on opposite sides of the at least one segmented member and the respective proximal ends of the bands are attached to the rotatable yoke.

In one embodiment, a proximal end of an upper band is connected with a first slider, and the first slider is connected with the rotatable yoke. In one embodiment, a proximal end of a lower band is connected with a second slider, and the second slider is connected with the rotatable yoke. The connection distance between the first and second sliders and

15

the yoke may be adjusted for controlling the tension applied onto the upper and lower bands.

In one embodiment, the first slider is adjusted in a proximal direction to impart a first tension on the upper band. Similarly, the second slider is adjusted in a proximal direction to impart a second tension on the lower band. In one embodiment, the second tension on the lower band is less than the first tension on the upper band.

In one embodiment, the cam engages a spring that normally urges the cam into one of two stable positions, a stable first position or a stable second position. In one embodiment, moving the cam to the second position increases the tension on the upper band while reducing the force on the lower band, which results in articulation of the at least one articulating segment at the distal end of the elongated shaft.

In one embodiment, at least one articulating segment is rotatably attached to the distal end of the elongated shaft so that rotation of the shaft is limited or controlled by features on the at least one articulating segment. In one embodiment, when the cam is in the second position, rotation of the distal end of the shaft is limited to a pre-defined angle by features on the at least one articulating segment. In one embodiment, the pre-defined rotation is preferably 60 degrees or any value between 0 and 90 degrees.

In one embodiment, an applicator instrument for dispensing surgical fasteners has an articulating shaft. In one embodiment, the applicator instrument includes a proximal shaft section having a proximal end, a distal end, and a longitudinal axis, and a handle attached to the proximal end of the proximal shaft section. The instrument preferably includes segmented members coupled with the distal end of the proximal shaft section. Each pair of segmented members accommodates a specific rotation limit. The instrument preferably has two articulation bands disposed within the segmented members. In one embodiment, a cam system is moved from a first position into a second position to tension the upper band while simultaneously releasing the lower band to force the segmented members into the articulated configuration. In one embodiment, the cam system is moved from the second position back to the first position to tension the lower band while simultaneously releasing the upper band to return the segmented members to a non-articulated position (e.g., straight).

In one embodiment, an over-center spring element automatically forces the articulating, segmented members into either the articulated or straight configuration. In one embodiment, the articulation system includes adjustment mechanisms for adjusting the level of tension applied to the upper and lower bands to provide appropriate tension that keeps the shaft at proper rigidity in both the articulated and straight configurations.

In one embodiment, the upper and lower band members may be attached to respective upper and lower sliders that may move axially for adjusting the respective tensions in the upper and lower bands and the rigidity of the segmented cannula. In one embodiment, the movement of the upper and lower sliders may be synchronized by coupling the sliders with a yoke giving a center of rotation located halfway between the upper and lower articulation bands. In one embodiment, the yoke is driven by a cam member that rotates back and forth (e.g., 90 degrees rotation). In one embodiment, the cam has a spiral slot which engages with a feature on the yoke. When the cam member reaches its limits of travel, the slope of the spiral slot is sufficiently steep (i.e. <10 degrees) to prevent back drive, i.e. the addition of external loads to the distal end of the articulating cannula is unable to force the cam member to rotate.

16

In one embodiment, the tension forces on the two bands may be adjusted by adjusting the distance between the sliders and rotating yoke.

In one embodiment, the articulating segmented members have a pin-less design, which reduces the risk of foreign matter (e.g., a pin) from falling into the abdominal cavity due to pin failure.

In one embodiment, the segmented members preferably have a channel or conduit for the surgical fasteners and the flexible member to pass through for firing a surgical fastener.

In one embodiment, the adjacent links of the segmented members are adapted to pivot relative to each other for articulating the segmented member. In one embodiment, each link has interlocking structures (e.g., wings) that maintain the rigidity and solid state of the segmented members when in the articulated or straight positions. In one embodiment, the interlocking structures fully restrict five degrees of freedom and only allow a limited range of rotation around one axis. In one embodiment, the links of the segmented members have pass ways or slots for receiving the upper and lower articulation bands.

In one embodiment, the distal ends of the upper and lower articulations bands are welded, joined, attached, hooked, crimped, or pinned onto a distal-most member of the series of segmented members.

In one embodiment, springs may be added to the tensioning adjustment means for tension force relief in the event excessive force is accidentally applied to the distal end of the articulating cannula. As such, less force would be required to displace the distal end of the articulating cannula assembly, resulting in less stress on the articulation bands.

In one embodiment, providing an applicator instrument having an articulating shaft provides many benefits. In one embodiment, providing an articulating shaft improves ergonomics during hernia surgery, enables ipsilateral (same side) mesh manipulation and fixation (reducing the required number of trocars), enhances maneuverability both inside and outside the body cavity, improves visualization of the fixation site, and reduces the length of surgical procedures.

In one embodiment, the articulation system requires only a one quarter turn (% turn) of the cam or articulation lever(s) for fully articulating the shaft, and only a one quarter return to return the shaft to a straight position/configuration.

In one embodiment, the articulation system provides a high degree of tension and rigidity, prevents cannula deformation during application, and prevents cannula angle change when the shaft is in the articulated position to provide for tip stability and accurate placement for the fixation of surgical fasteners.

These and other preferred embodiments of the present patent application will be described in more detail herein.

BRIEF DESCRIPTION OF THE DRAWING

FIG. 1A shows an applicator instrument for dispensing surgical fasteners having an elongated shaft, a housing connected to a proximal end of the elongated shaft, and a reconfigurable handle pivotally connected with the housing, the reconfigurable handle being in a pistol configuration, in accordance with one embodiment of the present patent application.

FIG. 1B shows the applicator instrument of FIG. 1A with the reconfigurable handle where the trigger is in a position in-line with the elongated shaft.

FIG. 2A shows a range of motion for the reconfigurable handle between the pistol configuration of FIG. 1A and the in-line configuration of FIG. 1B.

17

FIG. 2B shows the applicator instrument of FIG. 2A with the reconfigurable handle in the pistol configuration and the distal end of the elongated shaft in an articulated configuration.

FIG. 3 shows the applicator instrument of FIGS. 1A and 1B used for lateral access and midline access on a patient, in accordance with one embodiment of the present patent application. Lateral access is typical for a ventral or TAPP inguinal repair. Midline access is typical for a TEP inguinal repair.

FIG. 4A shows a cross-sectional view of the applicator instrument of FIGS. 1A and 1B with the reconfigurable handle in the pistol configuration.

FIG. 4B shows a cross-sectional view of the applicator instrument of FIGS. 1A and 1B with the reconfigurable handle in the in-line configuration.

FIG. 5 shows the reconfiguration mechanism and pivotal connection between housing and a reconfigurable handle of an applicator instrument, in accordance with one embodiment of the present patent application.

FIGS. 6A and 6A-1 show perspective views of a handle reconfiguration actuator and a slider used for changing the position of a reconfigurable handle relative to a housing, in accordance with one embodiment of the present patent application.

FIGS. 6B and 6B-1 show perspective views of the handle reconfiguration actuator and the slider of FIG. 6A with the handle reconfiguration actuator depressed for changing the position of the reconfigurable handle relative to the housing.

FIG. 7 shows a cross-sectional view of an applicator instrument including a positive stop provided on a gear for preventing squeezing of a trigger when a handle reconfiguration actuator is depressed, in accordance with one embodiment of the present patent application.

FIG. 8 shows a cross-sectional view of an applicator instrument including a positive stop provided on a gear for preventing the handle reconfiguration actuator from being depressed after the trigger has been squeezed for commencing a firing cycle.

FIG. 9A shows a cross-sectional view of an applicator instrument for dispensing surgical fasteners, in accordance with one embodiment of the present patent application.

FIG. 9B shows a perspective view of the applicator instrument of FIG. 9A.

FIGS. 10A and 10B show a flexible member, a storage reel and a drive wheel of an applicator instrument for dispensing surgical fasteners, in accordance with one embodiment of the present patent application.

FIG. 11 shows a flexible member and friction drive wheels of an applicator instrument for dispensing surgical fasteners, in accordance with one embodiment of the present patent application.

FIG. 12 shows a cross-sectional view of an applicator instrument for dispensing surgical fasteners including a storage reel, a drive wheel, and a proximal positive stop for halting rotation of the storage reel, in accordance with one embodiment of the present patent application.

FIG. 13A shows a cross-sectional view of an applicator instrument for dispensing surgical fasteners prior to the commencement of a firing cycle, in accordance with one embodiment of the present patent application.

FIG. 13B shows a cross-sectional view of the applicator instrument of FIG. 13A during a later stage of the firing cycle, in accordance with one embodiment of the present patent application.

FIGS. 14A and 14A-1 show a magnified view of the firing system shown in FIG. 13B.

18

FIGS. 14B and 14B-1 show the firing system of FIGS. 14A and 14A-1 during later stages of the firing cycle after a surgical fastener has been stripped from a cartridge.

FIG. 15A shows a cross-sectional view of an applicator instrument for dispensing surgical fasteners prior to the commencement of a firing cycle, in accordance with one embodiment of the present patent application.

FIG. 15B shows a cross-sectional view of the applicator instrument of FIG. 15A during a later stage of the firing cycle, in accordance with one embodiment of the present patent application.

FIG. 16A shows a magnified view of the firing system shown in FIG. 15B.

FIG. 16B shows the firing system of FIG. 16A during a later stage of the firing cycle.

FIGS. 16C and 16D show a one-way pawl for engaging supplemental teeth on a drive gear, in accordance with one embodiment of the present patent application.

FIG. 17 shows a spring-loaded cartridge containing a plurality of stacked surgical fasteners, in accordance with one embodiment of the present patent application.

FIGS. 18A-18D show a method of using a pusher secured to a distal end of a flexible member for removing a lowermost surgical fastener from a cartridge, in accordance with one embodiment of the present patent application.

FIGS. 19A-19H show a surgical fastener used to secure prosthetic devices to tissue, in accordance with one embodiment.

FIG. 20 shows the surgical fastener of FIGS. 19A-19H in a stacked array, in accordance with one embodiment.

FIGS. 21A-21D show a pusher used to dispense the surgical fastener of FIGS. 17-17H from an applicator instrument, in accordance with one embodiment.

FIGS. 22A-22D show the pusher of FIGS. 19A-19D engaged with the surgical fastener of FIGS. 17A-17H.

FIGS. 23 and 24 show a method of guiding a surgical fastener and a pusher through a channel of an elongated shaft of an applicator instrument, in accordance with one embodiment of the present patent application.

FIGS. 25A-25D show a cartridge system of an applicator instrument having an elevator for aligning a surgical fastener with a distal end of a flexible member, in accordance with one embodiment of the present patent application.

FIGS. 26A-26B, 26B-1, and 26C show a cartridge system of an applicator instrument having a racetrack element, in accordance with one embodiment of the present patent application.

FIGS. 27A-27C show a cartridge system of an applicator instrument having a rotary element, in accordance with one embodiment of the present patent application.

FIGS. 28A-28B show a side loaded cartridge system of an applicator instrument having a rotary drum, in accordance with one embodiment of the present patent application.

FIGS. 29A-29B show a top loaded cartridge system of an applicator instrument having a rotary drum, in accordance with one embodiment of the present patent application.

FIG. 30A shows the applicator instrument of FIG. 1A with the reconfigurable handle in the pistol configuration and the distal end of the elongated shaft in an articulated configuration.

FIG. 30B shows the applicator instrument of FIG. 1B with the reconfigurable handle in the in-line configuration and the distal end of the elongated shaft in an articulated configuration.

FIG. 31A shows an applicator instrument for dispensing surgical fasteners having an articulation system for articulation.

19

lating a distal end of an elongated shaft, in accordance with one embodiment of the present patent application.

FIG. 31B shows a distal end of an elongated shaft having an articulating member with links capable of pivoting relative to one another, the articulating member being in a straight configuration, in accordance with one embodiment of the present patent application.

FIG. 32A shows the articulation system of FIG. 31A with a cam plate rotated into a second position for articulating the distal end of the elongated shaft of FIG. 31B.

FIG. 32B shows the distal end of the elongated shaft of FIG. 31B with the articulating member in a fully articulated configuration.

FIGS. 33A-33C show an applicator instrument having an articulation system and a distal end of an elongated shaft being in a straight configuration, in accordance with one embodiment of the present patent application.

FIGS. 34A-34C show the applicator instrument of FIGS. 33A-33C with the elongated shaft being in a fully articulated configuration.

FIG. 35 shows a perspective view of an intermediate link of the articulation member of FIG. 31B.

FIG. 36 shows a cross-sectional view of an elongated shaft and a flexible member disposed within an elongated conduit of the elongated shaft, buckled under compressive load.

FIG. 37 shows a cross-sectional view of a distal end of an elongated shaft having a positive distal stop for halting distal movement of a flexible member, in accordance with one embodiment of the present patent application.

FIG. 38 shows a partial cross-sectional view of an applicator instrument having an articulation system with a tensioning spring coupled with a slider, in accordance with one embodiment of the present patent application.

FIG. 39 shows a partial cross-sectional view of an applicator instrument having an articulation system with a first tensioning spring coupled with a first slider and a second tensioning spring coupled with a second slider, in accordance with one embodiment of the present patent application.

FIGS. 40 and 41A-41D show a gear train for an applicator instrument, in accordance with one embodiment of the present patent application.

FIGS. 42A-42E show an applicator instrument, in accordance with one embodiment of the present patent application.

FIGS. 43A-43C show an applicator instrument, in accordance with one embodiment of the present patent application.

FIGS. 44A-44C show schematic views of a firing system for an applicator instrument, in accordance with one embodiment of the present patent application.

FIG. 45A shows an applicator instrument having a flexible member in an extended position, in accordance with one embodiment of the present patent application.

FIG. 45B shows an applicator instrument having a flexible member in a retracted position, in accordance with one embodiment of the present patent application.

FIGS. 46A-46E show a schematic of an applicator instrument having an energy storing assembly, in accordance with one embodiment of the present patent application.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

Referring to FIG. 1A, in one embodiment, an applicator instrument 100 for dispensing surgical fasteners preferably

20

includes an elongated shaft 102 having a proximal end 104 and a distal end 106. In one embodiment, surgical fasteners are advanced from the proximal end 104 to the distal end 106 of the elongated shaft 102 for dispensing the surgical fasteners one at a time from the distal end 106 of the shaft. In one embodiment, the distal end 106 of the elongated shaft 102 is capable of being articulated, as will be described in more detail herein.

In one embodiment, the applicator instrument 100 includes a housing 108 and a reconfigurable handle 110 that is coupled with a proximal end of the housing 108. The applicator instrument 100 includes a trigger 112 that may be squeezed toward the reconfigurable handle 110 for dispensing a surgical fastener from the distal end 106 of the elongated shaft 102.

In one embodiment, the applicator instrument 100 includes an articulation lever 114 that is accessible on the housing 108. The articulation lever 114 is desirably movable between a first position whereby the distal end 106 of the elongated shaft 102 is in a straight configuration, and a second position whereby the distal end 106 of the elongated shaft 102 is in an articulated configuration. The articulation lever 114 may be horizontal in the first position and vertical in the second position to provide a visual indicator as to whether the distal end of the elongated shaft is straight or articulated.

In one embodiment, the housing 108 of the applicator instrument 100 is adapted to receive a cartridge 116 that contains a plurality of surgical fasteners. In one embodiment, the surgical fasteners within the cartridge 116 are stacked atop one another and the cartridge is spring-loaded for urging the stacked surgical fasteners to move toward one end of the cartridge (e.g., in a downward direction). As will be described in more detail herein, the applicator instrument 100 preferably includes a flexible member that strips a single surgical fastener located at a lower end of a stack of surgical fasteners and pushes the surgical fastener through the elongated shaft 102 for being dispensed from the distal end 106 of the elongated shaft. In one embodiment, the flexible member serves as a structure for firing or dispensing a surgical fastener from a distal end of the elongated shaft.

In one embodiment, the applicator instrument 100 may be placed in either a pistol configuration or an in-line configuration, or at a plurality of positions located between the pistol and in-line configurations. FIG. 1A shows the applicator instrument 100 with the handle 110 in a pistol configuration. Referring to FIG. 1B, in one embodiment, the handle 110 may be selectively rotated relative to the housing 108 for placing the handle in an in-line configuration. The ability to move the handle between a pistol configuration, an in-line configuration, and any intermediate positions/angles between the pistol and in-line configurations may be particularly useful during various types of surgical procedures (e.g., hernia repair procedures) where ergonomics and instrument maneuverability are critical. In one embodiment, the handle 110 may be locked at a range of different angular orientations relative to the housing 108. In one embodiment, the handle may be rotated beyond the pistol or in-line configurations. In one embodiment, as the position of the handle is being reconfigured, a gear train in the handle 110 is preferably disengaged from a firing system in the housing so that reconfiguring the handle does not affect the trigger, the gear train, or the firing system.

Referring to FIG. 2A, in one embodiment, the handle 110 of the applicator instrument 100 is reconfigurable so that it may be positioned in either the pistol configuration or an in-line configuration. In one embodiment, the handle 110

21

may also be positioned at intermediate locations between the pistol configuration and the in-line configuration. In one embodiment, the trigger 112 is coupled to the handle 110 and moves with the handle as it rotates relative to the housing 108 between the pistol configuration and the in-line configuration.

In one embodiment, the lower end of the cartridge 116 that contains a stack of surgical fasteners is inserted into an opening in the housing 108. In one embodiment, the lower end of the cartridge 116 forms a snap-fit connection with the housing 108. In one embodiment, a cartridge release button 118 may be depressed for removing the cartridge 116 from its snap-fit connection with the housing 108. The cartridge release button 118 may be on either the cartridge 116 or the housing 108. In one embodiment, an applicator instrument does not have a cartridge release button on the housing and a release element is provided on the cartridge that is inserted into the housing 108. In one embodiment, a proximal end 120 of the housing 108 has a concave surface and a distal end 122 of the handle 110 has a convex surface that guides movement of the handle 110 relative to the housing 108 as it rotates between the in-line and the pistol configuration.

Referring to FIGS. 2A and 2B, in one embodiment, the applicator instrument 100 preferably includes the elongated shaft 102 that projects from a distal end of the housing 108. The articulation lever 114 may be rotated from the first position (i.e., horizontal) shown in FIG. 2A, whereby the distal end of the elongated shaft is straight, and the second position (i.e., vertical) FIG. 2B whereby the distal end 106 of the elongated shaft 102 is in an articulated configuration. In one embodiment, the distal end 106 of the elongated shaft 102 may be moved between the straight configuration and the articulated configuration with the handle 110 in either the in-line configuration (FIG. 2A) or the pistol configuration (FIG. 2B). A reconfiguration button 136 may be engaged for moving the handle 110 between the pistol and in-line configurations. The position of the handle may be locked in place by releasing the reconfiguration button 136.

Referring to FIG. 3, in one embodiment, the position of the handle 110 relative to the housing 108 may be selectively reconfigured for improving ergonomics and instrument maneuverability during a surgical procedure. In one embodiment, when the applicator instrument 100 is used for trans-abdominal pre-peritoneal (TAAP) inguinal and ventral repairs, a lateral access approach into the patient may be utilized. For lateral access, in order to improve ergonomics and instrument maneuverability, the handle 110 may be placed in the in-line configuration relative to the housing 108. Depending on the location of the target fixation site relative to the trocar, other handle positions may be used, as necessary. In contrast, for totally extra-peritoneal (TEP) inguinal repairs, where trocars are placed near the patient's mid-line, the handle 110 of the applicator instrument 100 may be placed in the pistol configuration relative to the housing 108 for improving ergonomics and instrument maneuverability. Depending on the location of the target fixation site relative to the trocar, other handle positions may be used, as necessary. Although the present invention is not limited by any particular theory of operation, it is believed that providing applicator instruments for dispensing surgical fasteners that have reconfigurable handles, with one or more positions between the pistol and in-line configurations, dramatically improves the maneuverability and ergonomics of the applicator instruments when used during laparoscopic procedures. Of particular benefit is the ability to achieve a neutral wrist angle for the surgeon. Reconfigurable handles also enable more accurate placement of surgical fasteners,

22

reduced fatigue, and enhanced accuracy when affixing surgical fasteners. Moreover, reconfigurable handles enable a surgeon to maintain the entire applicator instrument within a sterile field so that a surgeon may avoid moving any portion of the applicator instrument outside the sterile field as the surgeon maneuvers the applicator instrument into different orientations.

Referring to FIG. 4A, in one embodiment, the applicator instrument 100 preferably includes the housing 108 and the reconfigurable handle 110 coupled with a proximal end of the housing. The applicator instrument 100 includes the elongated shaft 102 that extends distally from a distal end of the housing 108. In one embodiment, the applicator instrument includes a gear train 124 that extends between the trigger 112 in the handle 110 and the housing 108 for activating a firing system to dispense a surgical fastener. In one embodiment, the applicator instrument has a firing system that meshes with the gear train 124. In one embodiment, the firing system preferably includes a storage reel 126 for storing a flexible member used to dispense a surgical fastener from a distal end of the elongated shaft 102, a drive wheel 128 that engages the flexible member for driving the flexible member toward the distal end of the elongated shaft 102 for dispensing a surgical fastener, and a constant torque spring for storing energy used to drive the drive wheel 128.

In one embodiment the housing 108 has an opening adapted to receive a cartridge 116 containing a plurality of surgical fasteners stacked in the cartridge. The surgical fasteners within the cartridge 116 are adapted to be dispensed one at a time from the distal end of the elongated shaft 102. During a firing cycle, each time the trigger 112 is squeezed toward the handle 110, a firing cycle commences for firing a single surgical fastener from the distal end of the elongated shaft 102. In one embodiment, the applicator instrument 100 includes a trigger return spring 130 having a lower end 132 connected with the handle 110 and an upper end 134 connected with the trigger 112. In one embodiment, the trigger return spring 130 normally urges the trigger to return to the open position shown in FIG. 4A.

In one embodiment, an applicator instrument may be capable of receiving different cartridges containing surgical fasteners having different properties. In one embodiment, the different properties for the surgical fasteners may relate to the material, absorption time, size, length, width, leg length, barb length, number of barbs on a leg, curved legs, straight legs, color, opacity, cartridge quantity, etc. In one embodiment, a first cartridge may contain surgical fasteners having a first property (e.g., legs having a first length) and a second cartridge may contain surgical fasteners having a second property (e.g., legs having a longer second length) that is different than the first property. The first cartridge may be inserted into the applicator instrument for dispensing the first surgical fasteners into the patient. After completing the fixation necessary with the first surgical fasteners, the first cartridge may be decoupled from the applicator instrument and replaced by the second cartridge for dispensing the second surgical fasteners into the patient. A portion of the applicator instrument (e.g., the elongated shaft), may remain inside the patient as the cartridges are changed, which preferably minimizes the time required to perform a surgical procedure and enhances sterile conditions. After exchanging cartridges, the applicator device is immediately ready to use the second set of surgical fasteners. Additional cartridges having additional properties may be used during a single surgical procedure. In one embodiment, multiple surgical fastener types may be provided in a single cartridge.

23

In one embodiment, the applicator instrument **100** preferably includes a handle reconfiguration actuator **136** (e.g., a depressible button) that is provided on the reconfigurable handle **110**. In FIG. 4A, the handle reconfiguration actuator **136** is depressed so that the handle **110** may be rotated between the pistol configuration shown in FIG. 4A and the in-line configuration shown in FIG. 4B. Referring to FIGS. 4A and 4B, in one embodiment, the trigger **112** is coupled with the reconfigurable handle **110** so that the trigger rotates with the handle as it moves between the pistol configuration and the in-line configuration.

In one embodiment, the handle **110** may be rotated relative to the housing **108** when the handle reconfiguration actuator **136** is depressed or moved. When the handle reconfiguration actuator **136** is not depressed, the handle **110** is locked in place relative to the housing **108** and may not move between the pistol and in-line configurations. In one embodiment, the gear train extends between the handle and the housing in order to actuate the firing system. The handle includes the trigger **112** for activating the gear train **124**. When the trigger is pulled, the gear train **124** is activated for commencing a firing cycle and allowing a surgical fastener to be dispensed from a distal end of the elongated shaft **102**. In one embodiment, the gear train **124** is initially disengaged when the trigger **112** is open. In one embodiment, when the trigger **112** is actuated (e.g., squeezed), the gear train **124** engages the firing system. Because the gear train **124** is initially disengaged, the handle **110** can be reconfigured between the pistol configuration and the in-line configuration without affecting the stroke of the trigger/gear train. In one embodiment, a one-way bearing meshes the gear train with the firing system during a first stage of a firing cycle, and the one-way bearing decouples the gear train from the firing system during a later stage of the firing cycle. In another embodiment, when the handle reconfiguration actuator **136** is depressed for rotating the handle **110**, a portion of the gear train located in the handle **110** is disengaged, allowing for the reconfiguration of the handle relative to the housing and not affecting the stroke of the trigger/gear train. In one embodiment, when the trigger **112** is squeezed for activating a firing cycle, the handle reconfiguration actuator **136** is blocked from being depressed so as to prevent reconfiguration of the handle relative to the housing during a firing cycle. In one embodiment, when the handle reconfiguration actuator **136** is depressed, the trigger and/or gear train **124** is blocked to prevent firing of a surgical fastener prior to completing the reconfiguration of the handle relative to the housing.

Referring to FIG. 5, in one embodiment, a proximal end of the housing **108** includes a series of spaced reconfiguration notches **138A-138G** that are utilized for positioning the handle **110** at different configurations/angles relative to the housing **108**. In one embodiment, the reconfiguration notches **138A-138G** are arrayed in an arc-shaped pattern at the proximal end of the housing **108**. In one embodiment, the applicator instrument **100** includes a slider **140** that extends between the handle reconfiguration actuator **136** and the reconfiguration notches **138A-138G** at the proximal end of the housing **108**. The slider **140** includes a proximal end **142** that is engaged by an underside of the handle reconfiguration actuator **136** and a distal end **144** that is adapted to be seated within the reconfiguration notches **138A-138G**. In one embodiment, when the handle reconfiguration actuator **136** is depressed, the distal end **144** of the slider **140** disengages from one of the reconfiguration notches **138A-138G**, which enables the handle **110** to be rotated relative to the housing **108**. When the handle reconfiguration actuator **136** is

24

released, the distal end **144** of the slider **140** is seated within one of the reconfiguration notches **138A-138G** for locking the position of the handle **110** relative to the housing **108** so that the handle may not be pivoted relative to the housing.

Referring to FIGS. 6A and 6A-1, in one embodiment, when the handle reconfiguration actuator **136** is in an extended, undepressed position, the distal end **144** of the slider **140** sits within the reconfiguration notch **138A** for locking the position of the handle **110** relative to the housing **108**. A slider return spring **146** normally urges the distal end **144** of the slider **140** to be seated within one of the reconfiguration notches **138A-138G** (FIG. 5).

Referring to FIGS. 6B and 6B-1, when the handle reconfiguration actuator **136** is depressed, and underside of the handle reconfiguration actuator engages the proximal end **142** of the slider **140** for overcoming the force of slider return spring **146** and shifting the distal end **144** of the slider **140** away from the reconfiguration notch **138A**. As the distal end **144** of the slider **140** moves away from the reconfiguration notch **138A**, the slider return spring **146** is compressed. With the handle reconfiguration actuator **136** depressed, the handle **110** may be rotated relative to the housing **108** for aligning the distal end **144** of the slider **140** with any one of the other reconfiguration notches **138A-138G** (FIG. 5). When the handle **110** has been positioned at a desired configuration relative to the housing **108**, the handle reconfiguration actuator **136** may be released, whereupon the slider return spring **146** urges the slider **140** and the distal end **144** of the slider **140** back into engagement with one of the reconfiguration notches **138A-138G** on the housing **108** for locking the handle in place. The handle **110** is locked in place relative to the housing **108** as long as the distal end **144** of the slider **140** is seated in one of the reconfiguration notches **138A-138G** (FIG. 5).

Referring to FIG. 7, in one embodiment, the handle reconfiguration actuator **136** has a locking projection **150** that extends from an underside thereof. In one embodiment, the gear train **124** preferably has a drive gear **177** having a raised ring **152** that projects above a major face of the drive gear. The raised ring **152** on the drive gear **177** is not continuous and includes a ring opening **154** that is normally aligned with the locking projection **150** of the handle reconfiguration actuator **136** when the trigger **112** is in an extended/open position. In one embodiment, when the handle reconfiguration actuator **136** is depressed for changing the configuration of the handle **110** relative to the housing **108**, the locking projection **150** extends into the ring opening **154** of the raised ring **152** of the drive gear **177**, which blocks the trigger **112** from being pulled toward the handle **110**. Thus, the firing system including the gear train **124** of the applicator instrument may not be activated when the handle reconfiguration actuator **136** is depressed for reconfiguring the position of the handle **110** relative to the housing **108**.

Referring to FIG. 8, in one embodiment, when the trigger **112** is squeezed toward the handle **110** for commencing a firing cycle, the handle reconfiguration actuator **136** may not be depressed for reconfiguring the position of the handle **110** relative to the housing **108**. In one embodiment, the drive gear **177** includes the raised ring **152** projecting from a major face thereof. As the trigger **112** is pulled toward the handle **110**, the drive gear **177** rotates in a counterclockwise direction so that the closed portion of the raised ring **152** is aligned with the locking projection **150** extending from an underside of the handle reconfiguration actuator **136**. As a result, the handle reconfiguration actuator **136** is blocked

25

from being depressed by the engagement of the locking projection 150 with the outer perimeter of the raised ring 152 of the drive gear 177.

Referring to FIGS. 9A and 9B, in one embodiment, an applicator instrument 100 for dispensing surgical fasteners includes a housing 108 and a handle 110 that may be reconfigured relative to the housing 108 for moving the handle 110 between a pistol configuration and an in-line configuration. The applicator instrument 100 includes a firing system for dispensing a surgical fastener from a distal end of an elongated shaft 102 that projects from a distal end of the housing 108. In one embodiment, the firing system is activated by a trigger 112 that may be squeezed toward the handle 110. In one embodiment, the firing system includes a gear train 124 that extends between the trigger 112 and the housing 108. In one embodiment, portions of the gear train may be disposed within both the handle 110 and the housing 108. The firing system for the applicator instrument 100 desirably includes a storage reel 126 that is adapted to store a flexible member 160 and a drive wheel 128 that engages the flexible member 160 for driving the flexible member toward the distal end of the elongated shaft 102. The flexible member preferably engages surgical fasteners for dispensing the surgical fasteners from the distal end of the elongated shaft. A cartridge 116 containing a plurality of surgical fasteners may be inserted into an opening in the housing 108 to provide surgical fasteners for the firing system. In one embodiment, a plurality of surgical fasteners are arrayed in a stack within the cartridge 116 and a lower-most one of the surgical fasteners in the stack is dispensed from a distal end of the elongated shaft 102 each time the trigger 112 is squeezed for activating a firing cycle. A distal-most end of the flexible member is adapted to strip the lower-most surgical fastener from the stack of surgical fasteners within the cartridge 116 and advance the singulated surgical fastener toward the distal end of the elongated shaft 102 for dispensing the surgical fastener from the applicator instrument. In one embodiment, the storage reel 126 rotates in a counterclockwise direction for winding the flexible member 160 onto the storage reel. During the firing cycle, the storage reel 126 rotates in a clockwise direction as the flexible member is driven in a distal direction by the drive wheel 128, also rotating in a clockwise direction.

Referring to FIGS. 10A and 10B, in one embodiment, the flexible member 160 has a proximal end 162 that is attached to the storage reel 126. When the storage reel 126 rotates in a counterclockwise direction, the flexible member 160 is wound about the storage wheel. When the storage wheel 126 rotates in a clockwise direction, the flexible member 160 is unwound from the storage reel 126 so that it may be advanced toward the distal end of the elongated shaft. In one embodiment, the storage reel 126 has a notch 164 (FIG. 10A) formed in the outer perimeter thereof. In one embodiment, the notch 164 may be engaged by a pivoting stop 170 located in the housing for preventing further clockwise rotation of the storage reel.

In one embodiment, the drive wheel 128 has a plurality of gear teeth 166 that project from an outer perimeter thereof that engage a series of openings 168 formed in the flexible member 160. In one embodiment, when the drive wheel 128 rotates in a clockwise direction (from the perspective shown in FIG. 10A), the teeth 166 engage the openings 168 on the flexible member 160 for driving the flexible member towards the distal end of the elongated shaft. In one embodiment, a constant torque spring 172 is connected between a spool 139 and the drive wheel 128. In one embodiment, retraction of the flexible member 160 rotates the drive wheel

26

128 in a counterclockwise direction (from the perspective shown in FIG. 10A), which, in turn, winds the constant torque spring 172 from the spool 139 onto the drive wheel 128. This action stores potential energy as the constant torque spring 172 wants to return to the spool 139.

Referring to FIG. 11, in one embodiment, rather than using the cogged drive wheel shown in FIG. 10, the firing system may include a pair of friction wheels 128A', 128B' having opposing friction surfaces 166A' that engage top and bottom surfaces of a flexible member 160' for driving the flexible member 160' toward a distal end of an elongated shaft. In one embodiment, the pair of friction wheels are geared to each other to reduce the potential for slippage.

Referring to FIG. 12, in one embodiment, the applicator instrument 100 includes the proximal, pivoting stop 170 adapted to engage the notch 164 formed on the outer perimeter of the storage reel 126. The proximal positive stop 170 engages the notch 164 for preventing further clockwise rotation of the storage reel 126 so as to halt further distal movement of the flexible member 160 (FIG. 10A) toward the distal end of the elongated shaft. In one embodiment, as the flexible member 160 is wound onto the storage reel 126, the proximal positive stop 170 is adapted to pivot away from the outer perimeter of the storage reel 126 when the storage reel is rotated in a counterclockwise direction, however, as the flexible member is driven toward the distal end of the elongated shaft, the proximal stop 170 is adapted to pivot into engagement with the notch 164 of the storage reel 126 for halting further clockwise rotation of the storage reel 126.

Referring to FIG. 13A, in one embodiment, at the commencement of a firing cycle, the trigger 112 is in an extended position, away from the handle 110. The flexible member 160 preferably extends to the distal end of the elongated shaft 102. Surgical fasteners (not shown) are desirably stacked within the cartridge 116, which, in turn, is inserted into the housing 108. The firing system includes a constant torque spring 172 (FIG. 10B) that is connected between a spool 139 (FIG. 10B) coupled with the storage reel 126 and the drive wheel 128.

Referring to FIG. 13B, in one embodiment, as a user squeezes the trigger 112 toward the handle 110, the gear train 124 between the trigger 112 and the storage reel 126 re-meshes and begins rotating the storage reel 126 in a counterclockwise direction for retracting the distal end of the flexible member 160 from the distal end of the elongated shaft 102 and winding the flexible member 160 onto the storage reel 126 so that the distal end of the flexible member 160 is proximal to the cartridge 116 and the surgical fasteners stacked within the cartridge. Retraction of the flexible member 116 rotates the drive wheel 128 in a counterclockwise direction, which, in turn, winds the constant torque spring 172 from the spool 139 (FIG. 10B) onto the drive wheel 128. This action stores potential energy in the constant torque spring so that the constant torque spring wants to return to the spool 139 (FIG. 10B).

Referring to FIG. 14A, in one embodiment, an insertion tool 174, adapted to engage a surgical fastener, is attached to the distal end of the flexible member 160. In FIG. 14A, the insertion tool 174 is positioned proximal to the cartridge 116 and the plurality of surgical fasteners that are stacked within the cartridge. In the stage of the firing cycle shown in FIG. 14A, the proximal end of the flexible member 160 has been wound onto the storage reel 126 and the constant torque spring 172 has been wound from the spool 139 (FIG. 10B) onto the drive wheel 128.

Referring to FIG. 14A-1, in one embodiment, when the trigger 112 has almost reached a fully squeezed position

(i.e., almost fully closed), the insertion tool **174** attached to the distal end of the flexible member **160** has been retracted to a position that is proximal to the cartridge **116** and the stack of surgical fasteners disposed within the cartridge. In one embodiment, the housing **108** preferably includes a proximal hard stop **173** that engages the insertion tool **174** for preventing further proximal movement of the insertion tool **174**. At this stage of the firing cycle, the teeth on a large drive gear **177** are about to disengage from the teeth on a smaller clutch gear **179** for decoupling the storage reel **126** from the large drive gear **177**.

Referring to FIGS. **14B** and **14B-1**, at the end of rotation of the large drive gear **177** of the drive train **124**, the large drive gear **177** disengages from the smaller clutch gear **179** for removing the tensile constraint on the flexible member **160**. In one embodiment, once the gear teeth on the large drive gear **177** no longer engage the gear teeth on the smaller clutch gear **179**, the drive wheel **128** and the storage reel **126** are decoupled from the large drive gear **177** and are free to rotate relative to the large drive gear **177**, at which point, the energy stored in the constant torque spring **172** (FIG. **14A**) is released for rotating the drive wheel **128** in a clockwise direction, which, in turn, drives the flexible member **160** and the insertion tool **174** toward the distal end of the elongated shaft **102**. At this stage, the constant torque spring **172** applies a constant force to the drive wheel **128**, resulting in constant acceleration of the drive wheel **128**, the flexible member **160**, the insertion tool **174**, and the storage reel **126**. As the drive wheel **128** rotates in a clockwise direction, the teeth **166** on the drive wheel **128** engages the openings on the flexible member **160** for driving the flexible member in a distal direction. As the insertion tool **174** is driven distally, it strips a surgical fastener from the bottom of the stack of surgical fasteners disposed within the cartridge **116** and advances the stripped surgical fastener toward the distal end of the elongated shaft **102** for being dispensed into tissue (e.g., to secure a surgical mesh to tissue). Referring to FIG. **14B-1**, in one embodiment, the pivoting stop **170** is seated in the notch **164** of the storage reel **126** for halting further clockwise rotation of the storage reel, which, in turn, halts distal movement of the flexible member **160**.

Referring to FIGS. **15A** and **15B**, in one embodiment, prior to the commencement of a firing cycle, the insertion tool **174** attached to the distal end of the flexible member **160** is located within the housing **108** of the applicator instrument **100** and is proximal to the surgical fastener stack **178** within the cartridge **116**. The surgical fastener stack **178** is located within the cartridge **116**. The flexible member **160** is wound onto the storage reel **126** and energy is stored in drive wheel **128** prior or during each firing cycle. This system may be referred to as having a “coiled snake” firing system. During the trigger actuation, the energy is released to accelerate the drive wheel **128** and flexible member **160** toward the distal end of the instrument.

Referring to FIGS. **15B** and **16A**, in one embodiment, the user squeezes the trigger **112** toward the handle **110** so that the gear train **124** rotates the drive wheel **128** in a counter-clockwise direction for winding the constant torque spring **172** from the spool **139** (FIG. **10B**) onto the drive wheel **128**.

Referring to FIG. **16B**, in one embodiment, during a later stage of the firing cycle, the gear train **124** may decouple the trigger from the storage reel **126**, thereby freeing the storage reel **126** to rotate so that the constant torque spring can rotate the drive wheel **128** in a clockwise direction for advancing the flexible member toward the distal end of the elongated shaft. In one embodiment, the constant torque spring **172** moves from the drive wheel **128** back to the spool **139** (FIG.

10B) of the storage reel **126** for rotating the drive wheel **128** in a clockwise direction and driving the flexible member **160** and the insertion tool **174** at the distal end of the flexible member **160** toward the distal end of the elongated shaft **102**. As the insertion tool is driven distally, the insertion tool **174** desirably strips a lower-most surgical fastener **176** from the bottom of the surgical fastener stack **178** within the cartridge **116** and advances the stripped surgical fastener toward the distal end of the elongated shaft **102**.

Referring to FIGS. **16C** and **16D**, in one embodiment, a one-way pawl **125** engages with a member of the gear train **124**. In one embodiment, a drive gear **177** has teeth **129** that are adapted to engage the pawl **125**. The pawl **125** preferably ensures that a user must fully squeeze and fully release the trigger. If the trigger is partially deployed in either direction, the pawl **125** will desirably engage the gear train and prevent the gear train from moving in the opposite direction. Once the trigger reaches the limits of its travel, the pawl **125** will be spring loaded via a pawl reset spring **127** to allow the pawl to reset for travel in the opposite direction. In one embodiment, a drive gear has teeth **129** that are adapted to engage the pawl **125** to prevent partial squeezing of the trigger.

Referring to FIG. **17**, in one embodiment, a cartridge **116** contains a stack of surgical fasteners **178**. An outer surface of the cartridge **116** includes a catch **180** that enables the cartridge **116** to be snap-fit into the housing **108** (FIG. **2A**) of an applicator instrument. In one embodiment, a cartridge release button **118** (FIG. **2A**) may be depressed for disengaging from the catch **180** so that the cartridge **116** may be removed from the snap-fit connection with the housing. In one embodiment, the user may depress the catch **180** so that the cartridge **116** may be removed from the snap-fit connection with the housing **108**. In one embodiment, the cartridge **116** includes a constant force spring **182** that urges the surgical fasteners **178** toward the lower end **184** of the cartridge **116**. In one embodiment, the constant force spring pulls on a cartridge sled **183**, which pushes on the stack of surgical fasteners.

Referring to FIG. **18A**, in one embodiment, the insertion tool **174** attached to the distal end of the flexible member **160** is adapted to strip a lower-most surgical fastener **176** from the bottom of the surgical fastener stack **178**. The insertion tool **174** desirably includes a fin **175** that projects above a main body of the insertion tool **174** at a distal-most end of the insertion tool for engaging a rear surface of the lower-most surgical fastener **176**. The insertion tool **174** is adapted to strip the lower-most surgical fastener **176** from the bottom of the stack and advance the surgical fastener through a conduit **105** of the elongated shaft **102** for dispensing the surgical fastener **176** from the distal-most end of the elongated shaft.

Referring to FIG. **18B**, in one embodiment, the fin **175** of the insertion tool **174** contacts the proximal or trailing end of the lower-most surgical fastener **176** for stripping the surgical fastener **176** from the bottom of the stack **178**. Referring to FIGS. **18C** and **18D**, in one embodiment, the flexible member **160** and the insertion tool **174** move distally for guiding the stripped surgical fastener **176** into the conduit **105** of the elongated shaft **102**. As shown in FIG. **18C**, ramped features **187** within the track simultaneously bias the tips of the surgical fastener and the inner crown of the surgical fastener. These ramped features ensure that the entire surgical fastener drops vertically with minimal angling. Angling of the surgical fastener should be avoided to reduce opportunities for jamming. The flexible member

29

160 continues to move distally until the insertion tool 174 dispenses the surgical fastener 176 from the distal-most end of the elongated shaft 102.

Referring to FIGS. 19A-19C, in one embodiment, a surgical fastener 220 for securing prosthetic devices (e.g., surgical mesh) to tissue preferably includes a first leg 222A having a proximal end 224A and a distal end 226A. In one embodiment, a first cross sectional area CS1 adjacent the proximal end 224A of the first leg 222A is greater than a second cross sectional area CS2 adjacent the distal end 226A of the first leg 222A. In one embodiment, the first leg 222A tapers inwardly between the proximal end 224A of the first leg 222A and the distal end 226A of the first leg 222A. In one embodiment, the largest cross sectional area of the first leg is adjacent the proximal end 224A of the first leg and the first leg tapers inwardly between the proximal end and the distal end thereof.

In one embodiment, the first leg 222A extends along a first longitudinal axis A_1 . In one embodiment, the first leg 222A includes an insertion tip 228A located at the distal end 226A of the first leg. In one embodiment, the insertion tip 228A has a distal-most point 229A. In one embodiment, the insertion tip 228A is asymmetrical so that it skews outwardly relative to the longitudinal axis A_1 of the first leg 222A.

In one embodiment, the surgical fastener 220 preferably includes a second leg 222B having a proximal end 224B and distal end 226B. The second leg 222B desirably has a first cross sectional area CS1' adjacent the proximal end 224B that is greater than a second cross sectional CS2' adjacent the distal end 226B. In one embodiment, the second leg 222B tapers inwardly between the proximal end 224B of the second leg 222B and the distal end 226B of the second leg 222B. In one embodiment, the second leg 222B has an insertion tip 228B located at the distal end thereof. In one embodiment, the insertion tip 228B has a distal-most point 229B. In one embodiment, the second leg 222B extends along a second longitudinal axis A_2 that is parallel to the first longitudinal axis A_1 of the first leg 222A. In one embodiment, the insertion tip 228B on the second leg 222B is asymmetrical and skews outwardly relative to the second longitudinal axis A_2 of the second leg 222B.

In one embodiment, the first leg 222A includes a distal barb 230A. In one embodiment, the distal barb 230A is positioned adjacent the distal end 226A of the first leg 222A and is proximal to the distal-most point 229A on the insertion tip 228A. The distal barb 230A preferably extends inwardly toward the second leg 222B. In one embodiment, the first leg 222A has a proximal barb 232A that also extends inwardly toward the second leg 222B. In one embodiment, the proximal barb 232A on the first leg 222A is located between the proximal end 224A of the first leg 222A and the distal barb 230A. In one embodiment, the proximal barb 232A is located about halfway between the proximal end 224A and the distal end 226A of the first leg 222A.

In one embodiment, the second leg 222B of the surgical fastener 220 preferably includes a distal barb 230B that extends inwardly toward the first leg 222A. In one embodiment, the distal barbs 230A, 230B on the respective first and second legs 222A, 222B oppose one another, extend toward one another, and are aligned with one another along the lengths of the respective first and second legs 222A, 222B.

In one embodiment, the second leg 222B includes a proximal barb 232B that extends inwardly toward the first leg 222A. In one embodiment, the proximal barbs 232A, 232B on the respective first and second legs 222A, 222B extend toward one another, oppose one another, and are

30

aligned with one another along the lengths of the respective first and second legs 222A, 222B.

In one embodiment, the surgical fastener 220 extends along a central axis A_3 that bisects the surgical fastener 220 into a first half including the first leg 222A with the associated barbs 230A, 232A, and a second half including the second leg 222B with the associated barbs 230B, 232B. The central axis A_3 is preferably parallel to both the first longitudinal axis A_1 of the first leg 222A and the second longitudinal axis A_2 of the second leg 222B. In one embodiment, the central axis A_3 bisects the surgical fastener 220 for splitting the surgical fastener into two evenly sized parts and is equidistant from the first longitudinal axis A_1 of the first leg 222A and the second longitudinal axis A_2 of the second leg 222B.

In one embodiment, the distal barb 230A on the first leg 222A has an inner tip 234A and the distal barb 230B on the second leg 222B has an inner tip 234B. The respective inner tips 234A, 234B define a distance D_1 that extends along an axis that is perpendicular to the central axis A_3 of the surgical fastener 220.

In one embodiment, the proximal barb 232A on the first leg 222A has an inner tip 236A and the proximal barb 232B on the second leg 222B has an inner tip 236B. The inner tips 236A, 236B define a distance D_2 that extends along an axis that is perpendicular to the central axis A_3 of the surgical fastener 220. In one embodiment, the distance D_1 between the inner tips 234A, 234B of the respective distal barbs 230A, 230B is greater than the distance D_2 between the inner tips 236A, 236B of the proximal barbs 232A, 232B. In one embodiment, the distance D_1 is about 0.030 inches and the distance D_2 is about 0.025 inches.

In one embodiment, the first and second insertion tips 228A, 228B are advanced into tissue followed by the first and second distal barbs 230A, 230B to form two spaced tissue openings. In one embodiment, the two tissue openings will be about 0.030 inches apart from one another, which is equal to the distance D_1 between the inner tips 234A, 234B of the first and second distal barbs 230A, 230B. In one embodiment, the first and second proximal barbs 232A, 232B are closer together than are the first and second distal barbs 234A, 234B, which enables the first and second proximal barbs 236A, 236B to grip onto the side walls of the tissue openings previously formed by the first and second distal barbs 230A, 230B. Thus, an important tissue anchoring benefit is gained by providing proximal barbs 232A, 232B that are closer together than the distal barbs 230A, 230B.

In one embodiment, the surgical fastener 220 may be made of absorbable and/or non-absorbable materials. Preferred absorbable materials include PDS, PDS/lactide-glycolide blends, PLA, etc. In one embodiment, each surgical fastener is sized to fit inside of a 5 mm outer diameter tube (a typical trocar cannula dimension). The surgical fastener is fabricated by molding, however, with small modifications, other processes such as casting, stamping, and machining may be used. In one embodiment, the surgical fasteners may be extruded into a general shape, and then formed. In one embodiment, the surgical fasteners may be printed using a 3-D printer.

Referring to FIGS. 19A-19D, in one embodiment, the surgical fastener 220 preferably includes a bridge 238 that interconnects the proximal ends 224A, 224B of the respective first and second legs 222A, 222B. The central axis A_3 of the surgical fastener 220 desirably bisects the bridge. In one embodiment, the bridge 238 includes a major surface 240 that extends adjacent the proximal ends 224A, 224B of

31

the first and second legs 222A, 222B, and a crown 242 that projects proximally from the major surface 240 to define a proximal-most portion of the surgical fastener 220. In one embodiment, the major surface 240 is flat. As will be described in more detail herein, the major surface 240 is preferably engaged by a distal end of a flexible member or by an insertion tool secured to the distal leading of a flexible member for applying an insertion force to the surgical fastener and controlling the orientation of the surgical fastener as the surgical fastener is dispensed from an applicator instrument. In one embodiment, the crown 242 includes a center section 244 that spans the thickness T_1 (FIG. 19D) of the surgical fastener 220 and first and second lateral flanges 246A, 246B that extend laterally from the center section 244.

Referring to FIGS. 19C and 19D, in one embodiment, the surgical fastener 220 has a length L_1 of about 0.248 inches, a width W_1 of about 0.160 inches, and a thickness T_1 of about 0.050 inches. In one embodiment, the width W_1 of the surgical fastener 220 is the same at both the proximal and distal ends thereof, and the width W_1 remains constant between the proximal and distal ends of the surgical fastener to aide with feeding. In other words, the width W_1 defined by the distance between the outer surfaces of the first and second legs 222A, 222B at the proximal ends 224A, 224B of the legs equals the distance between the outer surfaces of the insertion tips 228A, 228B at the distal ends of the first and second legs 222A, 222B. In one embodiment, the cross-sectional areas of the legs decrease when moving from the proximal ends to the distal ends of the respective first and second legs 222A, 222B, however, the width W_1 of the surgical fastener remains constant between the proximal and distal ends of the first and second legs.

In one embodiment, the center section 244 of the crown 242 has a width W_2 of about 0.057 inches and the laterally extending flanges 246A, 246B define a width W_3 of about 0.120 inches.

Referring to FIGS. 19D and 19E, the center section 244 of the crown 242 defines a thickness T_2 that equals the thickness T_1 of the surgical fastener 220. In one embodiment, the laterally extending flanges 246A, 246B have a thickness T_3 of about 0.020 inches.

Referring to FIGS. 19E and 19F, in one embodiment, the crown 242 at the proximal end of the surgical fastener 220 desirably includes the center section 244 and the first and second laterally extending flanges 246A, 246B that extend laterally from the center section 244. The major surface 240 of the bridge extends around the sides of the respective first and second laterally extending flanges 246A, 246B. In one embodiment, a first section 240A of the major surface 240 extends around the first laterally extending flange 246A and has a C-shape, and a second section 240B of the major surface 240 extends around the second laterally extending flange 246B and has a C-shape. The C-shaped first and second sections 240A, 240B of the major surface 240 have the same shape and configuration and oppose one another on opposite sides of the center section 244. In one embodiment, the C-shaped sections 240A, 240B are aligned with the respective longitudinal axes A_1 , A_2 of the first and second legs 222A, 222B (FIG. 19A).

Referring to FIG. 19G, in one embodiment, the first leg 222A of the surgical fastener 220 has a distal end 226A with the first insertion tip 228A having the distal-most point 229A. In one embodiment, the first insertion tip 228A is skewed outwardly relative to the longitudinal axis A_1 of the first leg 222A. The first leg 222A includes the distal barb 230A that extends inwardly toward the second leg 222B of

32

the surgical fastener 220. The second leg 222B has a distal end 226B with the second insertion tip 228B having the distal-most point 229B. In one embodiment, the second insertion tip 228B is skewed outwardly relative to the longitudinal axis A_2 of the second leg 222B. The second leg 222B includes the distal barb 230B that extends inwardly toward the first leg 222A. In one embodiment, the distal barbs 230A, 230B extend toward one another, oppose one another, and are aligned with one another adjacent the distal ends 226A, 226B of the respective first and second legs 222A, 222B. The central axis A_3 bisects the surgical fastener into a first half including the first leg 222A and a second half including the second leg 222B.

Referring to FIGS. 19D and 19G, in one embodiment, the first distal barb 230A and the first proximal barb 232A have respective thicknesses that are less than the thickness or diameter of the first leg 222A, and the second distal barb 230B and the second proximal barb 232B have respective thicknesses that are less than the thickness or diameter of the second leg 222B.

Referring to FIGS. 19G and 19H, in one embodiment, the surgical fastener 220 desirably includes a first major surface 250 that extends over a top side of the surgical fastener 220 and a second major surface 252 that extends over an underside of the surgical fastener 220. In one embodiment, the first major surface 250 is flat and extends over the center section 244 of the crown 242 and the first and second legs 222A, 222B of the surgical fastener. In one embodiment, the second major surface 252 is also flat and extends over opposite sides of the center section 244 of the crown 242 and the first and second legs 222A, 222B. The flat, first and second major surfaces 250, 252 may be used to control the orientation of the surgical fastener as it moves distally through the shaft of an applicator instrument.

In one embodiment, the first leg 222A desirably has an outer surface 227A that extends along the length of the first leg. In one embodiment, the outer surface 227A of the first leg 222A may include a flat surface that is used to control the orientation of the surgical fastener as it moves distally through the shaft of an applicator instrument. In one embodiment, the second leg 222B desirably has an outer surface 227B that extends along the length of the second leg. In one embodiment, the outer surface 227B of the second leg 222B may include a flat surface that is used to control the orientation of the surgical fastener as it moves distally thorough the shaft of an applicator instrument.

Referring to FIGS. 19H and 20, in one embodiment, the flat top and bottom major surfaces 250, 252 of the surgical fastener 220 enable a plurality of surgical fasteners to be stacked one atop another with the flat major surfaces of adjacent surgical fasteners engaging one another within the stack. FIG. 20 shows a stack of three surgical fasteners 220A-220C. The bottom surgical fastener 220A in the stack has a flat top surface 250A and a flat bottom surface 252A. The flat bottom surface 252B of the second surgical fastener 220B in the stack is in contact with the flat top surface 250A of the first surgical fastener 220A. The flat bottom surface 252C of the third surgical fastener 220C in the stack is in contact with the flat top surface 250B of the second surgical fastener 220B. As shown in FIG. 20, the flat major surfaces of the respective surgical fasteners are in contact with one another for maintaining the respective surgical fasteners in the stacked array. Although FIG. 20 shows three stacked surgical fasteners 220A-220C, in other embodiments, the number of surgical fasteners in a stack may be five, 10, 20, 30, or more. In one embodiment, the surgical fasteners may

be loaded into a cartridge 116 (FIG. 2A) to form a stack of deployable surgical fasteners.

Referring to FIGS. 21A-21D, in one embodiment, an insertion tool 260 is used for advancing surgical fasteners toward a distal end of an applicator instrument. In one embodiment, the insertion tool 260 preferably includes a proximal end 262 that may be coupled with a distal end of a flexible member 160 and a distal end 264 that is adapted to engage the proximal end of a surgical fastener. In one embodiment, the insertion tool 260 and the flexible member 160 are features on the same part.

Referring to FIGS. 19A, 19E, and 21A-21D, in one embodiment, the insertion tool 260 preferably includes opposing C-shaped projections 266A, 266B that oppose one another at the distal-most end of the insertion tool. In one embodiment, the C-shaped projections are designed to engage the C-shaped surfaces 240A, 240B of the bridge at the proximal end of the surgical fastener 220. In one embodiment, the insertion tool 260 includes a surface 268, such as a flat surface, adjacent the distal end of the insertion tool that is surrounded by the opposing C-shaped projections 266A, 266B. In one embodiment, the crown 242 at the proximal end of the surgical fastener is disposed within the space bounded by the C-shaped projections 266A, 266B. The distal surface 268 may or may not engage the center section 244 and the first and second laterally extending flanges 246A, 246B of the crown at the proximal end of a surgical fastener. In one embodiment, the center section 244 and laterally extending flanges 246A, 246B of the insertion control surface 242 has a height H_1 (FIG. 19C) and the C-shaped projections 266A, 266B have a height H_2 (measured from surface 68) that is greater than the height H_1 so that the distal surface 268 is spaced from the crown 242 when the C-shaped projections 266A, 266B engage the flat surfaces 240A, 240B. As a result, all of the insertion force transferred from the insertion tool 260 to the surgical fastener 220 is transferred via the C-shaped projections 266A, 266B of the insertion tool engaging the C-shaped surfaces 240A, 240B aligned with the proximal ends of the respective first and second legs 222A, 222B.

In one embodiment, the insertion tool 260 preferably includes a fin or stripper ramp 270 that extends above the C-shaped projections 66A, 66B at the distal end thereof. The stripper ramp 270 has a distal face 272 that lies in a plane with the distal faces 274A, 274B of the respective C-shaped projections 266A, 266B. In one embodiment, the insertion tool includes an attachment flange 275 at the proximal end 262 thereof that is coupled or attached with the distal end of a flexible member so that the insertion tool may move distally and proximally with the flexible member 160. The insertion tool may also be part of the flexible member.

In one embodiment, the insertion tool 260, coupled with the distal end of a flexible member 160 via the attachment flange 275, is advanced distally toward the trailing end of a surgical fastener whereupon the distal-most face 272 of the ramp 270 engages the crown 242 (FIG. 19B) at the proximal/trailing end of the surgical fastener. As the insertion tool advances distally, the proximal end of the surgical fastener is desirably directed into alignment with the major distal surface 268 at the distal end 264 of the insertion tool 260 so that the center section 244 and the first and second laterally extending flanges 246A, 246B (FIG. 19E) of the crown 242 are disposed between the opposing C-shaped projections 266A, 266B of the insertion tool 260.

Referring to FIGS. 21A-21D, in one embodiment, the insertion tool 260 has a flat top surface 290, a flat bottom surface 292, a first flat side surface 294A, and a second flat

side surface 294B. As will be described in more detail herein, in one embodiment, the flat surfaces 290, 292, 294A, and 294B are used to control the orientation of the insertion tool 260 as it moves distally in an applicator instrument.

Referring to FIGS. 19E and 21D, in one embodiment, when the insertion tool engages the proximal end of the surgical fastener 220, the lateral side surfaces 246A, 246B of the center section 244 may engage the ends 276 of the opposing C-shaped projections 266A, 266B for further controlling the orientation of the surgical fastener as it is advanced distally by the insertion tool 260.

Referring to FIGS. 19D and 22A-22D, in one embodiment, the insertion tool 260 may be advanced until the distal end 264 of the insertion tool captures the crown 242 (FIG. 19B) located at the proximal end of the surgical fastener 220. In one embodiment, the center section 244 and the first and second laterally extending flanges 246A, 246B (FIG. 19B) of the crown 242 are captured between the first C-shaped projection 266A and the second C-shaped projection 266B at the distal end 264 of the insertion tool 260. The distal faces 274A, 274B of the C-shaped projections 266A, 266B preferably abut against the C-shaped surfaces 240A, 240B (FIG. 19E) provided at the proximal ends of the respective first and second legs 222A, 222B. In one embodiment, the C-shaped projections 266A, 266B have a height H_2 (FIG. 21C) that is greater than 0.020 inches so that the major distal surface 268 of the insertion tool 260 does not engage the crown 242 at the proximal end of the surgical fastener. As a result, all of the insertion force that is transferred from the insertion tool 260 to the surgical fastener 220 is transmitted via the C-shaped projections 266A, 266B engaging the C-shaped surfaces 240A, 240B aligned with the proximal ends of the first and second legs 222A, 222B.

Referring to FIGS. 22A-22D, in one embodiment, when the crown of the surgical fastener 220 is engaged by the distal end of the insertion tool 260, the flat top surface 290 of the insertion tool 260 is aligned with the flat top surface 250 of the surgical fastener 220, the flat bottom surface 292 of the insertion tool 260 is aligned with the flat bottom surface 252 of the surgical fastener 220, the first flat side surface 294A of the insertion tool 260 is aligned with the flat side surface 227A of the first leg 222A of the surgical fastener 220, and the second flat side surface 294B of the insertion tool 260 is aligned with the flat side surface 227B of the second leg 222B of the surgical fastener 220.

Referring to FIG. 23, in one embodiment, an applicator instrument for dispensing surgical fasteners preferably has an elongated shaft 102 (FIG. 1A) with an elongated conduit 300 for guiding a surgical fastener and an insertion tool joined with the surgical fastener toward the distal end of the elongated shaft. In one embodiment, the elongated conduit 300 preferably includes a top wall 302, a bottom wall 304, and side walls 306A, 306B that extend between the top wall and the bottom wall. The top wall 302 of the elongated conduit 300 has an elongated top notch 308 formed therein and the bottom wall 304 has an elongated bottom notch 310 formed therein. In one embodiment, the elongated top and bottom notches 308, 310 oppose one another and are in alignment with one another.

Referring to FIG. 24, in one embodiment, the distal end of the insertion tool 260 engages the proximal end of the surgical fastener 220 for advancing/pushing the surgical fastener toward the distal end of the elongated shaft of the applicator instrument. In one embodiment, the flat top surface 250 of the surgical fastener 220 and the flat top surface 290 (FIG. 21A) of the insertion tool 260 are opposed by the top wall 302 of the elongated conduit 300, and the flat

35

bottom surface **252** of the surgical fastener **220** and the flat bottom surface **292** (FIG. **21A**) of the insertion tool **260** are opposed by the bottom wall **304** of the elongated conduit **300**. In addition, the flat side surface **227A** of the first leg **222A** of the surgical fastener **220** and the first flat side surface **294A** (FIG. **22C**) of the insertion tool **260** are opposed by the first side wall **306A** of the elongated conduit **300**, and the flat side surface **227B** of the second leg **222B** of the surgical fastener **220** and the second flat side surface **294B** (FIG. **22D**) of the insertion tool **260** are opposed by the second side wall **306B** of the elongated conduit **300**. The opposing top wall **302**, bottom wall **304**, and side walls **306A**, **306B** preferably guide and control the orientation of the surgical fastener **220** and the insertion tool **260** as the joined elements are driven distally through the elongated conduit **300**.

In one embodiment, as the insertion tool **260** moves distally through the elongated conduit **300**, the stripper ramp **270** slides through the elongated top notch **308** in the top wall **302** of the elongated conduit **300** and the attachment flange **275** slides through the elongated bottom notch **310** in the bottom wall **304** of the elongated conduit **300**. The registration of the stripper ramp **270** with the elongated top notch **308** and the attachment flange **275** with the elongated bottom notch **310** preferably provides further control over the orientation and stability of the surgical fastener **220** and insertion tool **260** as the joined elements move distally through the elongated conduit **300**.

Referring to FIG. **25A**, in one embodiment, an applicator instrument for dispensing surgical fasteners preferably includes a cartridge system **316** that utilizes a single linear path to strip a surgical fastener from the bottom of a surgical fastener stack and place the stripped surgical fastener into alignment with a distal end of a flexible member for dispensing the surgical fastener from the distal end of an elongated shaft. In one embodiment, prior to commencing a firing cycle, a surgical fastener stack **378** is arrayed within the cartridge **316**. A pusher **386** is located distal to the surgical fastener stack **378**. In one embodiment, an elevator **388** is aligned with a lower end of the cartridge **316**. The cartridge system includes a slide **390** that is in a proximal-most position. Referring to FIG. **25B**, in one embodiment, when a user squeezes the trigger of the applicator instrument, the slide **390** moves the pusher **386** proximally for pushing the lower-most surgical fastener **376** out of the cartridge **316** and into the elevator **388**.

Referring to FIG. **25C**, as the user finishes squeezing the trigger, the pusher **386** stops moving proximally once the lower-most surgical fastener **376** is fully inside the elevator **388**. The elevator **388** moves down so that the surgical fastener **376** is aligned with the distal end of the flexible member **360**. During this stage, the slide **390** moves proximally from the position shown in FIG. **25B**.

Referring to FIG. **25D**, when the applicator instrument fires, the slide **390** has moved all the way to the left and the elevator **388** is all the way down. The flexible member **360** moves along the lower guide **396** for advancing the surgical fastener **376** toward the distal end of the elongated shaft.

Referring to FIGS. **26A-26B**, **26B-1**, and **26C**, in one embodiment, an applicator instrument includes a cartridge system **416** and a firing system that has one path to strip a surgical fastener from a bottom of a surgical fastener stack during the insertion motion and a different path to follow on the retraction of the flexible member **460**. Referring to FIG. **26B-1**, in one embodiment, the cartridge system and the firing system utilize a racetrack arrangement **465** for providing the first path for distal movement of the flexible

36

member **460** and the second path on the return, proximal motion of the flexible member. In one embodiment, the leading end **460A** of the flexible member **460** is wider to engage with the upper path and a thinner section **460B** of the flexible member **460** just proximal to the leading end **460A** is thinner to allow the flexible member to pass between the first and second paths of the racetrack arrangement **465**.

Referring to FIGS. **27A-27C**, in one embodiment, an applicator instrument utilizes a cartridge system **516** having a rotary motion member **525** to strip a surgical fastener and place the stripped surgical fastener into proper alignment with a distal end of a flexible member **560** for advancement toward the distal end of an elongated shaft. Providing a cartridge system **516** with rotary motion member **525** enables tissue fasteners to be stacked in the cartridge in any orientation relative to the motion of the flexible member **560**. The rotary motion member preferably rotates the surgical fasteners **576** through any angle necessary to bring the surgical fasteners in line with the distal end of the flexible member **560** (90 degree rotation for example).

Referring to FIGS. **28A** and **28B**, in one embodiment, the cartridge system **616** includes a rotary drum **625** containing a plurality of surgical fasteners **676**. The cartridge **616** may be side loading for mounting onto the side of a housing **608** of an applicator instrument **600**. In one embodiment, a distal end of a flexible member **660** passes through an opening in the rotary drum **625** that contains a surgical fastener **676** for advancing the surgical fastener toward a distal end of an elongated shaft. In one embodiment, movement of the trigger rotates the cartridge **616** to present the next surgical fastener for insertion.

Referring to FIGS. **29A** and **29B**, in one embodiment, a cartridge **716** with a rotatory drum may be top mounted onto the housing **708** of an applicator instrument **700**. The distal end of a flexible member **760** advances through the rotatory drum **725** for engaging a surgical fastener **776** for being dispensed from a distal end of an elongated shaft **702**. In one embodiment, movement of the trigger rotates the cartridge **716** to present the next surgical fastener for insertion.

Referring to FIGS. **30A** and **30B**, in one embodiment, an applicator instrument **800** having one or more of the features disclosed herein includes an elongated shaft **802** having an articulating distal end **806**. The applicator instrument **800** includes a housing **808** and a handle **810** coupled with the housing. The handle **810** may be reconfigured between the pistol configuration shown in FIG. **30A** and the in-line configuration shown in FIG. **30B**, as described in detail herein. The applicator instrument **800** preferably includes an articulation lever **814** provided on the housing **808** that may be moved between a horizontal configuration for straightening the distal end **806** of the elongated shaft **802** and a vertical orientation for articulating the distal end **806** of the elongated shaft **802**. The elongated shaft **802** may be articulated with the applicator instrument in either the pistol grip configuration of FIG. **30A**, the in-line configuration of FIG. **30B**, or numerous positions therebetween.

Referring to FIG. **31A**, in one embodiment, the applicator instrument **800** preferably includes a housing **808** and an elongated shaft **802** projecting from a distal end of the housing **808**. In one embodiment, the applicator instrument **800** includes an upper articulation band **805A** and a lower articulation band **805B** that are coupled with an articulating member **815** (FIG. **31B**) located at the distal end **806** of the elongated shaft **802**.

The articulation system preferably includes an upper slider **820A** that is positioned over the top of the elongated shaft **802** and a lower slider **820B** that is positioned below

the elongated shaft. The upper slider **820A** is attached to the proximal end of the upper articulation band **805A** and the lower slider **820B** is attached to the proximal end of the lower articulation band **805B**. The articulation system includes a yoke **822** that is coupled with the upper and lower sliders **820A**, **820B**. In one embodiment, the yoke **822** is capable of rotating relative to the longitudinal axis of the elongated shaft **802**. An upper tension adjustment screw **825A** preferably connects the upper slider **820A** with an upper end of the yoke **822** and a lower tension adjustment screw **825B** preferably connects the lower slider **820B** with a lower end of the rotatable yoke **822**.

In one embodiment, the articulation system includes a cam plate **830** that is rotated by the articulation lever **814** (FIG. 30B) for moving the cam plate between a horizontal configuration associated with a straight elongated shaft and a vertical orientation associated with a fully articulated elongated shaft. The articulation system includes an over-center assembly **835** coupled with the distal end of the cam plate **830** for forcing the cam plate **830** into one of two positions, namely, the horizontal configuration associated with a straight elongated shaft and a vertical orientation associated with a fully articulated elongated shaft. If the cam plate **830** is at an intermediate position between a horizontal orientation and a vertical orientation, the over-center assembly **835** desirably forces the cam plate **830** to rotate into either the horizontal orientation or the vertical orientation. The mechanism to provide this force is a compression spring constrained between two struts. The upper strut is pivotally attached to the housing **808** and the lower strut is pivotally attached to the cam plate **830**. Rotating the cam plate **830** changes the force installed in the compression spring.

Referring to FIGS. 31A and 31B, in one embodiment, the upper and lower articulation bands **805A**, **805B** extend to the articulation member **815** at the distal end **806** of the elongated shaft **802**. In one embodiment, the articulation member **815** includes a plurality of articulating segments or links that are coupled together at the distal end **806** of the elongated shaft **802**, and that are adapted to pivot relative to one another. In one embodiment, the articulating member **815** includes a proximal articulation segment **840** attached to a distal end of the elongated shaft **802**, a plurality of intermediate articulating segments **842A-842D**, and a distal articulating segment **844** that extends to a distal-most end **806** of the elongated shaft **802**. In one embodiment, a distal end face of the distal articulating segment **844** preferably has casting **846** or projections formed thereon for engaging an opposing surface such as a surgical mesh positioned over tissue. The distal-most end of the distal articulating segment **844** has an opening **848** for dispensing a surgical fastener through the opening. The upper and lower articulating bands **805A**, **805B** desirably pass through the elongated shaft **802**, the proximal articulating segment **840**, and the intermediate articulating segments **842A-842D** for being affixed to respective upper and lower portions of the distal articulating segment **844**.

Referring to FIGS. 32A and 32B, in one embodiment, when the articulation lever **814** (FIGS. 30A and 30B) is moved into the vertical orientation, the cam plate **830** rotates in a counterclockwise direction whereupon it is urged by the over-center system **835** to remain in the vertical orientation. The rotation of the cam plate **830**, in turn, causes clockwise rotation of the yoke **822**, which causes the upper slider **820A** to move proximally and the lower slider **820B** to move distally. As the sliders **820A**, **820B** move, tension is increased in the upper articulation band **805A** while enabling the lower articulation band **805B** to move distally. As a

result of increasing the tension on the upper articulation band **805A** while enabling lower articulation band **805B** to move distally, the distal articulating segment **844** moves into the articulated configuration shown in FIG. 32B. For example, a minimum differential band tension is needed to provide a resistance to 2.0-2.5 lbs applied radially to the distal end **806** of the elongated shaft **802**. This amount of resistance ensures that the distal end of the instrument is sufficiently stiff to accommodate the needs of hernia repair procedures, including mesh manipulation and firing of the applicator.

Referring to FIGS. 33A-33C, in one embodiment, the articulation control lever **814** is in a horizontal configuration so that the cam plate **830** has a horizontal orientation and the over-center spring **835** urges the cam plate **830** to remain in the horizontal orientation. With the articulation control lever **814** in the horizontal orientation, the upper slider **820A** and the lower slider **820B** are in alignment with one another along the length of the elongated shaft **802**, and the rotatable yoke **822** has an upper end that is distal to a lower end thereof. With the articulation control lever **814** in the horizontal orientation, the upper articulation band **805A** experiences little or no tension compared to the lower articulation band **805B** so that the articulation member **814** at the distal end **806** of the elongated shaft **802** is in a straight configuration.

Referring to FIGS. 34A-34C, in one embodiment, when the articulation control lever **814** is rotated into a vertical orientation, the cam plate **830** is also in a vertical orientation and the over-center spring **835** urges the cam plate **830** to remain in the vertical orientation. As the cam plate rotates from the horizontal orientation (FIG. 33A) to the vertical orientation shown in FIG. 34A, the rotatable yoke **822** rotates in a clockwise direction so that the upper end of the yoke **822** is proximal to the lower end of the yoke. As a result, the upper slider **820A** moves proximally for tensioning the upper articulation band and the lower slider **820B** moves distally for providing little or no tension on the lower articulation band, compared to the upper articulation band. Referring to FIG. 34E, the tension on the upper articulation band articulates the articulation assembly **815** at the distal end **806** of the elongated shaft **802**.

Referring to FIG. 35, in one embodiment, an intermediate articulating segment **842** preferably has a proximal end **850** and a distal end **852**. The intermediate articulating segment **842** desirably has an upper channel **854** adapted to enable the upper articulating band **805A** (FIG. 31A) to pass therethrough and a lower channel **856** adapted to enable the lower articulation band **805B** (FIG. 31A) to pass therethrough. The intermediate articulating segment **842** also desirably includes a central passageway **858** through which a surgical fastener and the distal end of the flexible member **160** may pass for pushing a surgical fastener toward the distal end of the elongated shaft.

In one embodiment, each lateral side of the intermediate articulating member **842** has a cylindrical pocket **860** that enables adjacent articulating segments to mesh with and pivot relative to one another. In one embodiment, the cylindrical pocket **860** includes upper and lower wings **862A**, **862B** that control the pivoting range of an adjacent articulating member.

In one embodiment, the proximal end of the intermediate articulating segment **842** has a cylindrical shaped projection **864** with a T-shaped head **865** that is adapted to be seated within the cylindrical pocket **860** of an adjacent articulating segment. When the projection **864** is inserted into a pocket **860** of an adjacent segment, the T-shaped head **865** is seated

39

in a concave gap **861** of the pocket **860** and the wings **862A**, **862B** limit the pivoting movement of the adjacent articulating segments relative to one another.

FIG. **32B** shows how the projection **864** of a distal segment fits within the cylindrical pocket **860** of a proximal segment whereupon the segments may pivot relative to one another, with the degree of pivoting controlled by the upper and lower wings **862A**, **862B**. Additionally, the T-shaped head on the projection **864** engages with the wings to prevent the segments from separating from each other, especially when external stresses are applied to the distal end of the elongated shaft.

Referring to FIG. **36**, in one embodiment, an applicator instrument includes an elongated shaft **902** with a conduit **905** extending therethrough that is adapted to receive a flexible member **960** as the flexible member moves toward the distal end of the elongated shaft **902** for dispensing a surgical fastener. In one embodiment, the height of the conduit **905** is greater than the thickness of the flexible member **960** so that the flexible member **960** may buckle a controlled amount to act as a damper and/or limit the travel of the flexible member. In one embodiment, the taller sections of the conduit **905** relative to the thickness of the flexible member **960** may be contained within a designated or limited area of the elongated shaft **902** for providing the dampening effect.

Referring to FIG. **37**, in one embodiment, a distal end **1006** of an elongated shaft **1002** includes a distal positive stop **1015** that extends into the elongated conduit of the elongated shaft **1002**. In one embodiment, an insertion tool **1074** attached to a distal end of the flexible member **1060** preferably includes a rear attachment flange **1025** that is adapted to contact the distal positive stop **1015** for halting further distal movement of the insertion tool **1074** and the flexible member **1060**.

Referring to FIG. **38**, in one embodiment, an applicator instrument **1100** has one or more of the structural components described herein and preferably includes first and second sliders **1120A**, **1120B** coupled with a yoke **1122** for adjusting the tension on articulation bands. A tensioning spring **1185** is coupled with the second slider **1120B**. In one embodiment, the tensioning spring **1185** desirably provides tension force relief in the event that excessive tension force is applied to the lower articulation band secured to the articulation member at the distal end of the elongated shaft **1102**.

Referring to FIG. **39**, in one embodiment, an applicator instrument **1200** has one or more of the structural components described herein and preferably includes first and second sliders **1220A**, **1220B** coupled with a yoke **1222** for adjusting the tension on upper and lower articulation bands. A first tensioning spring **1285A** is coupled with the first slider **1220AB**. In one embodiment, the first tensioning spring **1285A** desirably provides tension force relief in the event that excessive tension force is applied to the upper articulation band secured to the articulation member located at the distal end of the elongated shaft **1202**. In one embodiment, the second tensioning spring **1285B** desirably provides tension force relief in the event that excessive tension force is applied to the lower articulation band secured to the articulation member at the distal end of the elongated shaft **1202**.

Referring to FIG. **40**, in one embodiment, a drive train **124** for an applicator instrument may include a trigger gear **113** attached to the trigger **112**, a drive gear **177** that meshes with the trigger gear **113**, a clutch gear **179** that meshes with the drive gear **177**, a mid gear **181** that may be coupled with

40

the clutch gear **179**, a one-way bearing **183**, and a clutch gear shaft **193** coupled with the clutch gear **179** for simultaneously rotating with the clutch gear. In one embodiment, the one-way bearing **183** is disposed on the clutch gear shaft **193**. In one embodiment, the gears preferably have teeth for meshing with other gears of the gear train **124**.

Referring to FIGS. **40** and **41A**, in one embodiment, the drive train for an applicator instrument preferably includes the drive gear **177** having teeth that mesh with teeth on the clutch gear **179**. When the trigger is fully extended in the ready to fire position, the drive gear **177** and the clutch gear **179** are disengaged via an interruption in the teeth pattern on the drive gear **177**. The drive train desirably includes the mid gear **181** having a central opening that receives the one-way bearing **183**, which, in turn, is mounted on the clutch gear shaft **193**.

Referring to FIGS. **40** and **41B**, in one embodiment, as the trigger **112** is squeezed, the trigger gear **113** rotates the drive gear **177** in the counterclockwise direction shown in FIG. **41B**. The teeth on the drive gear **177** engage the teeth on the clutch gear **179** for rotating the clutch gear **179** in the clockwise direction shown in FIG. **41B**. As the clutch gear **179** rotates clockwise, the clutch gear shaft **193** drives the one-way bearing **183** in a clockwise direction during which the one-way bearing **183** engages the mid gear **181** for rotating the mid gear in the clockwise direction shown in FIG. **41B**. In turn, the mid gear **181** drives the storage reel **126** (FIG. **14A-1**) to wind up the flexible member **160** on the storage reel and store energy in the constant torque spring **172** (FIG. **10B**). In one embodiment, the one-way bearing **183** only engages and drives rotation of the mid gear **181** when the clutch gear shaft **193** rotates in the clockwise direction shown in FIG. **41B**. The one-way bearing **183** preferably free wheels when the clutch gear shaft **193** rotates in a counterclockwise direction that is opposite the clockwise direction so that the clutch gear **179** and the mid gear **181** are decoupled from one another.

Referring to FIGS. **40** and **41C**, when the trigger **112** is almost fully squeezed (i.e., at the end of a trigger stroke), the interrupted tooth section **195** on the drive gear **177** results in the drive gear disengaging from the clutch gear **179**. This disengagement allows the mid gear **181** and the clutch gear **179** to move freely without any resistance from the trigger **112** and the gear train. At this stage, the energy that is stored in the constant torque spring **172** (FIG. **10B**) is released for driving the drive wheel **128** (FIG. **16A**) for advancing the flexible member **160** (FIG. **16B**) toward the distal end of the elongated shaft.

Referring to FIGS. **40** and **41D**, in one embodiment, after the trigger has been fully squeezed and the trigger return spring returns the trigger back to an open position, the trigger gear **113** rotates the drive gear **177** in a clockwise direction. In turn, the drive gear **177** rotates the clutch gear **179** in a counterclockwise direction. As the clutch gear **179** and the clutch gear shaft **193** rotate in the counterclockwise direction, the one-way bearing **183** will disengage from the clutch gear shaft **193**. Thus, the counterclockwise rotation of the clutch gear **179** will not be conveyed to the mid gear **181**, and therefore, the distal end of the firing system that includes the drive wheel **128** and the storage reel **126** will not be impacted by the trigger returning back to the open position. Thus, in one embodiment, the one-way bearing **183** disengages from the clutch gear shaft **193** when the clutch gear shaft rotates in a counterclockwise direction, whereupon the mid gear **181** will not be rotated by the one-way bearing **183**.

Referring to FIGS. **42A-42E**, in one embodiment, the drive wheel **128**, the storage reel **126**, and the spool **139**

41

(FIG. 10B) are assembled onto a carriage 165. In one embodiment, the carriage 165 is normally spring loaded distal but is held proximal by a catch 167. When the insertion tool 174 retracts, it releases the catch 167 and allows the carriage 165 to move distal, urged by a spring. When the carriage 165 moves distal, the storage reel 126 disconnects from the drive train 124 and the constant torque spring 172 accelerates the flexible member 160 and the surgical fastener toward the distal end of the elongated shaft 102 for being inserted into tissue. Once delivered, the flexible member experiences compressive loads due to the inertia of the drive wheel 128. The reaction force to these compressive loads generates a torque on the drive wheel 128, which allows the carriage 165 to move proximal and reset the catch 167. In one embodiment, inserting the cartridge 116 into the housing pushes the carriage 165 toward the proximal end of the housing, which engages/meshes the storage reel 126 with the drive train 124 so that the applicator instrument may be fired. In one embodiment, the cartridge 116 also engages the catch 167. Removing the cartridge 116 releases the catch 167 and allows the carriage 165 to move distal, disengaging the drive gear 165 and the storage reel 126 and preventing the device from being fired.

Referring to FIG. 42A, in one embodiment, the drive wheel 128, storage reel 126, and spool 139 (FIGS. 10A and 10B) are assembled onto a carriage 165. The carriage 165 is spring loaded distal but is held proximal by a catch 167.

Referring to FIGS. 42B and 42C, in one embodiment, when the insertion tool 174 fully retracts, it releases the catch 167 and allows the carriage 165 to move distal, urged by the spring.

Referring to FIGS. 42C and 42D, in one embodiment, when the carriage 165 moves distal, the storage reel 126 disconnects from the drive train 124 and the constant torque spring 172 accelerates the flexible member 160 and the surgical fastener toward the distal end of the elongated shaft 102 for being inserted into tissue.

Referring to FIG. 42E, in one embodiment, once delivered, the flexible member 160 experiences compressive loads due to the inertia of the drive wheel 128. The reaction force to these compressive loads generates a torque on the drive wheel 128, which allows the carriage 165 to move proximal and reset the catch 167.

In one embodiment, the drive wheel 128, storage reel 126, and spool are assembled onto a carriage. The carriage is spring loaded distal but is held proximal by a catch. When the insertion tool 174 retracts, it releases the catch and allows the carriage 165 to move distal, urged by the spring. When the carriage 165 moves distal, the storage reel 126 disconnects from the drive train 124 and the constant torque spring 172 (FIGS. 10A and 10B) accelerates the flexible member 160 and the surgical fastener toward the distal end of the elongated shaft 102 for being inserted into tissue. Once delivered, the flexible member experiences compressive loads due to the inertia of the drive wheel 128. The reaction force to these compressive loads generates a torque on the drive wheel 128, which allows the carriage to move proximal and reset the catch. As a refinement to the above, adding the cartridge 116 pushes the carriage 165 proximal engaging the drive train and storage reel into mesh (allowing the device to be fired). The cartridge 116 also engages the latch 167. Removing the cartridge 116 releases the latch 167 and allows the carriage 165 to move distal, disengaging the mid gear and tape gear and preventing the device from being fired.

In one embodiment, the applicator instrument may only be fired when the cartridge 116 is inserted into the applicator

42

instrument and cannot be fired when the cartridge 116 has been removed from the applicator instrument. Referring to FIGS. 43A-43C, in one embodiment, inserting the cartridge 116 into the applicator instrument pushes the carriage 165 toward the proximal end of the applicator instrument (to the right in FIGS. 43A-43C), which, in turn, couples and/or meshes the drive train 124 with the storage reel 126 (FIGS. 43B and 43C) for enabling the applicator instrument to be fired. In one embodiment, the cartridge 116 may also engage the catch 167. Referring to FIG. 43C, once the cartridge is in place, the cartridge no longer interacts with the carriage. In one embodiment, the cartridge 116 has a notch 159 formed on the right side of the cartridge, which allows the carriage 165 to translate proximally and distally, as necessary, during each firing stroke.

In one embodiment, removing the cartridge 116 releases the catch 167 for allowing the carriage 165 to move distal (to the left in FIGS. 43A-43C), thereby disengaging the drive train 124 from the storage reel 126 and preventing the applicator instrument from being fired.

Referring to FIG. 44A, in one embodiment, a firing system includes a flexible member 1360 that is initially retracted and held proximal by a power spring 1375. A friction drive wheel 1328 has potential energy stored therein. In one embodiment, the energy is stored via a constant torque spring 1372 pre-wound onto the drive wheel during assembly, such that the spring has adequate length to facilitate multiple cycles. Alternative stored energy means may include electro-mechanical, pneumatic, mechanical, motor driven, compressed gas, or other well-known energy sources. The friction drive wheel 1328 is held in place by a lever lock 1370, preventing clockwise rotation.

Referring to FIG. 44B, in one embodiment, squeezing a trigger of the applicator instrument engages an idler wheel 1375 with the flexible member 1360 and toggles the level lock 1370. At this stage, the constant torque spring 1372 has more force than the power spring 1375, and drives the flexible member 1360 distally (to the left in FIG. 44B).

Referring to FIG. 44C, in one embodiment, a proximal or trailing end of the flexible member 1360 includes a wedge 1361 that allows the spring loaded lever 1370 to re-engage with the friction drive wheel 1328. At this stage, the idler wheel 1375 separates from the flexible member 1360, which enables the power spring 1375 to retract the flexible member 1360.

In one embodiment, some of the energy stored in the constant torque spring 1372 is used on each firing cycle. In one embodiment, the constant torque spring 1372 is sufficiently long to provide enough energy for many firing cycles. This design has many benefits including that the force required of the user is very low (e.g., a user may only need to engage the idler wheel 1375 and release the lever lock 1370).

In an alternate embodiment, the constant torque spring is pre-wound onto the drive wheel during assembly to have sufficient stroke for multiple cycles. A latch holds the drive wheel in place, preventing rotation. During the trigger squeeze, the flexible member is wound onto the storage reel but is disengaged from the drive wheel, reducing the trigger force for the user. When the user has nearly completed the trigger stroke, the trigger releases a latch, which engages an idler with the flexible member and the drive wheel. The latch also releases the constraint preventing rotation on the drive wheel. This allows the drive wheel to accelerate due to the force applied by the constant torque spring. The flexible member extends distally, delivering the fastener. As the flexible member reaches the end of its travel, a feature on the

flexible member resets the latch and disengages the flexible member from the drive wheel.

Referring to FIGS. 44A-44C, in one embodiment, the constant torque spring 1372 is pre-wound onto the drive wheel during assembly to have sufficient stroke for multiple cycles. A latch holds the drive wheel in place, preventing rotation. During the trigger squeeze, the flexible member is wound onto the storage reel but is disengaged from the drive wheel, reducing the trigger force for the user. When the user has nearly completed the trigger stroke, the trigger releases a latch, which engages the flexible member with the drive wheel and releases the constraint preventing rotation on the drive wheel. This allows the drive wheel to accelerate due to the force applied by the constant torque spring. The flexible member extends distally, delivering the fastener. As the flexible member reaches the end of its travel, a feature on the flexible member resets the latch and disengages the flexible member from the drive wheel.

FIG. 45A shows an applicator instrument 1400 having one or more of the elements described herein. In FIG. 45A, the applicator instrument 1400 has a flexible member 1460 in an extended position for dispensing a surgical fastener. In FIG. 45B, the flexible member 1460 is in a retracted position. In one embodiment, a torsion spring is assembled with the storage reel 1426. The torsion spring is preferably pre-assembled with a torque greater than that required to wind the flexible member 1460 onto the storage reel. When the insertion tool 1474 is fully retracted, it preferably engages a proximal stop 1473 that prevents further proximal travel of the insertion tool 1474. At this point, the flexible member 1460 is stationary and further travel of the trigger 1412 and gear train 1424 only results in extra compression of the torsion spring. Such a system accounts for tolerances and ensures that the insertion tool 1474 is reliably retracted proximal of the cartridge 116 (FIG. 13B) on every firing cycle.

Referring to FIG. 46A, in one embodiment, an applicator instrument 1500 for dispensing surgical fasteners includes a firing system that incorporates the “coiled snake” concept. In one embodiment, the applicator instrument 1500 includes an actuator or trigger 1512 that is coupled with a gear train 1524, a torsion spring 1572, a pair of drive wheels 1528A and 1528B, and a flexible member 1560 that is advanced distally by the drive wheels.

Referring to FIG. 46B, in one embodiment, as the user squeezes the trigger 1512, one end of the torsion spring is held in place by a latch 1515 and the other end is rotated, compressing the torsion spring and storing potential energy. Referring to FIG. 46C, near the conclusion of the trigger compression stroke, a latch 1515 releases the energy that has been previously stored in the torsion spring 1572. In turn, the energy from the torsion spring 1572 drives the gear train 1524. Referring to FIG. 46D, in one embodiment, the stored energy is transmitted through the gear train 1524 for rotating the upper drive wheel 1528A in a clockwise direction and the lower drive wheel 1528B in a counterclockwise direction. Referring to FIG. 46E, in one embodiment, the rotating drive wheels 1528A and 1528B, which may be friction wheels or cogged wheels, drive the flexible member 1560 toward a distal end of a cannula for delivering a surgical fastener from the distal end of the cannula.

In one embodiment, an applicator instrument for dispensing surgical fasteners may have a firing system that incorporates the “coiled snake” concept disclosed herein whereby the firing system uses energy stored in a spring that is located distal to the gear train. In one embodiment, an applicator instrument for dispensing surgical fasteners may

have a firing system that incorporates the “coiled snake” concept disclosed herein whereby the firing system uses energy stored in a spring located prior or proximal to the gear train, with the energy being released through the drive train to drive the flexible member. In one embodiment, a user squeezes a trigger, handle and/or actuator for compressing a torsion spring. At the conclusion of the compression stroke, a latch is disengaged for releasing the energy stored in the torsion spring. In one embodiment, the energy stored in the compression spring is transmitted through a gear train to a set of drive wheels. The released energy rotates the drive wheels, which, in turn, drive the flexible member down a cannula to deliver the surgical fastener from the distal end of the cannula.

In one embodiment, the applicator instruments disclosed herein may be used during surgical procedures such as hernia repair procedures. In one embodiment, the patient is prepared for surgery in administering anesthesia. Abdominal access is prepared by inserting two or more ports through the abdominal wall and insufflating the abdominal cavity. The hernia site is assessed under direct visualization using a laparoscopic camera. Adhesions are reduced and the peritoneum sac is excised if appropriate. A mesh is placed into the abdominal cavity. The surgeon selects from a variety of cartridges to identify one appropriate for the type of hernia repair being performed. The variety may include different fastener designs, materials, or quantities. The cartridge is attached to the applicator instrument. The distal end of an elongated shaft is advanced through one of the port openings. During a surgical procedure, the distal end of the elongated shaft may be articulated if necessary to access areas requiring fixation, such as the ipsilateral side of the mesh, nearest the port. When the elongated shaft is passed through a port, the reconfigurable handle may be pivoted into a variety of configurations to improve the ergonomics and maneuverability of the surgeon, depending on the location of the target fixation site relative to the port. The applicator instrument may be used to dispense surgical fasteners for securing surgical mesh to tissue. Cartridges can be changed for reloading or changing the type of surgical fastener being dispensed by the applicator instrument. The articulated distal end is straightened before removing the device through the port.

In one embodiment, the applicator instruments disclosed herein may be used during robotic surgical procedures such as hernia repair procedures. In one embodiment, the patient is prepared for surgery in administering anesthesia. Abdominal access is prepared by inserting two or more ports through the abdominal wall and insufflating the abdominal cavity. A surgical robot is attached to the pre-placed ports. The hernia site is assessed under direct visualization using a laparoscopic camera. Adhesions are reduced and the peritoneum sac is excised if appropriate. A mesh is placed into the abdominal cavity. The surgeon selects from a variety of cartridges to identify one appropriate for the type of hernia repair being performed. The variety may include different fastener designs, materials, or quantities. The cartridge is attached to the applicator instrument. The distal end of an elongated shaft is advanced through one of the port openings. The housing assembly is attached to the arm of a surgical robot. During a surgical procedure, the distal end of the elongated shaft may be articulated if necessary to access areas requiring fixation, such as the ipsilateral side of the mesh, nearest the port. The applicator instrument may be used to dispense surgical fasteners for securing surgical mesh to tissue. Cartridges can be changed for reloading or changing the type of surgical fastener being dispensed by the

applicator instrument. The articulated distal end is straightened before removing the device through the port.

In one embodiment, the applicator instruments and surgical tools disclosed herein may be coupled with and/or be in communication with a robotic surgical system, such as the systems and devices disclosed in US 2014/0005662 to Shelton, the disclosure of which is hereby incorporated by reference herein. In one embodiment, the robotic surgical system may have a sterile barrier located between the applicator instruments and surgical tools and the robotic part of the robotic surgical system, whereby the applicator instruments and surgical tools are located in the sterile environment.

In one embodiment, a robotic surgical system may have a master controller and control systems such as the systems and devices disclosed in U.S. Pat. No. 7,524,320, the disclosure of which is hereby incorporated by reference herein. The master controller may have control elements (e.g., knobs, actuators) that are engaged by a surgeon and manipulated in space while the surgeon views a surgical site through a video monitor and/or stereo display. The master controller may include manual input devices that move with multiple degrees of freedom. In one embodiment, the master control has an actuator for actuating surgical tools (e.g., dispensing a surgical fastener).

In one embodiment, the robotic surgical system may include a robotic cart that is configured to actuate a plurality of surgical tools and/or instruments. Various robotic surgery systems and methods employing master controller and robotic cart arrangements are disclosed in U.S. Pat. No. 6,132,368, the disclosure of which is hereby incorporated by reference herein. In one embodiment, a robotic cart may include a base from which surgical tools are supported. In one embodiment, the surgical tools may be supported by a series of manually articulatable linkages, generally referred to as set-up joints, and a robotic manipulator. These structures may have protective covers extending over much of the robotic linkage. The protective covers may be optional, and may be limited in size or entirely eliminated to minimize the inertia that is encountered by servomotors used to manipulate such devices, to limit the volume of moving components so as to avoid collisions, and to limit the overall weight of the robotic cart. In one embodiment, the robotic cart may have dimensions suitable for transporting the cart between operating rooms. The robotic cart is preferably configured to pass through standard operating room doors and onto standard hospital elevators. The robotic cart preferably has a weight and includes one or more wheels that allow the cart to be easily moved and positioned adjacent an operating table.

Other embodiments may incorporate a wide variety of alternative robotic structures, including those described in U.S. Pat. No. 5,878,193, the disclosure of which is hereby incorporated by reference herein. Additionally, while the data communication between a robotic component and the processor of the robotic surgical system is described with reference to communication between the surgical tool and the master controller, similar communication may take place between circuitry of a manipulator, a set-up joint, an endoscope or other image capture device, or the like, and the processor of the robotic surgical system for component compatibility verification, component-type identification, component calibration communication, and confirmation of coupling of the component to the robotic surgical system.

In one embodiment, during a surgical procedure, a surgeon may input control commands to the master controller or a control unit of the robotic surgical system, which

“robotically-generates” output motions that are ultimately transferred to the applicator instruments disclosed herein. As used herein, the terms “robotically-generates” or “robotically-generated” refer to motions that are created by powering and controlling the motors of the robotic surgical system and other power driven components. These terms are distinguishable from the terms “manually-actuable” or “manually generated” which refer to actions taken by a surgeon that result in control motions that are generated independent from those motions that are generated by powering the motors of the robotic surgical system.

While the foregoing is directed to embodiments of the present invention, other and further embodiments of the invention may be devised without departing from the basic scope thereof, which is only limited by the scope of the claims that follow. For example, the present invention contemplates that any of the features shown in any of the embodiments described herein, or incorporated by reference herein, may be incorporated with any of the features shown in any of the other embodiments described herein, or incorporated by reference herein, and still fall within the scope of the present invention.

What is claimed is:

1. An applicator instrument for dispensing surgical fasteners comprising:

a housing;

an elongated shaft extending from the housing, the elongated shaft having a proximal end secured to the housing, a distal end, and a conduit extending between the proximal and distal ends of the elongated shaft; and a firing system disposed within the housing, the firing system comprising an insertion tool having a proximal end and a distal end, the insertion tool having a top surface that extends between the proximal and distal ends thereof and a stripper ramp that projects above the top surface of the insertion tool;

wherein the firing system is configured to be coupled to be a robotic arm that is operable for activating the firing system to commence a firing cycle, wherein during the firing cycle the firing system drives the insertion tool toward the distal end of the elongated shaft, and

wherein the conduit has a top wall that extends between the proximal and distal ends of the elongated shaft and a top notch formed in the top wall, and wherein the stripper ramp of the insertion tool slides through the top notch formed in the top wall of the conduit for guiding movement of the insertion tool as the firing system drives the insertion tool toward the distal end of the elongated shaft.

2. The applicator instrument as claimed in claim 1, wherein the housing includes a cartridge opening, and wherein the applicator instrument further comprises:

a cartridge configured to be inserted into the cartridge opening;

a plurality of surgical fasteners pre-loaded into the cartridge;

a ramp located between the cartridge opening and the proximal end of the elongated shaft for guiding one of the surgical fasteners removed from the cartridge into axial alignment with the conduit of the elongated shaft.

3. The applicator instrument as claimed in claim 2, wherein after the cartridge is inserted into the cartridge opening and prior to commencement of the firing cycle, the surgical fasteners are proximal to a distal end of the housing and are not in alignment with the conduit of the elongated shaft, and wherein during the firing cycle the firing system

47

guides the removed surgical fastener against the ramp and into axial alignment with the conduit of the elongated shaft.

4. The applicator instrument as claimed in claim 3, wherein the cartridge opening is located adjacent the proximal end of the elongated shaft.

5. The applicator instrument as claimed in claim 4, wherein the cartridge comprises:

a cartridge body having an upper end and a lower end, wherein the lower end of the cartridge body is insertable into the cartridge opening;

the surgical fasteners being arranged in a stack that extends between the upper and lower ends of the cartridge body;

a constant force spring that is configured to urge the stack of surgical fasteners toward the lower end of the cartridge body for being fed into the firing system.

6. The applicator instrument as claimed in claim 5, wherein the firing system comprises a flexible firing member having a distal end that is secured to the insertion tool.

7. The applicator instrument as claimed in claim 6, wherein the surgical fastener insertion tool is configured to remove one of the surgical fasteners from a lower end of the end stack of surgical fasteners and drive the removed surgical fastener toward the distal end of the elongated shaft.

8. The applicator instrument as claimed in claim 6, wherein the firing cycle comprises:

an energy storing stage during which energy is built up in a constant torque spring of the firing system; and

an energy releasing stage during which the built up energy is released from the constant torque spring to drive the flexible firing member, the surgical fastener insertion tool, and the removed surgical fastener toward the distal end of the elongated shaft.

9. The applicator instrument as claimed in claim 8, wherein at a beginning of the energy storing stage the surgical fastener insertion tool is adjacent the distal end of the elongated shaft, wherein at an end of the energy storing stage the surgical fastener insertion tool is disposed within the housing and is proximal to the stack of surgical fasteners, and wherein during the energy releasing stage the surgical fastener insertion tool strips the removed surgical fastener from the lower end of the stack of surgical fasteners and drives the removed surgical fastener toward the distal end of the elongated shaft.

10. The applicator instrument as claimed in claim 9, wherein the firing system includes the constant torque spring for storing the energy during the energy storing stage, and wherein the applicator instrument further comprises:

a gear train coupled with the robot arm, wherein during the energy storing stage of the firing cycle the gear train meshes with the firing system for enabling the energy to be stored in the constant torque spring.

11. The applicator instrument as claimed in claim 1, wherein the insertion tool has a bottom surface and first and second side surfaces that extend between the top and bottom surfaces of the insertion tool, and wherein the insertion tool further comprises an attachment flange adjacent the proximal end of the insertion tool that extends below the bottom surface of the insertion tool.

12. The applicator instrument as claimed in claim 11, wherein the elongated conduit includes a bottom wall that opposes the top wall of the elongated conduit, the bottom wall extending between the proximal and distal ends of the elongated shaft, the elongated conduit having first and second side walls that extend between the top and bottom walls of the elongated conduit, the elongated conduit including a bottom notch formed in the bottom wall of the

48

elongated conduit for guiding movement of the insertion tool as the firing system drives the insertion tool toward the distal end of the elongated shaft.

13. The applicator instrument as claimed in claim 1, wherein the elongated shaft has a distal shaft section that is configured to move between a straight configuration and an articulated configuration, and wherein the robotic arm is coupled with the elongated shaft for controlling movement of the distal shaft section between the straight configuration and the articulated configuration.

14. An applicator instrument for dispensing surgical fasteners comprising:

a distal housing assembly including a housing having a cartridge opening and an elongated shaft extending from the housing, the elongated shaft having a proximal end secured to the housing, a distal end, and an elongated conduit extending between the proximal and distal ends thereof;

a cartridge inserted into the cartridge opening, the cartridge containing a plurality of surgical fasteners that are pre-loaded into the cartridge; and

a firing system disposed within the housing for dispensing the surgical fasteners, the firing system comprising:

a flexible member having a distal end configured to reciprocate between the proximal and distal ends of the elongated shaft; and

an insertion tool having a proximal end and distal end, wherein the proximal end of the insertion tool is secured to the distal end of the flexible member, the insertion tool having a top surface that extends between the proximal and distal ends thereof and a stripper ramp that projects above the top surface of the insertion tool;

wherein the firing system is configured to be coupled to a robotic arm that is operable for activating the firing system to commence a firing cycle, wherein during a first stage of the firing cycle the stripper ramp removes one of the surgical fasteners from the cartridge and during a later stage of the firing cycle the firing system drives the insertion tool and the removed surgical fastener toward the distal end of the elongated shaft, wherein the elongated conduit has a top wall with an elongated top notch formed in the top wall, and wherein during the later stage of the firing cycle the stripper ramp slides through the elongated top notch formed in the top wall of the elongated conduit as the firing system drives the insertion tool and the removed surgical fastener toward the distal end of the elongated shaft.

15. The applicator instrument claimed as in claim 14, further comprising:

a ramp located between the cartridge opening of the housing and the proximal end of the elongated shaft for guiding the removed surgical fastener into axial alignment with the elongated conduit of the elongated shaft, wherein after the cartridge is inserted into the cartridge opening and prior to commencement of the firing cycle, the surgical fasteners that are pre-loaded into the cartridge are proximal to a distal end of the housing and are not in alignment with the conduit of the elongated shaft, and wherein during the firing cycle the firing system guides the removed surgical fastener against the ramp and into axial alignment with the conduit of the elongated shaft.

16. The applicator instrument claimed as in claim 14, wherein the elongated conduit has a bottom wall with an elongated bottom notch formed in the bottom wall, wherein

49

the insertion tool has an attachment flange adjacent the proximal end of the insertion tool that slides through the elongated bottom notch during the later stage of the firing cycle, and wherein the elongated top and bottom notches guide proximal and distal movement of the insertion tool within the elongated conduit.

17. The applicator instrument as claimed in claim 14, wherein the removed surgical fastener comprises:

- a first leg having a proximal end, a distal end, and a first longitudinal axis that extends between the proximal and distal ends of the first leg;
- a second leg having a proximal end, a distal end, and a second longitudinal axis that extends between the proximal and distal ends of the second leg;
- a bridge interconnecting the proximal ends of the first and second legs;

wherein the distal end of the insertion tool comprises a distal face located below the stripper ramp, and wherein the housing comprises an inclined surface located adjacent a proximal end of the elongated conduit for vertically shifting a position of the removed surgical fastener from a first position in which the removed surgical fastener is in contact with the stripper ramp to a second position in which the removed surgical fastener is not in contact with the stripper ramp and is in alignment with the elongated conduit of the elongated shaft.

18. The applicator instrument as claimed in claim 17, wherein:

the bridge of the surgical fastener comprises a major surface that faces away from the distal ends of the first and second legs, and a crown that extends proximally from the major surface of the bridge, wherein the crown has a cruciform shape including a center section that is centered on the major surface of the bridge, first and second flanges that extend laterally from opposite sides of the center section, a first C-shaped section that surrounds the first flange, and a second C-shaped section that opposed the first C-shaped section and that surrounds the second flange; and

the insertion tool comprises first and second C-shaped projections on opposite ends of the distal face that oppose one another and that extend distally from the distal face, wherein after the surgical fastener has been vertically shifted into the second position the first C-shaped projection engages the first C-shaped section of the major surface that surrounds the first flange of the crown, the second C-shaped projection engages the second C-shaped section of the major surface that surrounds the second flange of the crown, and the center section of the crown is disposed between the first and second C-shaped projections.

19. The applicator instrument as claimed in claim 14, wherein the elongated shaft has an articulated distal shaft section that is movable between a straight configuration and a curved configuration, and wherein the robotic arm is coupled with the elongated shaft for controlling movement of the articulated distal shaft section between the straight configuration and the curved configuration.

20. An applicator instrument for dispensing surgical fasteners comprising:

- a distal housing assembly including a housing and an elongated shaft extending from the housing, the elongated shaft having a proximal end secured to the housing, a distal end, and an elongated conduit extending between the proximal and distal ends of the elongated shaft; a surgical fastener disposed within the housing;

50

an inclined surface located adjacent a proximal end of the elongated conduit for guiding the surgical fastener into alignment with the elongated conduit; and

- a firing system disposed within the housing, the firing system comprising an insertion tool having a distal end with a distal face and a stripper ramp at the distal end of the insertion tool that projects above the distal face; wherein the firing system is configured to be coupled to a robotic arm that is operable for activating the firing system to commence a firing cycle, wherein during the firing cycle the firing system is configured to drive the insertion tool toward the distal end of the elongated shaft; and

wherein the firing cycle comprises:

- a first stage during which the stripper ramp engages a proximal end of the surgical fastener, the surgical fastener is proximal to the inclined surface, and the surgical fastener is not in alignment with the elongated conduit,
- a second stage during which the stripper ramp remains in contact with the proximal end of the surgical fastener and a distal end of the surgical fastener is in contact with the inclined surface, and
- a third stage during which the distal face of the insertion tool is in contact with the proximal end of the insertion tool, the stripper ramp is not in contact with the proximal end of the surgical fastener, and the surgical fastener is in alignment with the elongated conduit.

21. The applicator instrument as claimed in claim 20, wherein the elongated conduit has a top wall that extends between the proximal and distal ends of the elongated shaft and a top notch formed in the top wall, and wherein during the third stage of the firing cycle the stripper ramp slides through the top notch for guiding distal movement of the insertion tool as the insertion tool is driven toward the distal end of the elongated shaft.

22. An applicator instrument for dispensing surgical fasteners, comprising:

- a housing;
- an elongated shaft extending from the housing, the elongated shaft having a proximal end secured to the housing, a distal end, and a conduit extending between the proximal and distal ends of the elongated shaft;
- a firing system disposed within the housing, the firing system comprising an insertion tool having a proximal end and a distal end, the insertion tool having a top surface that extends between the proximal and distal ends thereof and a stripper ramp that projects above the top surface of the insertion tool;

wherein the firing system is configured to be coupled to be a robotic arm that is operable for activating the firing system to commence a firing cycle, wherein during the firing cycle the firing system drives the insertion tool toward the distal end of the elongated shaft,

wherein the conduit has a top wall that extends between the proximal and distal ends of the elongated shaft and a top notch formed in the top wall, and wherein the stripper ramp of the insertion tool slides through the top notch formed in the top wall of the conduit for guiding movement of the insertion tool as the firing system drives the insertion tool toward the distal end of the elongated shaft; and

- a handle rotatably coupled to the housing for movement relative to the housing to each of a plurality of operative positions, the handle including an actuator coupled to

51

the firing system so that operation of the actuator commences the firing cycle.

23. The applicator instrument as claimed in claim **22**, wherein the handle includes a locking element for selectively securing the handle in at least one of a first position 5 relative to the housing comprising a pistol configuration, a second position relative to the housing comprising an in-line configuration, and a third position relative to the housing comprising an inverted pistol configuration.

24. The applicator instrument as claimed in claim **23**, 10 wherein the locking element includes a reconfiguration button on a proximal handle assembly, the reconfiguration button being actuatable to unlock the handle for rotation relative to the housing.

* * * * *

15

52